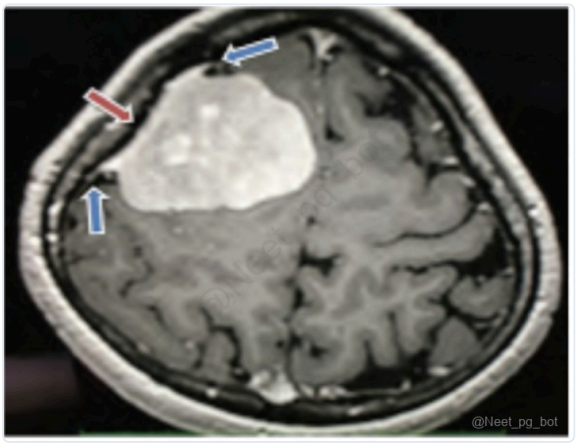
Radiology - Previous Year Questions

Marrow Q-Bank Collection • NEET PG / INI-CET / AIIMS (2017-2025)

Total Questions: 130
High-Yield Study Edition

Q1. Radiology NEET 2025 ID: MH0135

1. A female presented with recurrent headaches. Identify the lesion in the given MRI.



A Glioma

B ✓ Meningioma

C Ependymoma

D Pilocytic astrocytoma

Correct Answer: Meningioma

The most probable diagnosis of the above patient is meningioma. Meningioma shows a 'dural tail' extending from the tumour along the dura mater. It is a well-defined, hyperintense and homogeneously enhanced mass that arises from the meningocytes or arachnoid cap cells of the meninges. They mostly arise in the supratentorial region, but can also be seen in the infratentorial region. Meningomas are more commonly seen in females and are associated with the NF-2 mutation. Many small meningiomas are found incidentally and are entirely asymptomatic. Most common presentations include headache, paresis and change in mental status. Other options: Option A: Gliomas are intra-axial tumours. Option C: Ependymoma arise from the neoplastic transformation of the ependyma. They are usually intraventricular, but extraventricular rests of ependymal cells may give rise to cerebral hemisphere tumours. Option D: Pilocytic astrocytoma is a benign astrocytoma that typically occurs in children and young adults. MRI reveals a cystic mass.

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References: Grainger & Allison's Diagnostic Radiology 7th edition • Page no 1415, 1420, 1427, 1430
Online Resource • https://radiopaedia.org/articles/meningioma#

Q2. Radiology NEET 2025 ID: MH0143

2. What is the most appropriate statement regarding the given investigation?

A Invasive and can be therapeutic treatment for bladder calculi

B ✓ Non invasive and used to diagnose ureteropelvic junction obstruction

C Non invasive and gold standard for bladder cancer

D Requires percutaneous access to the renal pelvicalyceal system

Correct Answer: Non invasive and used to diagnose ureteropelvic junction obstruction

The given image shows Intravenous Urography (IVU). It is non-invasive and used to diagnose ureteropelvic junction obstruction. IVU is a non-invasive radiographic study of the renal parenchyma, pelvicalyceal system, ureters and urinary bladder using intravenous contrast medium. IVU can help to identify: * Ureteric obstruction, Ureteropelvic junction obstruction * Papillary necrosis * Anatomical variants such as a horseshoe kidney * The course of the ureters * Renal function Intravenous access is required for the injection of a contrast agent. It can be one of two main forms: bolus or drip infusion urography.

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References: Grainger & Allison's Diagnostic Radiology 7th edition • Page no 764
Online Resource • https://radiopaedia.org/articles/intravenous-urography

Q3. Radiology NEET 2025 ID: MH0498

3. A 55-year-old male presented with a pulsative swelling in the neck. The swelling was compressible and refilled on removing of pressure. The digital angiography image is given below. Identify the condition.

A Carotid aneurysm

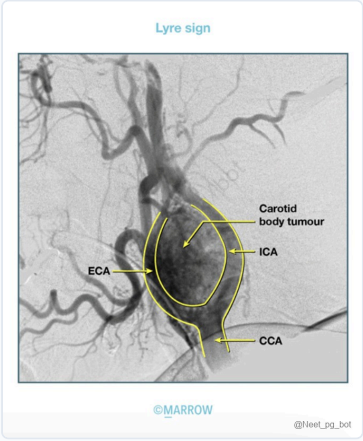
B ✓ Carotid body tumour

C Strawberry hemangioma

D AV fistula

Correct Answer: Carotid body tumour

The given clinical features, along with the image showing splaying of internal and external carotid arteries, suggest the diagnosis to be a carotid body tumour, also called a chemodectoma. Carotid body tumors are mostly benign, neurogenic tumors or paragangliomas arising from the chemoreceptor cells of the carotid bulb. They are related to catecholamine-secreting neuroendocrine tumors like pheochromocytomas and occasionally occur concurrently with them. Familial cases are caused by mutations in the SDH gene. The pathophysiology is believed to be chronic hypoxia in high-altitude states. Clinical features are a painless, slow-growing, pulsatile lateral neck mass, usually in the 5th decade, firm in consistency moves from side to side but not up to down. The tonsil may be displaced medially. A bruit may be associated with the tumor and sometimes the mass can be emptied and slow refilling occurs. On the angiogram, splaying of the internal and external carotid arteries at the common carotid bifurcation can be seen. This is known as Lyre's sign, as shown below:



FNAC is absolutely contraindicated. The Shamblin classification is utilised to assess the resectability of these tumors. Other Options: Option A: Carotid aneurysm is a swelling in the neck at the level of the thyroid cartilage, below the angle of the mandible. It is pulsatile, smooth, soft, nontender, horizontally mobile, with bruit. Neurological features due to embolic episodes, hoarseness of voice, Horner's syndrome, and dysphagia due to swelling extending into the tonsillar bed may also be present. Option C: Strawberry hemangiomas are capillary hemangiomas that may be present at birth or present a few weeks after birth. Option D: AV fistula is an abnormal communication between an artery and a vein.

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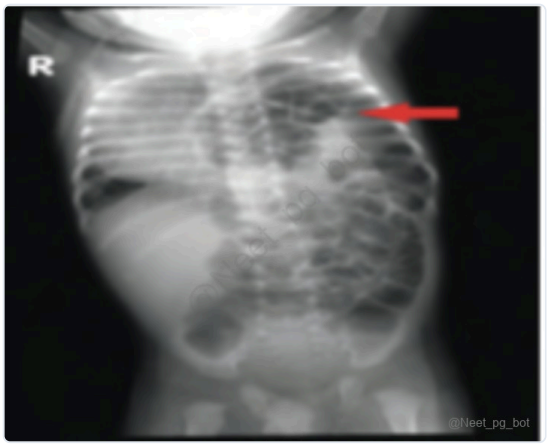
References: Bailey and Love's short practice of Surgery - 28th Edition • Page no: 809-810
P L Dhingra Diseases of Ear Nose and Throat and Head and neck surgery - 8th Edition • Page no: 463, 519
SRB Manual of surgery, 6th edition • Page no: 196, 197, 206, 210

Q4. Radiology NEET 2025 ID: MH0154

4. A newborn is admitted to the NICU after 24hour with severe respiratory distress. What is the diagnosis from the given xray?



A Congenital pulmonary airway malformation

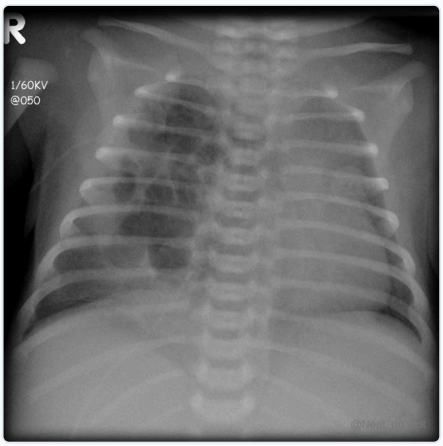
B Loculated pneumothorax

C ✓ Congenital diaphragmatic hernia

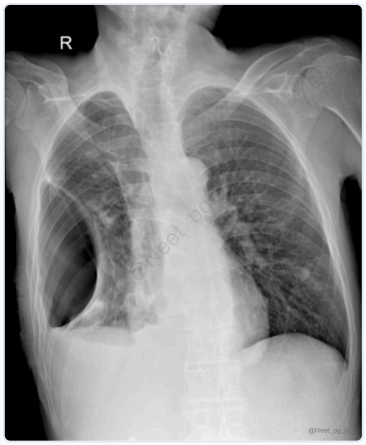
D Congenital lobar emphysema

Correct Answer: Congenital diaphragmatic hernia

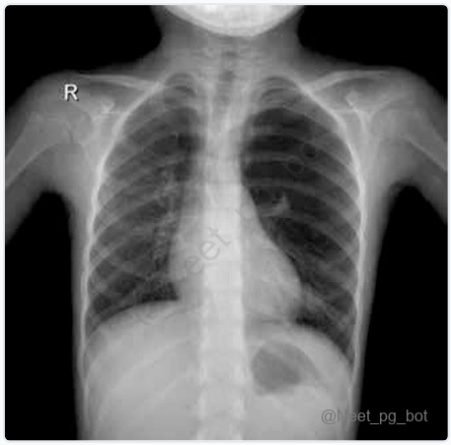
The above X-ray chest shows intra-thoracic bowel shadows along with mediastinal shift suggestive of congenital diaphragmatic hernia. Congenital diaphragmatic hernia (CDH) is a birth defect in the diaphragm. Malformation of the diaphragm allows the abdominal organs to push into the chest cavity, hindering proper lung formation and resulting in pulmonary hypoplasia. The most common type of CDH is a Bochdalek hernia, a posterolateral defect which usually involves the left side. Other options: Option A: Congenital pulmonary airway malformation (CPAM) is a group of cystic and non-cystic lung lesions resulting from early airway maldevelopment. They can communicate with the airways and are frequently infected.



Option B: Loculated pneumothorax is a form of pneumothorax where a pocket of pleural air is trapped within a localised area.

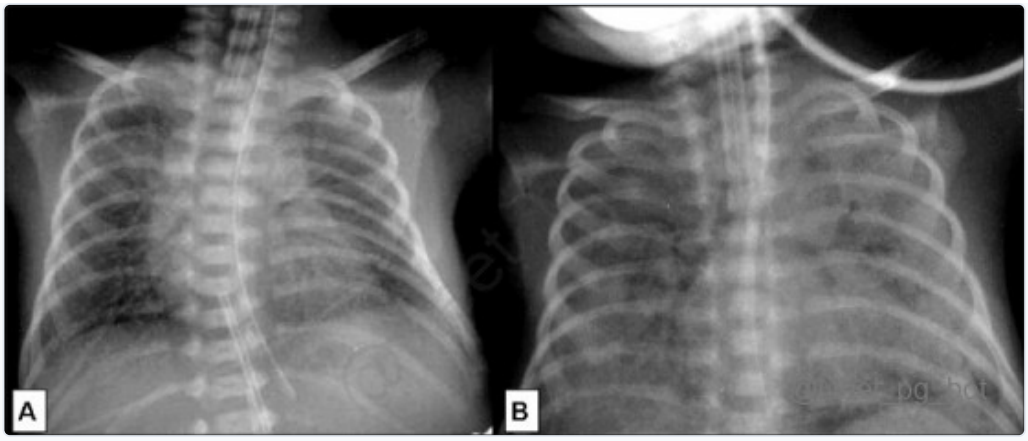


Option D: Congenital lobar emphysema is a bronchial abnormality that results in a check valve mechanism causing progressive hyperinflation of the affected lobe.

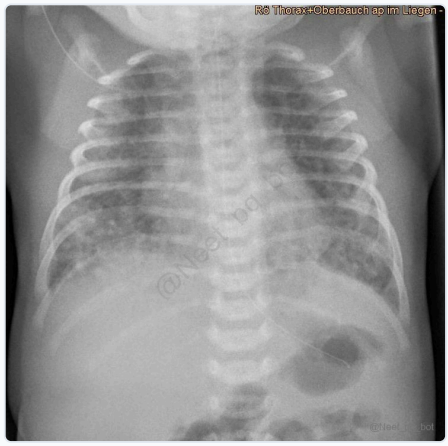


Pearl PM0028: Radiologic features of the major respiratory distress disorders in newborns Congenital Diaphragmatic Hernia * Mediastinal shift * Bowel loops in the left hemithorax

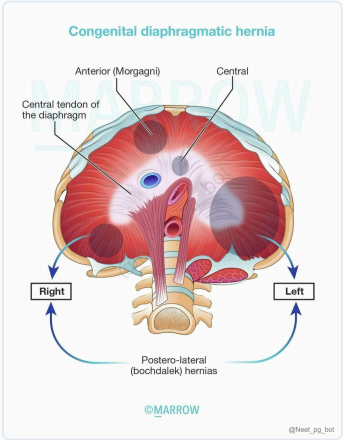
Respiratory Distress Syndrome * Low lung volume * Symmetric ground-glass opacity (reticulogranular pattern) * Air bronchogram * White-out lung (severe disease)



Chest radiograph (A) on Day 2: shows reticulogranular opacities throughout both lungs with prominent air bronchograms. The cardiac silhouette is preserved and both lungs are hypoaerated. Chest radiograph (B) on Day 9 (untreated): shows more severe changes. There is obliteration of the cardiac and diaphragmatic contours due to marked opacification, giving a complete 'white out' appearance Meconium Aspiration Syndrome * Hyperinflation * Patchy infiltrates * Coarse streaking



Pearl PM2134: Congenital diaphragmatic hernias (CDH) It is due to the failure of the closure of the pleuroperitoneal canal in the developing fetus (Normally the pleuroperitoneal cavities separate by 8-10 weeks of gestation). Left-sided hernias are more common(80%). The two well-recognized congenital sites where abdominal viscera can herniate into the chest are: * The foramen of Morgagni: a hernia in the anteromedial part of the diaphragm with a defect between the sternal and costal attachments. The most commonly involved viscus is the transverse colon. * The foramen of Bochdalek: through the dome of the diaphragm posterolaterally CDH causes pulmonary hypoplasia (bilateral) by compressing the developing lung. These lungs have smaller bronchi, with less bronchial branching and less alveolar surface area than lungs in normal infants. Pulmonary hypertension may also develop due to increased thickness of arteriolar smooth muscle and also by multiple local and systemic vasoactive factors. Clinical features: * respiratory distress at birth * grunting respiration * chest retractions * dyspnea * cyanosis * scaphoid abdomen * decreased breath sounds along with bowel sound over the chest with CDH * shifting of heart sounds to the right (for left-sided CDH)



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References: Grainger & Allison's Diagnostic Radiology 7th edition • Page no 1792, 1793
Online Resource • https://radiopaedia.org/articles/loculated-pneumothorax

Q5. Radiology NEET 2025 ID: MH0148

5. A 50-year-old female has complaints of right flank pain since the past 6 months. Identify the lesion.

A Gall bladder cancer

B Retroperitoneal tumor

C Liver abscess

D ✓ Right renal neoplasm

Correct Answer: Right renal neoplasm

The given clinical scenario of a 50-year-old female with complaints of right flank pain for the past 6 months, and the coronal section of the CT abdomen showing a right renal mass, suggests the lesion to be a right renal neoplasm. Renal neoplasm is usually seen in individuals aged 50 to 70 years and usually presents with flank pain, palpable mass in the flank or hematuria. CT is commonly used for the evaluation of a renal mass. On non-contrast CT, lesions are of soft tissue attenuation, while larger lesions frequently have areas of necrosis. Some lesions also have calcification. The image below shows a normal CT section of the abdomen.

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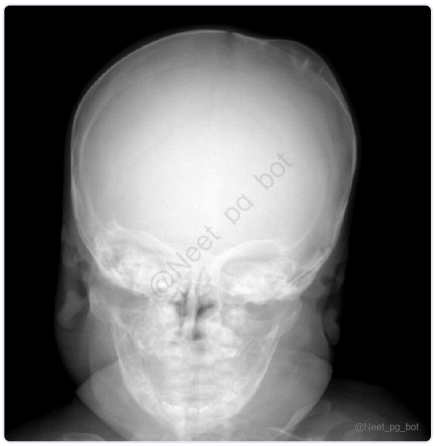
References: Grainger & Allison's Diagnostic Radiology 7th edition • Page no 723, 726, 727
Online Resource • https://radiopaedia.org/articles/renal-cell-carcinoma-1

Q6. Radiology NEET 2023 ID: MF8231

A 2-month-old infant is brought to the OPD with a parietal swelling present since birth. The X-ray image is given below. What could be the probable diagnosis?



A Subgaleal hematoma

B ✓ Cephalhematoma

C Caput succedaneum

D Enecephalocele

Correct Answer: Cephalhematoma

In the given clinical picture, the swelling over the infant's head is likely to be a **cephalhematoma**.

Cephalohematoma is a **collection of blood** in between the pericranium and the flat bone **(subperiosteal)** of the skull, usually unilateral and over a parietal bone. It is **never present at birth** but gradually develops after 12–24 h. The swelling is **limited by the suture lines** of the skull as the pericranium is fixed to the margins of the bone.

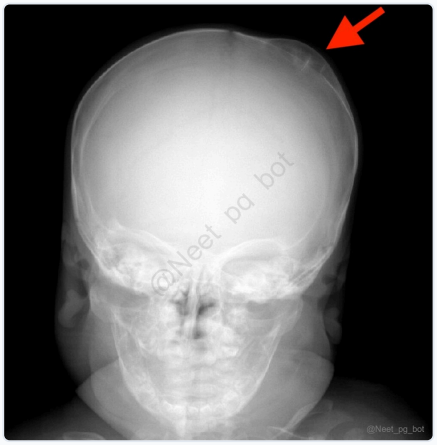
The arrow mark in the image represents cephalohematoma.

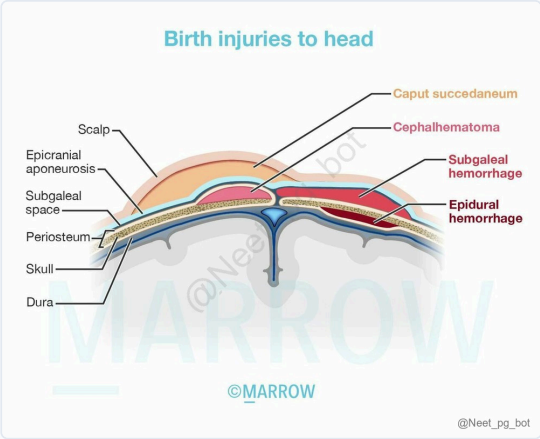
Note: Cephalhematoma is more likely to be identified soon after birth, and its presence as a defined area of swelling that does not cross suture lines is more consistent with this diagnosis.

Subgaleal hematoma occurs **between the galea aponeurotica and the periosteum.** It is commonly caused by birth trauma, particularly forceps or vacuum-assisted delivery. Subgaleal hematoma can cross suture lines and typically presents as a soft, fluctuant swelling on the scalp.

Caput succedaneum is an **edema** of the **scalp** that can **cross suture lines**. It is usually **present at birth** or develops shortly after due to pressure on the presenting part of the head during labor. It generally **resolves within a few days**.

Encephalocele is a neural tube defect that results in a **sac-like protrusion of the brain and its meninges** through an opening in the skull. It is a congenital condition and is typically diagnosed at birth or via prenatal ultrasound.





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References: DC Dutta's Textbook of Obstetrics - 8th edition, Page no: 558
DC Dutta's Textbook of Obstetrics 9th edition, Page no: 452

Q7. Radiology NEET 2023 ID: MF8079

A patient presented with abdominal pain and sterile pyuria, and the x-ray showed the following features. What is the likely diagnosis?

A ✓ Putty kidney

B Nephrocalcinosis

C Staghorn calculus

D Psoas calcification

Correct Answer: Putty kidney

A patient with **abdominal pain** and **sterile pyuria,** with the given x-ray showing **diffuse alveolar calcification** in the **right renal region** point towards a diagnosis of a **putty kidney**.

The putty kidney is a characteristic finding of genitourinary tuberculosis (TB), specifically **renal TB.** The given image shows a homogenous and ground-glass-like calcification of the kidney appearing like putty. It is caused by the dissemination of the organism through the bloodstream and is thus always a secondary TB. There is either reinfection or reactivation of old TB.

Genitourinary TB is a cause of **sterile pyuria**. Diagnosis is confirmed with a tuberculin test, or at least 3 consecutive early-morning specimens of urine are examined for acid-fast bacilli. They are stained with a Ziehl-Neelsen stain and subsequently cultured.

CT is the most sensitive modality and is the preferred means of visualizing renal calcifications. CT urography is done to assess the various stages of renal TB and its manifestations, which are as follows:

- Early:
  - Papillary necrosis
- Progressive:
  - Multifocal strictures
  - Generalized or focal hydronephrosis
  - Mural thickening and enhancement
  - Poorly enhancing renal parenchyma
- End-stage:
  - Progressive hydronephrosis
  - Amorphous dystrophic calcification that involves the entire kidney - **putty kidney**

Treatment involves short-course therapy with antitubercular drugs like pyrazinamide, isoniazid, and rifampicin. Partial nephrectomy, nephrectomy, or reconstructive surgery is often done.

Option B: Nephrocalcinosis - This condition is characterized by the deposition of calcium in the kidney parenchyma or tubules, often seen in hypercalcemic conditions like hyperparathyroidism. However, it would not explain the clinical presentation of sterile pyuria.

Option C: Staghorn calculi are large kidney stones that fill the renal pelvis and extend into the renal calyces, often caused by urinary tract infections with organisms such as *Proteus*, *Pseudomonas*, and *Staphylococcus*. While these calculi can cause abdominal pain, they usually do not result in sterile pyuria, as they are associated with infection.

The image below shows a staghorn calculus of the left kidney.

Option D: Psoas calcification - This refers to calcification within the psoas muscle and is often seen after acute renal failure and rhabdomyolysis. It is not associated with sterile pyuria and is not the cause of the patient's symptoms.

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References: Online Resource, https://radiopaedia.org/articles/putty-kidney
Bailey & Love's Short Practice of Surgery - 27th Edition, Page 1045, 1406
Online Resource, https://pubmed.ncbi.nlm.nih.gov/2229553/
Online Resource, https://www.uptodate.com/contents/nephrocalcinosis
Online Resource, https://openi.nlm.nih.gov
Grainger and Allison, textbook of medical imaging, 7th edition, Page no 748

Q8. Radiology NEET 2023 ID: MF8204

A woman with endometrial carcinoma is undergoing radiotherapy. Which of the following is true?

A ✓ Intensity is inversely proportional to the square of the distance from the source

B Small blood vessels are most radioresistant

C Rapidly proliferating cells are most radioresistant

D Small intestinal mucosa is most radio-resistant

Correct Answer: Intensity is inversely proportional to the square of the distance from the source

The true statement is that the **intensity** of radiation is **inversely proportional** to **the square of the distance from the source** in brachytherapy.

Radiotherapy is a common treatment modality for various cancers, including endometrial carcinoma. It works by **targeting rapidly dividing cancer cells with ionizing radiation**, which can **damage the DNA** of these cells and **lead to cell death.** Radiotherapy can be **delivered externally** (**external beam** radiotherapy) or internally (brachytherapy). In **brachytherapy**, **radioactive sources** are **placed within or adjacent to the tumor**, leading to a high radiation dose delivered directly to the tumor while **minimizing damage to surrounding healthy tissue**.

Other options:

Option B: Larger blood vessels are more radioresistant than microvasculature.

Option C: Permanent non-dividing cells are more radioresistant than rapidly proliferating cells.

Option D: Small intestinal epithelial cells (intestinal crypts), which divide rapidly with no differentiation are the most radiosensitive.

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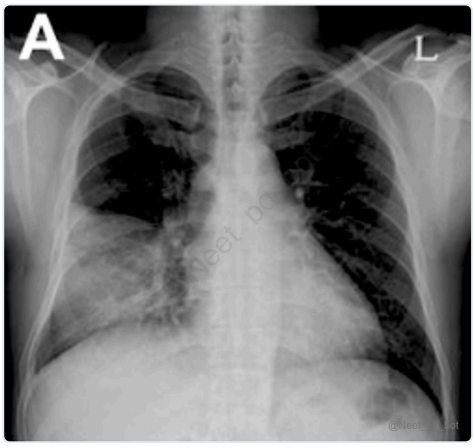
References: Grainger & Allison's Diagnostic Radiology 6th edition, Pg 1741
Basic Radiotherapy Physics and Biology, 2014 edition, Page no. 267, 268

Q9. Radiology NEET 2023 ID: MF8222

A patient presents with a history of fever and cough for the past 3-5 days. His examination findings reveal crepitations. Chest X-ray is given below. What is the probable diagnosis?



A Apical segment of Right lower lobe consolidation

B ✓ Right middle lobe consolidation

C Posterior segment of right lower lobe consolidation

D Loculated pleural effusion

Correct Answer: Right middle lobe consolidation

The given clinical scenario of **fever**, **cough**, and **crepitations** suggests features of **pneumonia**. The **chest X-ray** showing **opacities** with **obscuration** of the **right cardiac border** is suggestive of **right middle lobe consolidation**.

This **loss of clear demarcation** between the normal lung and its surrounding soft structures (heart, diaphragm) is called the **silhouette sign**. It occurs due to **loss of contrast** between lungs and soft tissue when the lung parenchyma is no longer filled with air. This sign may be seen in consolidation, atelectasis, or mass lesions.

In the following images:

- Left image - red line indicates the **normal** demarcation between the right heart border and middle lobe of the right lung
- Right image - **loss of demarcation** i.e, positive **silhouette sign** due to **right middle lobe consolidation**

Positive silhouette sign and corresponding anatomical site of pathology:

- **Right heart border -** right middle lobe or medial right lower lobe
- **Left heart border -** lingula segments of the left upper lobe
- Left hemidiaphragm or descending aorta - left lower lobe
- **Aortic knuckle -** left upper lobe (posterior segment)
- Right hemidiaphragm - right lower lobe
- Right paratracheal stripe - right upper lobe

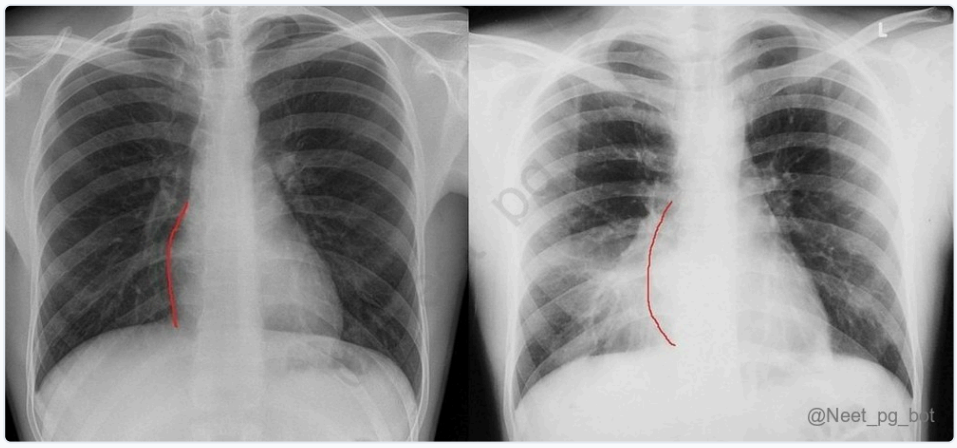
Other options:

Apical segment of Right lower lobe consolidation: In this case, the chest X-ray would show an opacity or consolidation in the apical segment of the right lower lobe. The consolidation would be located in the lower part of the right lung, towards the apex. The right heart border, however, would remain visible, as it is not adjacent to the apical segment of the right lower lobe.

Posterior segment of right lower lobe consolidation: A chest X-ray revealing consolidation in the posterior segment of the right lower lobe would show an opacity in the lower part of the right lung, more posteriorly. This would not affect the visibility of the right heart border, as it is not adjacent to the posterior segment of the right lower lobe. In a posteroanterior (PA) chest X-ray, this consolidation may be difficult to appreciate, but it would be more evident in a lateral view.

Loculated pleural effusion: In a chest X-ray showing a loculated pleural effusion, there would be a fluid collection within the pleural space. This fluid collection would appear as a well-defined opacity, typically with a convex border facing the lung parenchyma. It may appear as a lenticular shape, conforming to the space between the lung and chest wall or diaphragm. The

loculated pleural effusion would not create a silhouette sign with the right heart border, as it is located in the pleural space rather than the lung parenchyma.



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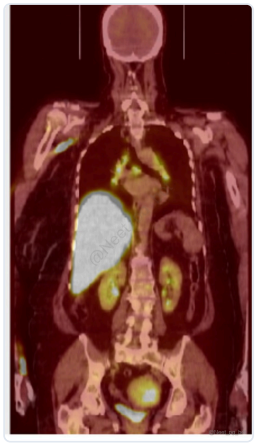
References: Online Resource, https://radiopaedia.org/articles/lobar-consolidation
Grainger & Allison's Diagnostic Radiology 7th edition, Page no: 174, 171
Online Resource, https://radiopaedia.org/articles/silhouette-sign-x-rays

Q10. Radiology NEET 2022 ID: MF5405

Identify the imaging modality given below.



A Cerebral blood flow scan

B ✓ PET CT

C Sestamibi scan

D Full body MDCT

Correct Answer: PET CT

The above imaging modality is a **PET CT** scan.

PET (**positron-emission tomography**) CT scan is a **hybrid** technique that combines PET and CT scan that is used for **targeted imaging** of tumor physiology and biology. It is based on the principle that **tumor cells** generally have **high metabolic activity** compared with normal tissues.

A small dose of the radiopharmaceutical agent **18F-fluorodeoxyglucose (FDG)**. This enters cells and is phosphorylated to **FDG 6-phosphate** which cannot be metabolized further. Since it is a highly polar molecule, it **cannot** easily **diffuse back** out of the cell. After intravenous injection of FDG, it continuously **accumulates** in the metabolically active cell.

The main **advantage** of PET-CT is the **identification** of distant **metastatic disease** not seen on CT alone. The disadvantages of PET scans are **radiation burden**, **poorer spatial** resolution when used alone, and **high cost.**

Other options:

Option A: **Cerebral blood flow** scan refers to the **perfusion MRI** that uses gadolinium as contrast and provides information about the cerebral blood flow and volume.

Option B: The **sestamibi scan** is used for **parathyroid localization** and is now considered the most **accurate** and **reliable** method for imaging the parathyroid glands. Sestamibi **accumulates** in **mitochondria.** Parathyroid **adenomas** often have a **high** concentration of **oxyphilic cells** with **high mitochondrial content**. These retain sestamibi tracer and adenomas are therefore associated with a **slow washout** when compared with the thyroid gland.

Option D: The **full body MDCT** scan refers to the **multidetector CT** scan which is an advanced CT imaging technique that uses multiple detector elements that provide improvement in speed, quality, and scope.

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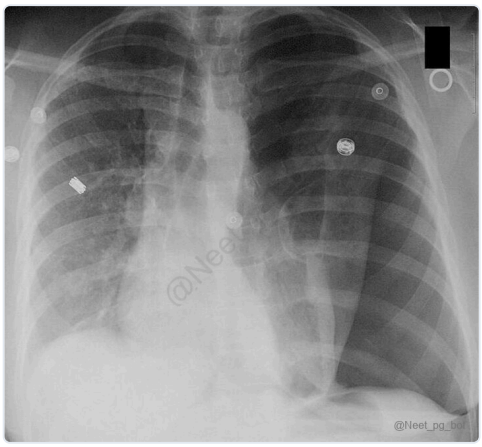
References: Bailey & Love's Short Practice of Surgery - 26th Edition, Page 827,1091
Grainger and Allison, textbook of medical imaging, 7th edition, Page 1642,1643
Harrison's Principles of Internal Medicine 20th Edition, Page 3037
Textbook of Radiology and Imaging: Volume 2, Page 1837

Q11. Radiology NEET 2022 ID: MF5404

A patient is brought to the casualty following a road traffic accident. On examination, the patient is conscious, BP is 90/60 mm Hg, respiratory rate is 40 breaths per minute, and pulse rate is 120 bpm. The X-ray is shown below. What is the next best step in management?



A ✓ Chest tube insertion

B Pericardiocentesis

C Thoracotomy

D Pleurodesis

Correct Answer: Chest tube insertion

The given **X-ray** features showing a well-demarcated area **devoid** of **lung markings** on the **left** side with **tracheal deviation** and the **mediastinal shift** to the **right side,** along with the clinical scenario is suggestive of **tension pneumothorax**. The next best step in the management is **chest tube insertion**.

Tension pneumothorax is a **clinical diagnosis,** and treatment should never be delayed while waiting for radiological confirmation. Imaging modalities used are ultrasound–extended focused assessment with sonar for trauma (**eFAST**), chest x-ray, and spiral CT.

A tension pneumothorax develops when there is a tear either in the **parietal** or **visceral pleura**. This tear acts as a **one-way valve** i.e., air enters from the lung or chest wall through it during inspiration but has no way to escape during expiration. This ultimately leads to the **collapse** of the **affected lung**.

It usually occurs following **penetrating** or blunt **chest trauma,** iatrogenic lung injury, or mechanical positive pressure ventilation.

It is characterized by:

- Chest pain
- **Tachypnea** and acute dyspnea
- Sudden onset **hemodynamic compromise** (tachycardia, hypotension, low O_2 saturation, loss of consciousness)
- Neck vein distention
- **Tracheal deviation** away from the side of the injury
- Decreased chest movements on the affected side
- **Hyper-resonance** on percussion on the affected side
- **Absence** or **decreased breath sounds** on the affected side

The X-ray below shows a **left-sided tension pneumothorax**. There is a large, well-demarcated area devoid of lung markings on the left side. The yellow arrows are pointing to the tracheal deviation and the mediastinal shift to the right side.

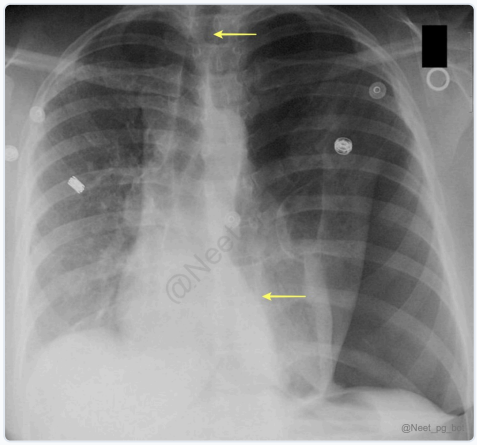
Management of tension pneumothorax:

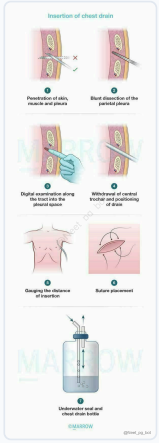
- **Immediate** management- **needle decompression** or **needle thoracostomy** i.e. insertion of a wide-bore needle in the 5th ICS anterior to the midaxillary line
- **Definitive** management- **tube thoracostomy** or **chest tube insertion** in the 5th ICS in the **triangle of safety,** identified as a triangle that lies anterior to the midaxillary line above the nipple, below, and lateral to the pectoralis major muscle.

The images below show the triangle of safety and technique of chest drain insertion.

Other options:

- **Pericardiocentesis** is done in cases of **cardiac tamponade** which is characterized by the presence of hypotension, distended jugular veins, and muffled heart sounds (**Beck's triad**). The **breath sounds** are **absent** in **tension pneumothorax** while they are **present in cardiac tamponade.**
- **Surgery** for pneumothorax can be performed by video-assisted thoracoscopic surgery **(VATS)** or as an open procedure **(thoracotomy)** in cases of spontaneous pneumothorax **failing** to settle despite chest drainage. Pleural adhesion is achieved by chemical **pleurodesis** where talc is used and is insufflated into the chest cavity.





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References: Advanced Trauma Life Support - Student Course Manual, 10e, Page no: 65,66
Bailey & Love's Short Practice of Surgery - 28th Edition, Page no: 319, 374

Q12. Radiology INI-CET 2023 ID: MF9219

A patient presented with dysuria and pain in the back. His lumbar spine X-ray shows the below finding. What is the likely diagnosis?

A Wilson's disease

B Addison's disease

C Cystic fibrosis

D ✓ Hyperparathyroidism

Correct Answer: Hyperparathyroidism

The given clinical scenario and X-ray showing **diffuse calcification of both kidneys** are suggestive of **nephrocalcinosis** which is seen in **hyperparathyroidism**.

Nephrocalcinosis is the **deposition of calcium** in the kidney parenchyma or tubules which is seen in conditions causing **hypercalcemia** like **hyperparathyroidism**.

Clinical features of hyperparathyroidism:

- Bones:
  - Pathological fractures
  - Subperiosteal bone resorption - 2nd and 3rd digits, most common in the radial aspect
  - Salt and pepper skull
  - Osteitis fibrosa cystica or VonReckling Hausen's disease of bone
- Stones
  - Multiple and recurrent renal stones - most common
  - Nephrocalcinosis
- Abdominal groans - colicky due to pancreatitis
- Psychiatric overtones - raised PTH causing CNS depression and psychiatric symptoms
- Band keratopathy
- Features of hypercalcemia -
  - Vomiting
  - Confusion
  - ECG changes: increased PR interval, shortened QT interval

Investigations done in primary hyperparathyroidism:

- **Ultrasonography** of the neck is a safe and widely used technique for the localization of abnormal parathyroid glands.
- Nuclear medicine scanning with **radiolabeled Sestamibi** is also a widely used technique. Sestamibi accumulates in parathyroid adenomas. This radionuclide is concentrated in the thyroid and parathyroid tissue but usually washes out of normal thyroid tissue in under an hour. It persists in abnormal parathyroid tissue.
- **Dual-energy radiographic absorptiometry** is a useful tool to demonstrate skeletal involvement in primary hyperparathyroidism.

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References: Sabiston Textbook of Surgery: The Biological Basis of Modern Surgical Practice, 20e, Page no: 927
Online Resource, https://emedicine.medscape.com/article/127351-overview#a3
Online Resource, https://radiopaedia.org/cases/nephrocalcinosis-3
Harrison's Principles Of Internal Medicine - 19th Edition, Page No: 313

Q13. Radiology INI-CET 2023 ID: MF9200

Identify the finding in the given ultrasound image.

A Mercedes-Benz sign

B Gall bladder polyp

C Gall bladder cancer

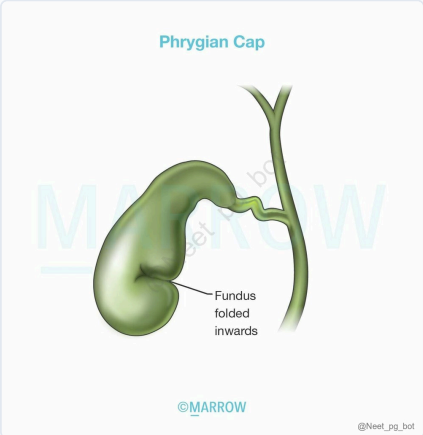
D ✓ Phrygian cap

Correct Answer: Phrygian cap

The ultrasound examination of the gallbladder shows a **Phrygian cap**. It is a **normal anatomical variant.**

Phrygian cap refers to an anatomical variant of the gallbladder, where, the **fundus of the gallbladder folds on itself**, or is separated from the body by a partial or complete septum. A true septum is uncommon and is usually an **apparent septum** due to the gallbladder wall folding. It is an **incidental finding** that may be seen on CT, MRI, or ultrasound. It causes no symptoms.

The **Phrygian cap** is named so, due to the resemblance of the ultrasonographic appearance to a hat or cap that is worn by the inhabitants of ancient Phrygia circa 1200 BC.



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References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no. 656
Harrison's Principles of Internal Medicine, 20th Edition, Page no. 2423
Grainger & Allison's Diagnostic Radiology Essentials 2nd edition, Page no. 360, 364
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4956632/

Q14. Radiology INI-CET 2023 ID: MF9206

Banana sign on USG is associated with which of the following conditions?

A ✓ Spina bifida

B Omphalocele

C Downs syndrome

D Klinefelter syndrome

Correct Answer: Spina bifida

The banana sign on USG is seen in **spina bifida**.

Banana sign is an abnormal anterior **curvature** of the **cerebellum** along with effacement of the cisterna magna that produces this appearance. It is seen in **spina bifida** and is also associated with **Arnold-Chiari** malformation.

Spina bifida is a congenital anomaly occurring due to incomplete development of the neural tube. The most commonly cited signs present in spina bifida are the banana sign and the lemon sign.

The **lemon sign** occurs due to the overlapping of the frontal bones due to the posterior shift of intracranial contents as shown below.

Other options:

Option B: **Omphalocele** is a congenital midline anterior abdominal wall defect through which the solid abdominal viscera or bowel herniates. The herniated abdominal contents are covered by a membranous sac as shown below.

The image below shows an ultrasound image of the fetus suggestive of an omphalocele (asterisk).

Option C: **Nuchal translucency (NT)** is used to antenatally screen **Downs syndrome (Trisomy 21)** during the first trimester. NT is the **maximum thickness** of the **translucent area** between the skin and soft tissue over the **fetal spine** at the back of the **neck**. When the NT measurement is increased, it is a marker for **fetal aneuploidy** and **structural anomalies**. The ultrasound image below shows increased NT thickness and absent nasal bone of a fetus with Trisomy 21 at 11 weeks gestation.

Lemon sign
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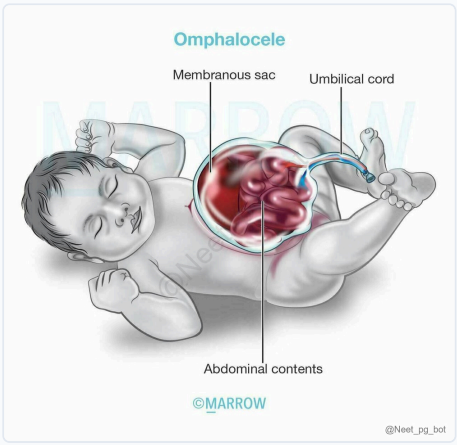
Omphalocele
Membranous sac
Umbilical cord
Abdominal contents
©MARROW
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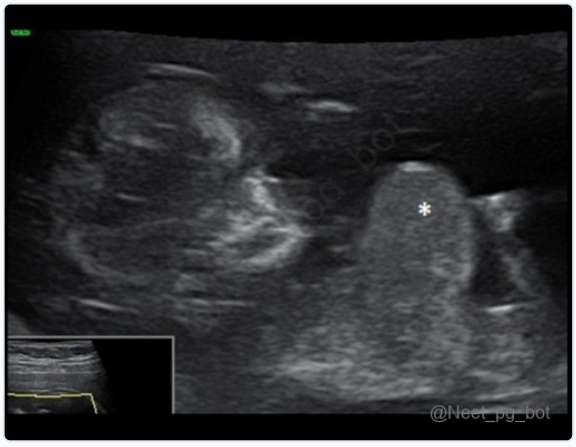
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8.3cm / 1.4 / 47Hz TIs 0.2
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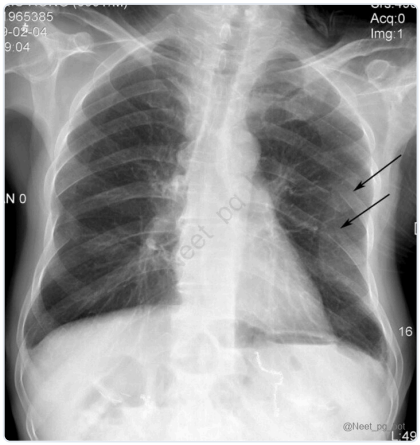
References: Williams Obstetrics 25th Ed, Page no 193
DC Dutta's Textbook of Obstetrics - 8th edition, Page no 375, 377, 471, 567, 570, 736
Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK559265/
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7864317/#B33

Q15. Radiology INI-CET 2023 ID: MF9215

Identify the fracture site in the X-ray shown below.



A 5th, 6th rib

B 5th, 6th & 7the rib

C 7th and 8th rib

D ✓ 6th and 7th rib

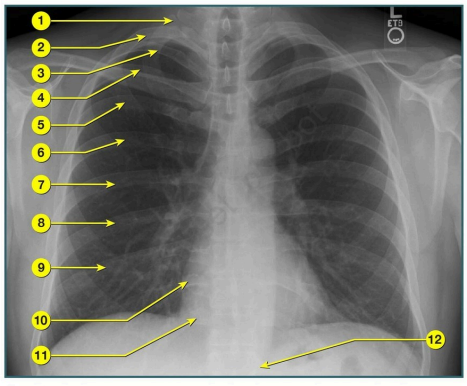
Correct Answer: 6th and 7th rib

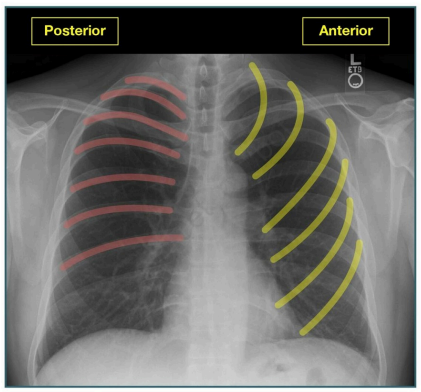
The given X-ray shows the **fracture** of the **6th and 7th rib**

In the chest X-ray, ribs appear as **light grey strips**. The **anterior part** of the rib is **more oblique** and the medial part cannot be demonstrable as it ends with costal cartilage. The **posterior part** of the rib is **more horizontal** and the attachment of the rib with the vertebra can be easily made out.

Normal Chest X-ray Rib Identification

The anterior(yellow) and posterior(red) ribs are highlighted in the image given below.





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References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no. 186
Online Resource, https://glassboxmedicine.com/2019/02/10/radiology-normal-chest-x-rays/

Q16. Radiology INI-CET 2023 ID: MF9197

Which of the following is not an advantage of USG over Mammography?

A ✓ Can be used for screening mammographically undetectable clinically non palpable breast mass

B Useful in young females with dense breast

C Differentiates between solid and Cystic lesions in the Breast

D Guided biopsy can be done

Correct Answer: Can be used for screening mammographically undetectable clinically non palpable breast mass

USG cannot be used for **screening** mammographically undetectable clinically nonpalpable breast mass. Although ultrasound has its advantages in certain scenarios, **mammography** remains the **primary screening tool** for breast tumor detection.

Mammography is complementary to physical examination and can detect clinically non-palpable breast mass. The primary purpose of mammography is to **detect small breast tumors**. While mammography is typically the initial screening tool for breast cancer detection, there are specific scenarios where ultrasound may be preferred or used in conjunction with mammography. Advantages of USG over mammography include:

- USG is preferred in **pregnant** women, **lactating** women, and **young women** since breast tissue is **dense** (Option B) in these populations.
- USG can **differentiate** between **solid** and **cystic lesions** (Option C) of the breast and is able to detect abnormal axillary lymph nodes.
- Ultrasound can assist in **guided breast biopsy** (Option D) procedures.

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References: Sutton's Radiology and Imaging for Medical Students 7th edition, Page no 1475
Online Resource, https://www.uptodate.com/contents/breast-imaging-for-cancer-screening-mammography-and-ultrasonography

Q17. Radiology INI-CET 2023 ID: MF9203

An MRCP image is shown below. Which among the following is the most likely diagnosis?

A ✓ Choledocholithiasis

B Cholecystitis

C Pancreatic duct stricture

D Malignancy

Correct Answer: Choledocholithiasis

The given **MRCP image** shows **multiple round filling defects within** the **common bile duct** indicative of **choledocholithiasis**.

Choledocholithiasis or common bile duct stones are classified based on their point of origin. Choledocholithiasis is mostly clinically silent but may present with biliary colic and even obstructive jaundice. **Management of choledocholithiasis** includes **ERCP +/- sphincterotomy** followed by laparoscopic cholecystectomy.

MRCP uses the fluid which is present **in the biliary and pancreatic ducts** as a contrast agent and does not require external contrast material administered into the ductal system. It is a **non-invasive** investigation for evaluating extrahepatic, intrahepatic bile duct, and pancreatic duct. **No interventions** are performed under MRCP.

MRCP uses **T2 weighted MRI** and it recognizes slow-flowing or stationary fluid within ducts, which appear brighter relative to adjacent solid tissues. MRCP can be used to evaluate various conditions of the pancreaticobiliary ductal system, examples include:

- Congenital anomalies of the cystic and hepatic ducts
- Post-surgical biliary anatomy
- Pancreas divisum
- Anomalous pancreaticobiliary junction
- **Choledocholithiasis**
- Biliary structures
- Chronic pancreatitis
- Pancreatic cystic lesions
- Trauma to the biliary system

MRCP in **benign strictures** would typically show **smooth elongated tapering of the bile duct**, while **malignancy-associated strictures** typically show **abrupt shouldering**.

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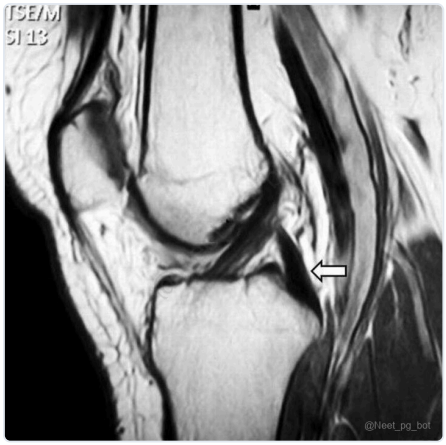
References: Sabiston Textbook of Surgery 20th Edition, Page no 1494 - 1497
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5608926/

Q18. Radiology INI-CET 2023 ID: MG1155

The image given below shows the MRI of the knee. Identify the marked structure.



A ACL

B ✓ PCL

C MCL

D Patellar tendon

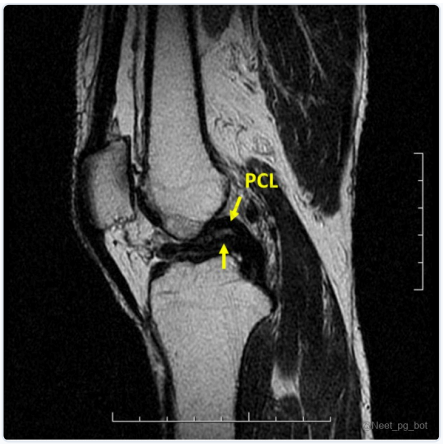
Correct Answer: PCL

In the above given image of an MRI of the knee joint, the arrow shows the **posterior cruciate ligament.**

The posterior cruciate ligament (PCL) is the only **primary restraint** to **posterior translation** throughout the entire **knee range of motion.** It prevents the internal rotation of the tibia on the femur when there is varus stress on the knee.

It acts as a **secondary restraint** to valgus/varus, e**xternal rotation.**

Anterior cruciate ligament	Posterior cruciate ligament
Attached to the anterior part of the intercondylar area of the tibia	Posterior part of an intercondylar area of the tibia
Direction – Upwards, backwards & laterally & attached to posterior aspect of medial surface of lateral condyle.	Upwards, forwards & medially & attached to the anterior aspect of the lateral surface of the medial condyle.
Prevents posterior dislocation of femur on tibia and forwards dislocation of tibia on femur.	Prevents anterior dislocation of femur on tibia and posterior dislocation of tibia on femur
Limits hyperextension of knee joint(taut during extension)	Limits hyperflexion of knee joint. (taut during flexion)
Weaker	Stronger



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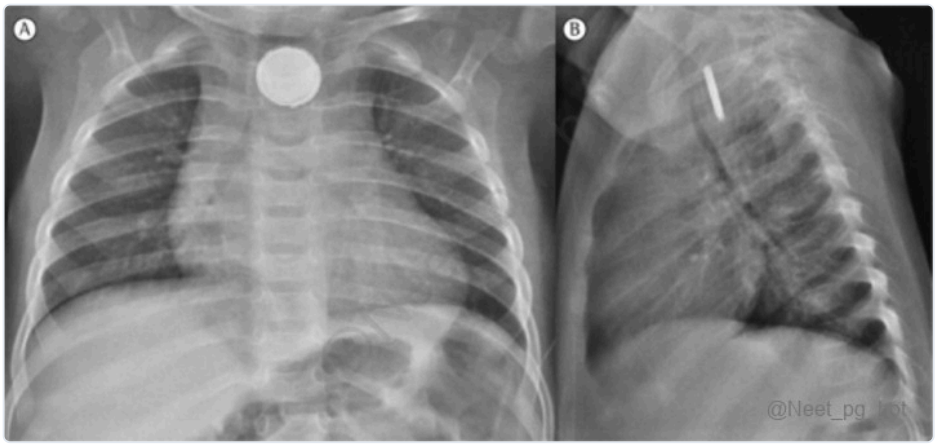
References: Essential Orthopaedics by Maheswari, 5th edition, Page no 356
Apley's System Of Orthopaedics And Fractures - 10th Edition, Page no 576

Q19. Radiology INI-CET 2023 ID: MG1027

A child was brought to the emergency with drooling and choking following ingestion of a foreign body. Based on the given X-ray, what object was most likely swallowed by the child?



A Cap of a cool drink bottle

B Coin

C ✓ Button battery

D Metal ball

Correct Answer: Button battery

The presence of a **halo sign** or **double ring sign** is indicative of the foreign body being a **button battery**. Furthermore, the presence of **irregularities** along the **periphery** of the battery indicates the **possibility of corrosion**.

Button batteries and sharp objects in the oesophagus and foreign bodies associated with **respiratory symptoms** all mandate urgent endoscopic removal.

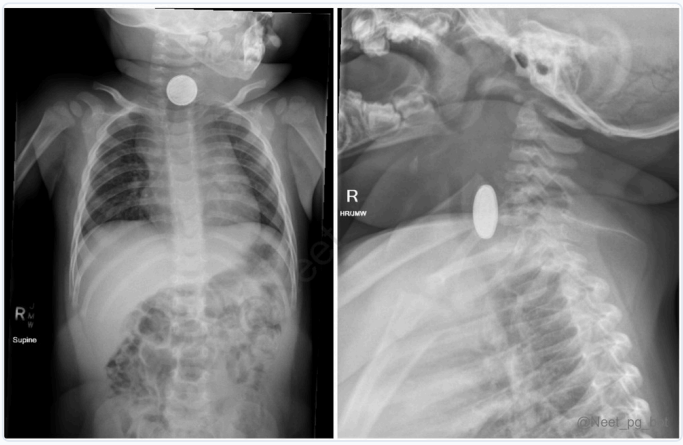
Button batteries, in particular, are very dangerous foreign bodies, which must be expediently removed because they can cause:

- **Mucosal injury** in 1-hour contact time
- Involve **all layers** of the **oesophagus** in 4 hours

Injury is thought to be secondary to electrochemical burns from electrical discharge. **Chemical burns** from electrolyte alkali leakage and pressure necrosis also may contribute to mucosal injury.

Asymptomatic blunt objects and coins lodged in the oesophagus can be observed for up to 24 hr in anticipation of passage into the stomach.

Another AP and lateral X-ray demonstrating the presence of a button battery in the oesophagus is as shown below.



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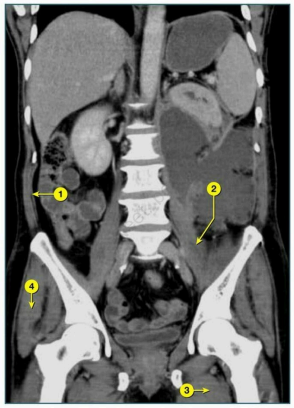
References: Online Resource, https://radiopaedia.org/cases/button-battery-oesophageal-impaction-1
Grainger & Allison's Diagnostic Radiology 7th edition, Page No. 1790
Nelson Textbook of Pediatrics 20th Edition, Page No. 672

Q20. Radiology NEET 2021 ID: MF1617

A 15 year old patient with history of spinal tuberculosis was brought to the hospital with low grade fever, dull backache and persistent flexion of hip joint. Hip joint motion was limited and painful. Identify the muscle affected in the CT image.



A 1

B ✓ 2

C 3

D 4

Correct Answer: 2

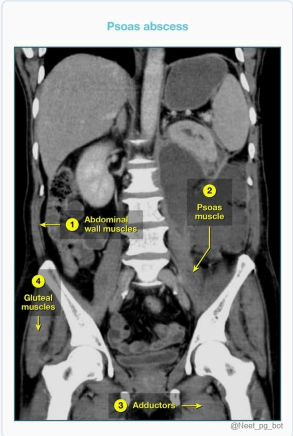
The clinical vignette of a patient with low-grade fever, dull backache, and persistent flexion of the hip joint with a history of spinal tuberculosis is suggestive of **psoas abscess** involving the **psoas muscle**, marked **2** in the given image**.** CT scan shows a mass in the psoas muscle lateral to the vertebral column which is diagnostic of the same.

Psoas abscess occurs due to a collection of pus in the iliopsoas compartment. The compartment is retroperitoneal and hence, the abscess is also called a **retroperitoneal** abscess. The compartment contains iliacus and iliopsoas muscles. The psoas compartment muscles lie in close proximity to major abdominal organs and is richly supplied by iliolumbar, external iliac and common femoral arteries. Owing to these reasons, there is hematogenous spread of infections like **tuberculosis** leading to abscess.

Clinical presentation is with back pain, fever, and lassitude. The hip may be in fixed flexion or there may be pain on passive extension.

CT is performed for diagnosis. Treatment is with percutaneous CT-guided drainage and antibiotic therapy.

Other muscles marked in the image are depicted below.



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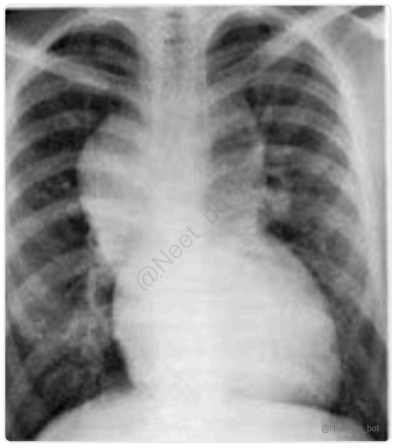
References: Online Resource, https://radiopaedia.org/cases/psoas-abscess-3
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5848408/

Bailey & Love's Short Practice of Surgery - 27th Edition, Page no 1065

Q21. Radiology NEET 2021 ID: MF1357

A 12 week old child presented with fever, cyanosis and fatigue. His chest X ray is shown below. What is the most likely diagnosis?



A ✓ Total anomalous pulmonary venous circulation

B Tetrology of Fallot

C Transposition of great arteries

D Pericardial effusion

Correct Answer: Total anomalous pulmonary venous circulation

The given X-ray shows **figure of 8** or **snowman appearance** characteristic of **total anomalous pulmonary venous circulation (TAPVC)**.

TAPVC refers to a condition in which pulmonary veins instead of draining into the left atrium drain into the right atrium directly or indirectly. This results in anomalous drainage of pulmonary venous blood flow into the systemic venous circulation. It is a cyanotic heart disease with features of congestive heart failure. The patients are irritable and have failure to thrive.

The following types are seen:

- **Supracardiac (most common)**: Common pulmonary vein drains into superior vena cava or the left innominate vein
- **Cardiac:** Pulmonary veins drain into the right atrium or coronary sinus.
- **Infracardiac** (always obstructive and most dangerous): Pulmonary veins drain into portal vein or inferior vena cava.

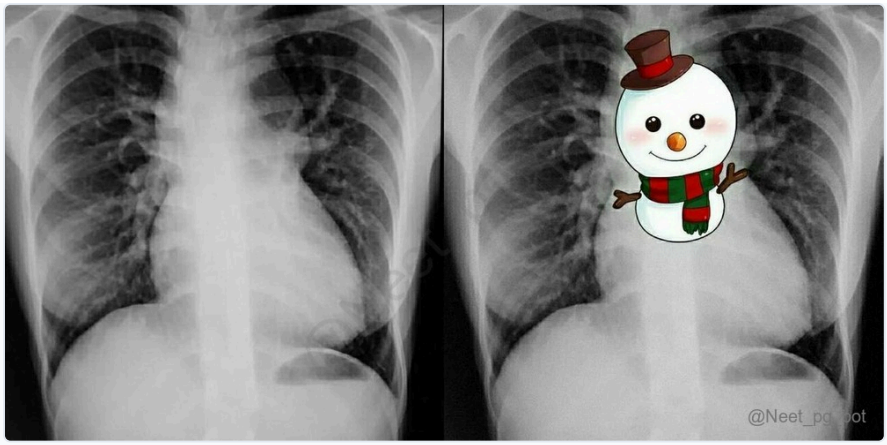
Snowman appearance is classically seen in **supracardiac variant** of TAPVC. The dilated vertical vein on the left, brachiocephalic vein on top, and the superior vena cava on the right form the head of the snowman; the body of the snowman is formed by the enlarged right atrium. It is shown in the image below.

Other options:

Option B: Tetrology of Fallot comprises pulmonary stenosis, ventricular septal defect, overriding of the aorta over the ventricular septum, and right ventricular hypertrophy. On chest X-Ray, the cardiac silhouette appears like a boot ("coeur en sabot").

Option C: In the transposition of great arteries, the aorta arises from the right ventricle and the pulmonary trunk from the left ventricle. The X-ray shows cardiomegaly with cardiac contours which appear like an egg on a string.

Option D: Pericardial effusions occur when excessive fluid gets collected in the pericardial space. The X-ray shows globular enlargement of the cardiac shadow giving a water bottle configuration.



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References: Nelson textbook of pediatrics 21st edition, Page no. 2398, 2408, 2412, 2396-2397,
Nelson Textbook of Pediatrics 20th Edition, Page no. 2227 - 2228
Ghai Essential Pediatrics 9th Edition, Page no. 423
Online Resource, https://radiopaedia.org/articles/snowman-sign-total-anomalous-pulmonary-venous-return-1
Online Resource, https://radiopaedia.org/articles/pericardial-effusion
Online Resource, https://radiopaedia.org/articles/transposition-of-the-great-arteries

Q22. Radiology NEET 2021 ID: MF1366

A patient following a skid resulting in a motor traffic accident was brought to the emergency room 2 hours later. On examination, he was stable with GCS 15/15. The pupil was reactive to light. Tenderness and bruising over the left lower chest wall with petechiae was seen. Severe tenderness was elicited in the left hypochondriac region and BP- 90/50 mm of Hg. What is the best investigation used in the ER?

GCS- Glasgow coma scale

FAST- Focused Assessment with Sonography for Trauma

- A ✓ FAST
- B X-ray
- C Diagnostic peritoneal lavage
- D CT scan

Correct Answer: FAST

FAST (Focused Assessment with Sonography for Trauma) is the **best investigation** that can be performed in the emergency room (ER) for this patient with a blood pressure of 90/50 mm Hg who has sustained blunt abdominal trauma.

Focussed assessment with sonography for trauma is a non-invasive investigation that can be performed rapidly in the emergency room. It helps in identifying an abnormal collection of fluid or blood. FAST is indicated in:

- Haemodynamically unstable patients who had a blunt abdominal trauma
- Patients who sustained penetrating abdominal trauma with no other indications for immediate laparotomy (signs of haemodynamic instability, peritoneal irritation, peritoneal penetration and gunshot wound with transperitoneal trajectory).

FAST includes an examination of **four regions**:

- Pericardial sac
- Hepatorenal fossa
- Splenorenal fossa, and
- Pouch of Douglas

It is an operator-dependent procedure with limited visualization of the retroperitoneal structures.

Other options:

Option B - A chest **X-Ray** is used to exclude hemothorax or pneumothorax in a **hemodynamically normal** patient with penetrating trauma above the umbilicus or a suspected thoracoabdominal injury. Since the given patient is alert, awake, and with no mention of any pelvic pain or tenderness, a pelvic X-Ray is not indicated either.

Option C - **Diagnostic peritoneal lavage** is an invasive procedure and requires surgical expertise. For these reasons, it is done only in centres where FAST and/ CT are not available.

Option D - **CT scan** requires transporting the patient and is time-consuming. Hence, it cannot be performed in the ER in a haemodynamically unstable patient.

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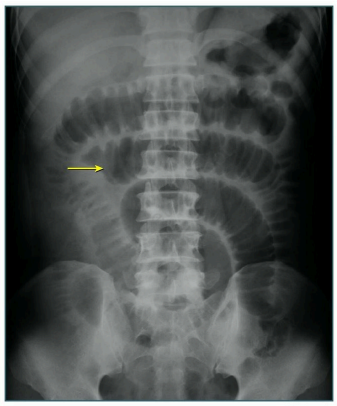
References: Advanced Trauma Life Support ATLS Student Course Manual 2018, Page no. 89 - 92
Current Diagnosis and Treatment : Surgery - 14th Edition, Page no. 210

Q23. Radiology NEET 2021 ID: MF1355

A patient presents with abdominal pain and distension. An X ray was taken which showed dilated bowel loops as given in the image below. Identify the segment involved.



A ✓ Jejunum

B Ileum

C Duodenum

D Transverse colon

Correct Answer: Jejunum

The given clinical scenario and X-ray suggest **intestinal obstruction** where the involved segment is the **jejunum**. The obstructed bowel loops are central in location and show **valvulae connivente** (mucosal folds that cross the entire diameter of the bowel) which is characteristic of jejunal involvement. The gas bubbles trapped between the valvulae connivente give a **string of beads** appearance.

Radiological features of bowel obstruction

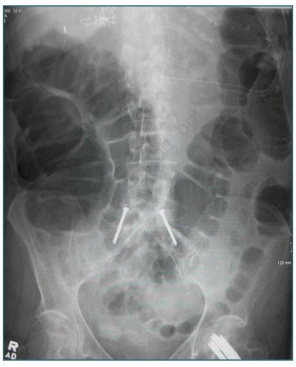
- Small bowel - The involved dilated segments are straight and usually centrally located.
- Jejunum - It is characterized by the presence of valvulae connivente with gas bubbles trapped between them. This gives a **string of beads/ ladder effect.**
- Ileum - The distal ileal obstruction is usually described as featureless.
- Caecum - A rounded gas shadow in the right iliac fossa is suggestive of caecal obstruction.
- Large bowel - The presence of haustrations is a feature of large bowel obstruction. Haustral folds are irregular and do not cross the entire diameter of the bowel. These are absent in the caecum.

In intestinal obstruction, fluid levels appear as the obstruction progresses. When the fluid levels are prominent, it indicates that the obstruction is advanced. The number of fluid levels also increases if the lesion is more distal.

The **most reliable criterion** for the diagnosis of small bowel obstruction is the **transition zone,** seen on **CT imaging**. It identifies a definite point of obstruction. Dilated small bowel loops are seen proximal to the site of obstruction and collapsed loops are seen distal to the obstruction.

Valvulae connivente in jejenal obstruction

Large bowel obstruction -
X-Ray appearance



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References: Bailey & Love's Short Practice of Surgery - 27th Edition, Page no. 1288
Grainger & Allison's Diagnostic Radiology 6th edition, page no. 597 - 599, 672

Q24. Radiology NEET 2021 ID: MF1367

A child with acute lymphoblastic leukemia undergoes prophylactic irradiation prior to autologous haematopoietic stem cell transplantation. Which of the following will be the least affected?

A Spermatogonia

B Intestinal epithelial cells

C ✓ Neurons

D Bone marrow/ erythroid precursor cells

Correct Answer: Neurons

Neurons are **least susceptible** to radiation. The radiosensitivity of biological tissue is directly proportional to its mitotic activity and inversely proportional to its degree of differentiation. This is known as the **law of Bergonie and Tribondeau**.

Hence, **hematopoietic cells** are the **most radiosensitive**, and the **central nervous system** is the **least sensitive** to radiation.

Acute Radiation Syndromes (ARS) is an acute illness that is caused by exposure of the entire body to a high dose of penetrating radiation in a very short period of time. It is also known as **radiation toxicity** or **radiation sickness.** It consists of three syndromes:

Syndrome	Radiation dose
Hematopoietic (Bone marrow)	>0.7 Gy
Gastrointestinal	>10 Gy
Cardiovascular/ Central nervous system	>50 Gy

Stages of ARS:

- Prodromal stage - This stage starts within minutes of exposure and may continue for a few days. Symptoms include nausea, vomiting, and anorexia.
- Latent stage - It begins within a few hours and continues up to a few weeks. The patient generally feels healthy.
- Manifest illness stage - This stage may continue up to months and symptoms vary according to the syndrome.
- Recovery/ death - The recovery process occurs over two years. Recovery cannot be expected in cardiovascular and central nervous system syndromes, which lead to death.

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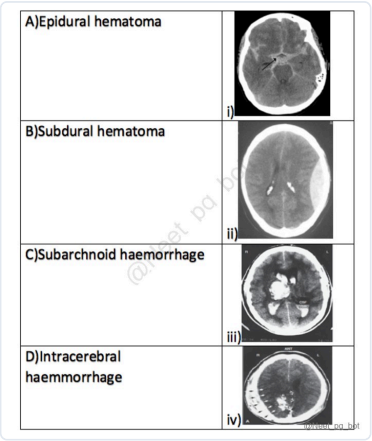
References: Online Resource, https://www.cdc.gov/nceh/radiation/emergencies/arsphysicianfactsheet.htm
Online Resource, https://radiopaedia.org/articles/law-of-bergonie-and-tribondeau

Q25. Radiology NEET 2019 ID: MC4621

Match the following intracranial bleed:



A A -i B-ii C- iii D- iv

B ✓ A -ii B-iv C-i D-iii

C A-iv B-iii C -ii D-i

D A -iv B-ii C-iii D-i

Correct Answer: A -ii B-iv C-i D-iii

Epidural hematoma:

Noncontrasted axial computed tomography scan showing a **lenticular** area of increased density in left parietal region.

- Typically occur in the groove of the middle meningeal artery associated with temporal bone fractures that have resulted in a tear of middle meningeal artery.
- They tend to not to cross suture lines of the skull.

Subdural hematoma:

A non contrasted computed tomography scan of acute subdural hematoma shows a **cresentric** area of increased density in the right posterior parietal region between brain and the skull. An area of intraparenchymal haemorrhage also seen. In addition, mass effect causes a midline shift to left.

- They can cross suture lines but they do not cross the tentorium or falx.
- Can be acute or chronic

Subarachnoid haemorrhage:

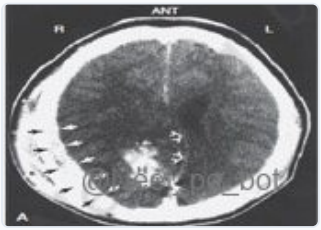
A non contrasted axial computed tomography scan shows the blood as areas of increased density near the base of the brain shows blood in **Texaco star pattern** formed blood radiating from suprasellar cistern into the sylvian fissures and anterior interhemispheric fissure.

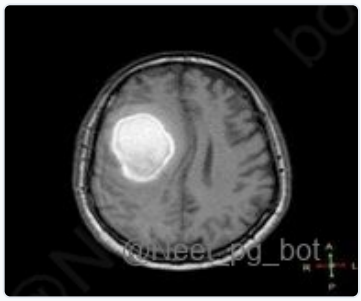
- Most often accompanied by severe sudden onset headache.
- Can really be visualised in acute stage, when blood is radiographically dense (whiter) than the CSF.

Intracereberal haemorrhage:

Non contrast computed tomography shows a large area of fresh blood appearing white in the region of right thalamus. Blood is denser than CSF so settles dependently.

This settling is not seen with subarachnoid or interparenchymal blood.





@Neet_pg_bot

References: Essentials of Radiology, 3e, Page nos 18-21

Q26. Radiology NEET 2019 ID: MA8460

The following appearance in IVP is caused by:

A Horse shoe kidney

B ✓ Ureterocoele

C Bladder tumour

D Posterior urethral valve

Correct Answer: Ureterocoele

The given intravenous pyelography (IVP) depicts the **cobra head** or **adder head** deformity or **spring onion** sign seen in **bilateral ureterocele.** These signs refer to a **dilated, contrast-filled distal ureter** surrounded by a thin rim of the ureteric wall and bladder mucosa.

Ureterocoele is a **cystic enlargement** of the intramural portion of the ureter, more commonly seen in **females** than males. It is thought to be caused due to ureteric orifice atresia. They are similar to ectopic ureters, which typically drain the upper pole of the kidney and are associated with dysplastic or non-functional renal tissue.

If they present clinically, ureteroceles often manifest with **urinary tract infections** (UTIs) in **children**, while in **adults,** they are more likely to present with **stones** in the lower ureter. Ureterocoeles can enlarge and obstruct the bladder neck or the contralateral ureteric orifice.

The classic sign of a ureterocele on intravenous urography is a ‘cobra head’, ‘adder head’, or 'spring onion sign'. On ultrasonography, it can be seen as a **fluid-filled cystic mass** in the bladder.

The treatment of simple ureteroceles is **surgical excision with re-implantation** of the ureter.

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References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no 758, 760
Bailey & Love's Short Practice of Surgery - 28th Edition, Page no. 1467

Q27. Radiology NEET 2019 ID: MA0586

Which of the following investigations is shown in the radiograph?

A Barium swallow

B ✓ Barium meal

C Barium meal follow through

D Barium enema

Correct Answer: Barium meal

The given image is a double-contrast radiograph of a **barium meal**. It is used to examine the lower half of the esophagus, stomach, and entire duodenum. It is used in identifying ulcers, small fleshy lumps (polyps), tumours.

GI Imaging	Preparations of barium	Structures best visualized
Barium swallow	Barium paste	Esophagus
Barium meal	95% BaSO4	Stomach and duodenum
Barium meal follow-through	50% BaSO4	Small bowel
Barium enema	25% BaSO4	Large bowel

Barium meal follow-through is useful in visualising small bowel pathology and evaluating the function of the **small bowel**. It is performed following a barium meal examination where X-rays are taken 10-15 minutes after the intake of barium. This allows time for the barium to reach the small intestine. It is inferior to enteroclysis enema, where contrast is administered through a nasogastric tube.

The below radiograph is a **double-contrast barium enema** with a normal **large bowel** structure. The contrast is administered through a rectal tube and a series of radiographs are obtained.

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References: Davidson's Principles and Practice of Medicine, 24th edition, Page 773

Q28. Radiology NEET 2019 ID: MA3899

What is the best imaging study for earliest diagnosis of cerebral infarct?

A FLAIR MRI

B Non-contrast CT

C CECT

D ✓ Diffusion weighted MRI

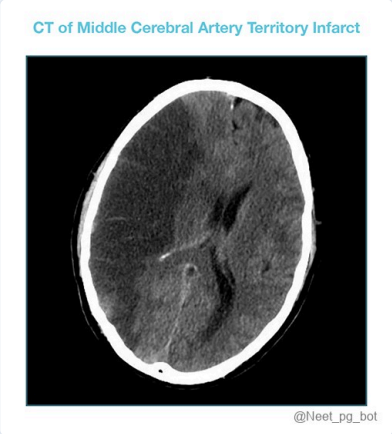
Correct Answer: Diffusion weighted MRI

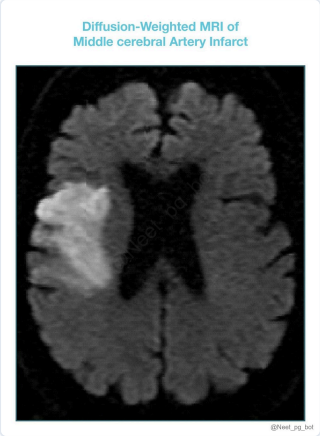
The best imaging modality for the early diagnosis of **cerebral infarction** is **diffusion-weighted MRI (DWI)**, which can detect infarcts as early as 30 minutes of onset and identify cytotoxic oedema within a few hours of ischemic stroke.

However, the initial investigation in suspected stroke cases is **non-contrast CT** due to its rapid acquisition time, widespread availability, and cost-effectiveness. It helps quickly rule out hemorrhagic stroke and facilitates the timely administration of intravenous tPA in eligible patients.

Acute MCA territory infarct on CT is shown below:

The image below is of a DWI MRI of a patient clearly showing the extent of the infarct.



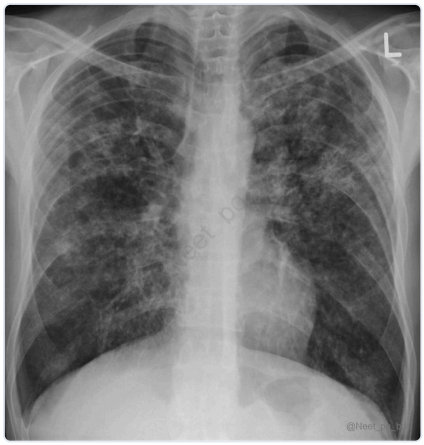


@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no. 1375
Online Resource, Shafaat O, Sotoudeh H. Stroke Imaging. [Updated 2023 May 1]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK546635/

Q29. Radiology NEET 2019 ID: MC7580

During evaluation of PUO, a patient's chest X-ray is as seen below. What is your diagnosis?



A ✓ Miliary tuberculosis

B Pulmonary edema

C Necrobiotic nodules

D Tuberculoma

Correct Answer: Miliary tuberculosis

The given chest radiography image shows **miliary tuberculosis.**

It is the widespread dissemination of Mycobacterium tuberculosis, via **hematogenous spread.**

Classic miliary TB is defined as **millet-like** (mean, 2 mm; range, **1-5 mm**) seeding of TB bacilli with uniform size and distribution in the lung, as seen on chest radiography. This particular sign is called the **miliary shadow** which is also seen in sarcoidosis and histoplasmosis.

Sputum-smear microscopy is negative in most cases.

Physical findings include hepatomegaly, splenomegaly, and lymphadenopathy

Bronchoalveolar lavage and transbronchial biopsy are used to provide bacteriologic confirmation

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References: Harrison's Principles of Internal Medicine, 21st Edition, Page no: 1366, 1367
Grainger & Allison's Diagnostic Radiology 7th edition, Page no: 117

Q30. Radiology NEET 2019 ID: MC7597

Identify the structure marked in the following MRI scan:

A Midbrain

B ✓ Cerebellum

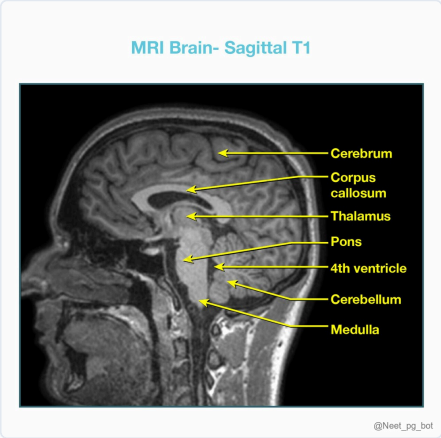
C Brainstem

D Cerebrum

Correct Answer: Cerebellum

The given image shows an MRI of the sagittal section of the brain. The area marked with a red arrow is the cerebellum.

The other areas also visualized in the image are :



@Neet_pg_bot

References: Gray's Anatomy: The Anatomical Basis of Clinical Practice, 41st Edition, Page no 331

Q31. Radiology NEET 2019 ID: MC7519

Which of the following is a water soluble contrast?

A ✓ Iohexol

B Barium

C Bromium

D Calcium

Correct Answer: Iohexol

Iohexol is a nonionic, **water-soluble** radiographic contrast medium.

It is used to visualize the following structres:

- Arteries
- Veins
- Ventricles of the brain
- The urinary system
- Joints

It can be used during computer tomography (CT scan) as well. It can be given orally, intravenously, or can be injected into the body cavity.

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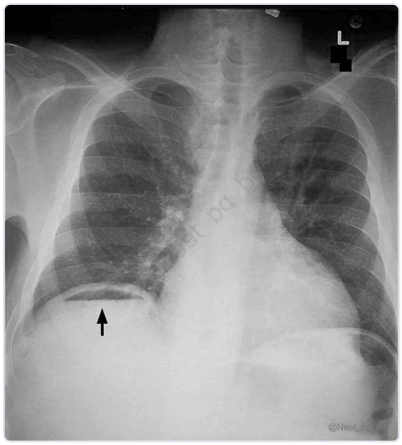
References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 44

Q32. Radiology NEET 2019 ID: MC9589

All of the following could be the cause of the following X-ray finding except:



A Duodenal ulcer perforation

B Post laparoscopy

C Chilaiditi syndrome

D ✓ Acute intestinal obstruction

Correct Answer: Acute intestinal obstruction

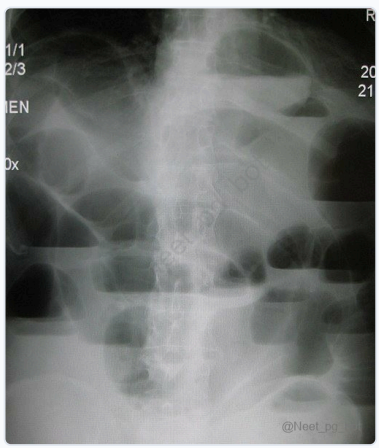
The given X-ray shows **air under the diaphragm**, indicative of **pneumoperitoneum**, which is **not** caused by **acute intestinal obstruction.**

Causes of pneumoperitoneum include:

- Bowel perforation: Gastric and posterior duodenal perforations in long-standing cases of peptic ulcers., ischemic bowel, necrotizing enterocolitis
- Small bowel perforation secondary to trauma or penetrating injuries
- Post-surgery/laparoscopy
- Causes other than perforation are:
  - **Chilaiditi syndrome** - the intestine is seen between the liver and diaphragm giving a false impression of free air
  - **Sub-phrenic abscess**
  - **Postoperatively,** air may be present in the abdomen for up to 7 days

An intestinal obstruction would show **multiple dilated loops** on X-ray with possible **air-fluid levels**. Only when the obstruction becomes severe and there is a perforation, pneumoperitoneum occurs.

The image given below is an abdominal radiograph showing multiple dilated loops with air-fluid levels suggestive of intestinal obstruction.

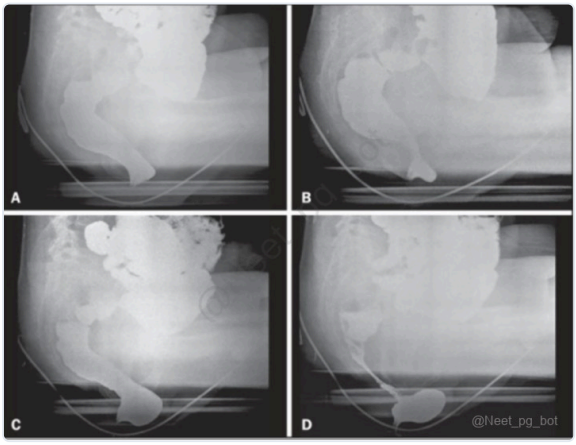


@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no. 463-466
Online Resource, Chapman BC, McIntosh KE, Jones EL, Wells D, Stiegmann GV, Robinson TN. Postoperative pneumoperitoneum: is it normal or pathologic? J Surg Res. 2015 Jul;197(1):107-11. doi: 10.1016/j.jss.2015.03.083. Epub 2015 Apr 1. PMID: 25940159.

Q33. Radiology INI-CET 2025 ID: MG9298

1. Identify the investigation:



A MR defecogram

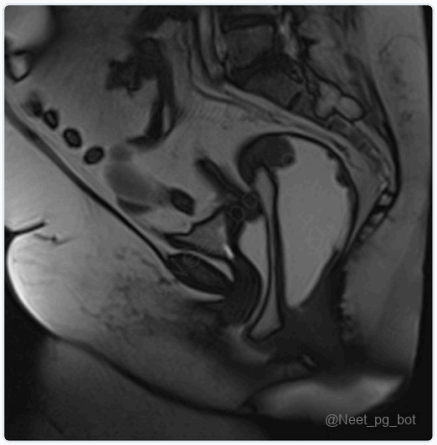
B Barium swallow

C CT angiography

D ✓ Xray defecogram

Correct Answer: Xray defecogram

The given image shows X-ray Defecography. X-ray Defecography is a fluoroscopic technique to evaluate pelvic floor prolapse, incomplete or obstructed defaecation/constipation, rectal prolapse and faecal incontinence. The technique traditionally involves fluoroscopy and barium. Other options: Option A: MR defecogram is a better technique that is being used more commonly. It is shown in the image given below.



Option B: Barium swallow is used to assess the oesophagus and stomach. Option C: CT angiography is used to assess the vasculature.

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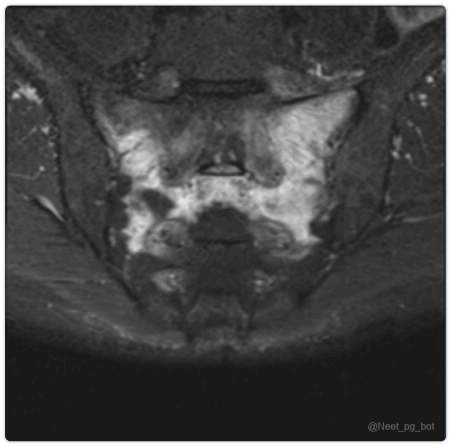
References: Grainger & Allison's Diagnostic Radiology 7th edition • Page no 596
Online Resource • https://radiopaedia.org/articles/evacuation-proctography

Q34. Radiology INI-CET 2025 ID: MG9297

2. A 74-year-old female complains of lower back pain. STIR sequence MRI performed showed the Honda H sign. What is the likely diagnosis?



A ✓ Sacral insufficiency due to sacral compression fracture

B Multiple myeloma

C Osteomyelitis

D Transient marrow edema syndrome

Correct Answer: Sacral insufficiency due to sacral compression fracture

The given clinical scenario of a female complaining of lower back pain and STIR sequence MRI showing the Honda H sign suggests the most probable diagnosis to be Sacral insufficiency due to sacral compression fracture. Sacral insufficiency fractures are a subtype of stress fractures, which are the result of normal stresses on abnormal bone, most frequently seen in patients with osteoporosis. They are usually seen in elderly females who present with low back pain without any history of significant trauma. Risk factors include osteoporosis, rheumatoid arthritis, Paget's disease, osteomalacia, diabetes mellitus and history of radiation therapy. A typical H-sign or Honda sign (uptake in H pattern) on Tc-99m MDP scan is shown below.

Other options: Option B: Multiple myeloma presents with lytic punched-out lesions, mostly in the skull. Option C: Osteomyelitis is characterised by the presence of sequestrum, involucrum, and cloacae. Option D: Transient marrow edema syndrome is a diagnosis of exclusion and generally presents in the vertebral bodies and bones of the extremities. It is characterized by pain and increased interstitial fluid within bone marrow without an obvious cause.

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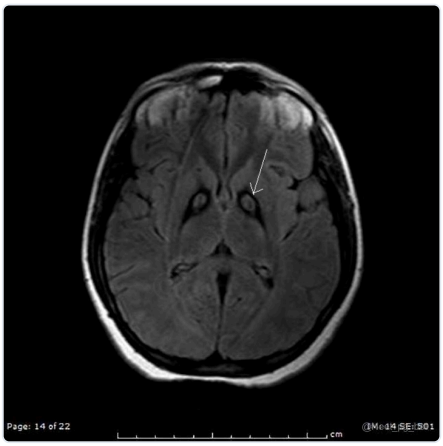
References: Grainger & Allison's Diagnostic Radiology 7th edition • Page no 1093
Online Resource • https://radiopaedia.org/cases/honda-sign?lang=gb
Online Resource • https://radiopaedia.org/articles/sacral-insufficiency-fracture-1?lang=gb
Online Resource • https://www.ncbi.nlm.nih.gov/books/NBK559176/

Q35. Radiology INI-CET 2025 ID: MG9295

3. A 12 year old was brought with rigidity and cognitive impairment. T2W MRI showing tiger eye sign as shown in the image below. Which of the following is the cause?



A ✓ Pantothenate kinase deficiency

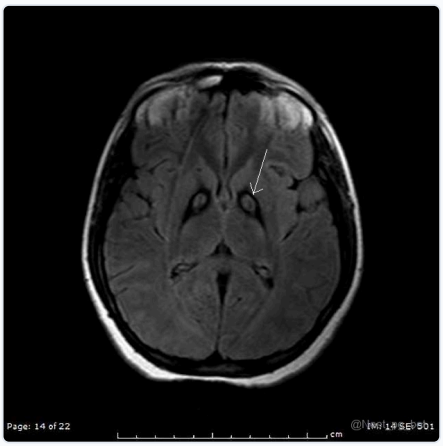
B Wilsons disease

C Krabbe disease

D Huntington Chorea

Correct Answer: Pantothenate kinase deficiency

The above clinical scenario, along with the MRI image showing the eye of the tiger sign, is suggestive of Pantothenate kinase-associated neurodegeneration (PKAN), formerly known as Hallervorden-Spatz syndrome. Pantothenate kinase deficiency is the cause. The eye of the tiger sign refers to an abnormal, low T2 signal intensity on MRI (due to abnormal accumulation of iron) in the globus pallidus, with a longitudinal stripe of high signal intensity (due to gliosis and spongiosis). This is an autosomal recessive disorder that typically manifests in childhood, leading to spasticity, Parkinsonism, and dystonia. It is classified as a subtype of neurodegeneration with brain iron accumulation (NBIA). The image below shows an MRI depicting the eye of a tiger sign.



Other options: Option B: Wilson's disease is an autosomal recessive disorder that occurs due to mutations in the ATP7B gene. Neurological features include dysarthria, dystonia, rigidity, tremors, and choreiform movements. These occur due to copper deposition in the basal ganglia. Option C: Krabbe disease is an autosomal recessive neurodegenerative disorder that occurs due to a deficiency of galactocerebrosidase. Option D: Huntington's chorea is a trinucleotide repeat disorder. The disease predominantly affects the striatum but progresses to involve the cerebral cortex and other brain regions.

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References: Harrison's Principles of Internal Medicine, 21st Edition • Page no: 3235, 3260, 3404, 3409
Online Resource • https://www.ncbi.nlm.nih.gov/books/NBK430689/
Online Resource • https://radiopaedia.org/articles/pantothenate-kinase-associated-neurodegeneration

Q36. Radiology INI-CET 2025 ID: MG9290

4. A 15-year-old boy presents with recurrent seizures. A CT done shows the following findings. What is the diagnosis?

A ✓ Neurocysticercosis

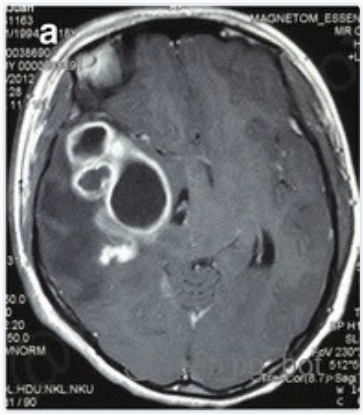
B Tuberculoma

C Brain abscess

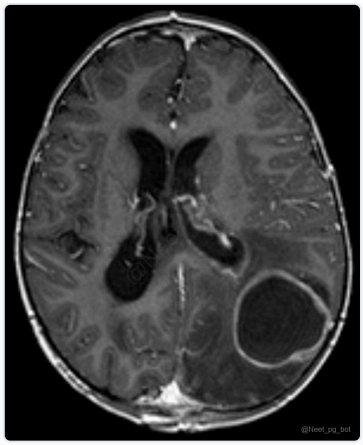
D Cerebral metastasis

Correct Answer: Neurocysticercosis

The given clinical scenario, along with CT showing ring-enhancing cystic lesions in the brain with an eccentric dot indicating scolex, suggests the most probable diagnosis to be neurocysticercosis. Neurocysticercosis is a form of cysticercosis caused by Taenia solium. It usually presents with seizures and symptoms of increased intracranial pressure, including headache, nausea, vomiting, vision changes, dizziness, ataxia, or confusion. Neurocysticercosis larvae can spread to the brain parenchyma, subarachnoid space and ventricles. The four stages of parenchymal neurocysticercosis are: * Vesicular stage- This stage consists of a viable larva with a scolex (worm head). The cysts are thin-walled with no or little enhancement, and there is no perilesional oedema * Colloidal vesicular stage- Ring-enhancing lesions with perilesional oedema and mass effect are seen * Granular nodular stage- thick enhancing cyst walls and progressive surrounding oedema are seen * Calcified nodular stage- The oedema resolves, and small calcified lesions of 2–10 mm diameter are seen Other options: Option B: Tuberculoma (parenchymal granulomas) can mostly occur at the junction of white and grey matter. On CT, they appear as small, rounded, solid lesions that show homogenous enhancement on contrast CT and show ring enhancement when central caseation or liquefaction occurs. MRI brain shows a conglomerate or lobulated irregular shape, ring-enhancing lesion.



Option C: Brain abscess is seen as a rim-enhancing lesion along with a history of fever.



Option D: Cerebral metastasis appears as solid rather than cystic masses.

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References: Grainger & Allison's Diagnostic Radiology 7th edition • Page no 1483, 1484, 1493, 1495
Online Resource • https://radiopaedia.org/articles/cyst-with-dot-sign-neurocysticercosis

Q37. Radiology INI-CET 2025 ID: MG9288

5. A 35-year-old patient presented with colicky pain and is suspected to have ureteric stone. Which of the following is the best diagnostic investigation in this case?

A ✓ NCCT KUB

B CECT abdomen

C USG KUB

D Ureteroscopy

Correct Answer: NCCT KUB

NCCT KUB (Non-contrast CT of the kidney, ureters, and bladder) is the best diagnostic modality for the clinical evaluation of ureteric stones. NCCT is the preferred investigation for evaluation of: * Renal stones except for indinavir stones * Ureteric stones * Head injury * Salivary gland stones Other options: Option B: Contrast-enhanced CTs abdomen are not done for renal and ureteric stones, as non-contrast CTs provide enough visualisation. Additionally, the use of contrast material adds an additional risk of developing contrast-induced kidney injury. Option C: A USG KUB is less sensitive compared to an NCCT KUB, is operator-dependent, and it can be difficult to visualise mid-ureteric stones. Option D: Ureteroscopy is an invasive test and is used for therapeutic intervention, not initial diagnosis.

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References: Bailey & Love's Short Practice of Surgery - 28th Edition • Page no: 1475
Grainger & Allison's Diagnostic Radiology 7th edition • Page no: 769

Q38. Radiology NEET 2020 ID: MA0570

A 56-year-old female patient presented with progressive dysphagia for solids, loss of weight, and loss of appetite. A barium swallow done is given below. What is the most probable diagnosis?

A ✓ Carcinoma esophagus

B Diffuse esophageal spasm

C Esophageal leiomyoma

D Esophageal web

Correct Answer: Carcinoma esophagus

A patient with the given history and barium swallow findings of **irregular narrowing**, **mucosal destruction**, and **shouldered margins** of esophagus, are suggestive of **carcinoma esophagus**.

Esophageal carcinoma is more common in men, with tobacco use and alcohol being the strongest risk factors. The squamous cell carcinoma variant usually affects the **upper 2/3rd**, and the adenocarcinoma variant usually affects the **lower 1/3rd** of the esophagus. Clinical features include **progressive dysphagia** (most common), regurgitation of food, vomiting, odynophagia and weight loss.

On barium swallow, carcinoma esophagus presents with irregular narrowing and ulceration of esophagus. **Flexible endoscopy** with **biopsy** is preferred for confirmation of diagnosis. **Contrast-enhanced CT and PET** can be used to assess distant metastasis. **Endoscopic ultrasonography (EUS)** is the investigation of choice to assess tumor invasion and lymph node metastasis.

Other options:

Option B: Diffuse esophageal spasm results in the **corkscrew** appearance of the esophagus on barium study.

Option C: Esophageal leiomyoma appears as a smooth, **semilunar filling defect** on barium swallow. it is seen as a discrete ovoid mass that is well outlined by barium.

Option D: Esophageal webs occur in the upper and middle esophagus as a thin membrane that can be identified by the **translucent** part on contrast radiology.

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References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no 485-486

Q39. Radiology NEET 2020 ID: MD0670

A 30-year-old woman presents with persistent hypertension, chest pain and cramping in her legs, particularly during exertion. A chest radiograph was taken, which reveals the condition, as shown in the image below. What is the likely underlying cause of her hypertension?

A ✓ Coarctation of aorta

B Essential hypertension

C Aortic aneurysm

D Aortaarteritis

Correct Answer: Coarctation of aorta

This clinical scenario and the image showing the **figure of 3 sign** point towards **coarctation of aorta** as the cause of hypertension in this patient.

The figure of 3 sign refers to the indentation of the aorta at the coarctation site. There is pre and post-stenotic dilatation along with left paramediastinal shadow, as shown below.

Other chest X-ray features in coarctation of the aorta are dilated left subclavian artery, dilated ascending aorta and **notching of ribs** 4-8 due to rib erosion by dilated collateral vessels.

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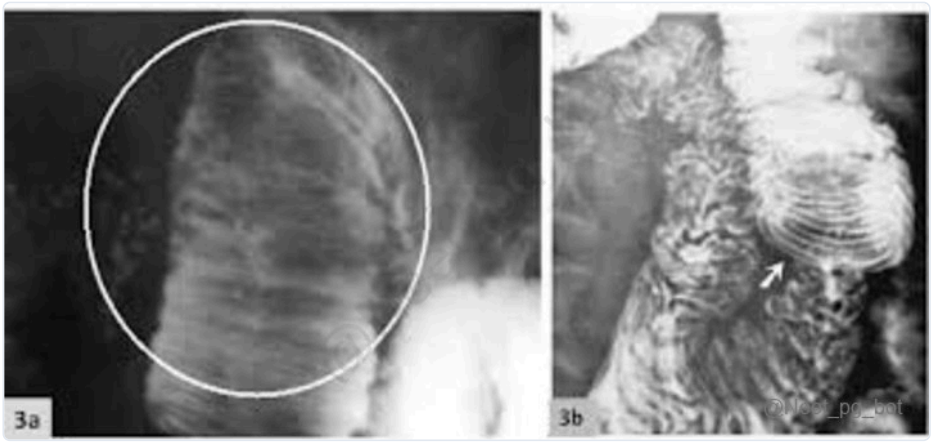
References: Harrison's Principles of Internal Medicine, 20th Edition, Pg 1837
Nelson textbook of pediatrics 21st edition, Pg 2391

Q40. Radiology NEET 2020 ID: MD0810

A 51-year-old man complains of abdominal pain and blood-stained stools. The barium study of his intestines shows the following finding. What is the most likely diagnosis?



A ✓ Intussusception

B Volvulus

C Diverticulitis

D Ulcerative colitis

Correct Answer: Intussusception

The given clinical scenario and the radiological images are suggestive of **intussusception**.

In intussusception, the occluding mass prolapsing into the lumen has been termed the **coiled spring appearance** (barium in the lumen of the intussusceptum and in the intraluminal space). In **adults**, it develops due to a **tumor** of the bowel.

The intussusception appears as a **sausage-shaped mass** in a longitudinal plane or a **target mass** in the transverse plane on CT and USG.

Ultrasound signs include:

- Target sign (also known as the doughnut sign)
- Pseudokidney sign- this is seen in the longitudinal section of the bowel
  - the edematous bowel resembles renal parenchyma and
  - the mesentery which is dragged into the intussusception resembles renal hilum
- Crescent in a doughnut sign.

Note: The colographic appearance of intussusception is also described as a claw sign, pincer defect, shouldering effect, and coiled-spring pattern.

Option B: Volvulus presents with constipation, abdominal bloating, nausea, and/or vomiting. It shows **coffee bean sign**

Option C: Diverticulitis presents with pain in the left iliac fossa

Option D: Ulcerative colitis causes **backwash ileitis**

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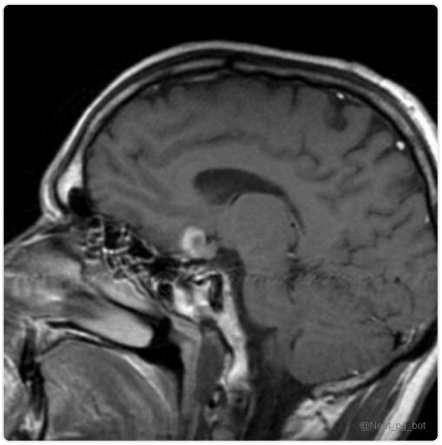
References: Grainger & Allison's Diagnostic Radiology 6th edition, Pg no 599
Online Resource, https://radiopaedia.org/cases/coiled-spring?lang=us

Q41. Radiology NEET 2020 ID: MD0813

A cystic lesion in the suprasellar region with calcification is seen on the MRI. What is the most likely diagnosis?



A ✓ Craniopharyngioma

B Pituitary adenoma

C Meningioma

D Oligodendroglioma

Correct Answer: Craniopharyngioma

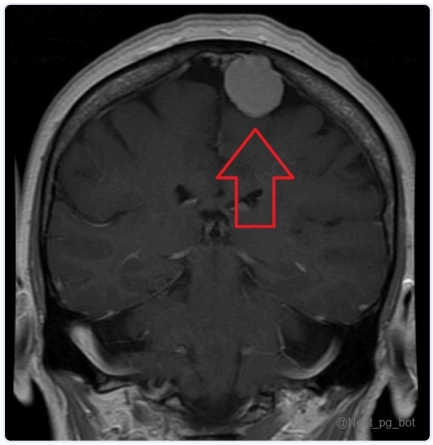
A cystic lesion in the suprasellar region with calcification is suggestive of **craniopharyngioma.** It is derived from **Rathke's pouch** and arises near the pituitary stalk, commonly extending into the suprasellar cistern.

Other options:

Option B: **Pituitary adenomas** are seen in the sellar region. Microadenomas are best shown on contrast-enhanced images. Macroadenomas are seen as **snowman or figure of eight** appearance.

Option C: **Meningioma** is a supratentorial tumor, mainly in the parasagittal region. A classical dural tail may be seen in intracranial meningiomas but is not pathognomic. The MRI image below shows a **well-defined**, **hyper-intense**, and **homogeneously enhanced mass** close to the midline suggestive of meningioma.

Option D: **Oligodendroglioma** is exclusively found in the cerebral hemispheres, most commonly in the frontal lobes. Upto 90% of oligodendrogliomas contain visible calcification on CT. The CT image below shows a calcified cortical mass that suggests oligodendroglioma.



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References: Grainger & Allison's Diagnostic Radiology 7th edition, Pg 1418, 1430, 1435, 1437, 1438
Online Resource, Di Muzio B, Guan H, Elfeky M, et al. Snowman sign (pituitary macroadenoma). Reference article, Radiopaedia.org (Accessed on 02 Jan 2025) https://doi.org/10.53347/rID-38698

Q42. Radiology NEET 2020 ID: MD0745

Personal monitoring of radiation is done by:

- **A** ✓ TLD badge
- **B** Collimators
- **C** Linear accelerator
- **D** Grid

Correct Answer: TLD badge

Personal monitoring of radiation is done by a **TLD badge.**

The TLD badge (thermoluminescent dosimeter) should be positioned according to local or national practices. In India, one TLD badge has to be worn at the **chest level inside** the lead **apron** if the apron is used. An additional **wrist badge** has to be worn for procedures requiring hands close to the primary beam.

The thermoluminescent material used is **calcium sulfate** doped with **dysprosium** or **lithium fluoride**. Radiation exposure is monitored every **three months**. The maximum dose limit for occupational radiation workers is 30 mSv in a year, provided the cumulative dose in a block of 5 years does not exceed 100 mSv.

The given image shows a TLD badge.

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References: Basic Radiotherapy Physics and Biology, 2014 edition, Page no. 162
Online Resource, https://www.aerb.gov.in/images/PDF/DiagnosticRadiology/EEI.pdf

Q43. Radiology NEET 2020 ID: MD0671

The appearance of USG at saphenofemoral junction, showing common femoral vein, common femoral artery and the great saphenous vein is called as:

A ✓ Mickey mouse sign

B String sign

C Tillaux sign

D Stemmer sign

Correct Answer: Mickey mouse sign

The **normal anatomy** of the femoral vessels and great saphenous vein at the **saphenofemoral junction** on USG is called the **mickey mouse sign.**

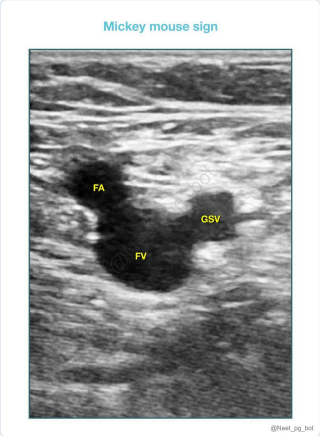
The given images below show the mickey mouse sign (FV- femoral vein; FA- femoral artery; GSV- great saphenous vein).

Other options:

Option B**: String sign** (of Kantor) on barium studies appears as a thin, thread-like stream of contrast passing through the stenosed bowel segment, particularly in Crohn's disease.

Option C: **Tillaux sign** is said to be positive if the mass/swelling is mobile perpendicular to the mesentery attachment line. Tillaux's sign is seen in a mesenteric cyst.

Option D: **Stemmer's sign** is the inability to tent the skin at the base of the toes due to skin thickening, as seen in lymphedema.



@Neet_pg_bot

References: Bailey & Love's Short Practice of Surgery - 28th Edition, Page No. 1031, 1032
Harrison's Principles of Internal Medicine, 21st Edition, Page No. 2119
Online Resource, Amini B, Niknejad M, Yap J, et al. String sign (bowel). Reference article, Radiopaedia.org (Accessed on 12 Dec 2024) https://doi.org/10.53347/rID-1375
Online Resource, Almasarei H, Degefu D, Knipe H, Mickey Mouse sign (ultrasound). Reference article, Radiopaedia.org (Accessed on 12 Dec 2024) https://doi.org/10.53347/rID-56729
SRB Manual of surgery, 6th edition, Page No. 569, 570

Q44. Radiology NEET 2020 ID: MD0903

Plethoric lung fields are seen in:

A ✓ Total Anomalous Pulmonary Venous Connection

B Tricuspid atresia

C Ebstein anomaly

D Tetrology of Fallot

Correct Answer: Total Anomalous Pulmonary Venous Connection

Plethoric lung fields are seen in **Total Anomalous Pulmonary Venous Connection** (TAPVC). In this condition, pulmonary venous obstruction results in pulmonary arterial hypertension, resulting in plethoric lung fields.

	Increased pulmonary blood flow (pulmonary plethora)	**Decreased** pulmonary blood flow
Acyanotic	Ventricular septal defect, Atrial septal defect, Patent ductus arteriosus	—
Cyanotic	TAPVC	Ebstein anomaly, Tetrology of Fallot, Tricuspid atresia

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References: Ghai Essential Pediatrics - 10th Edition, Page no. 440
Grainger & Allison's Diagnostic Radiology 7th edition, Page No 289, 294
Online Resource, Knipe H, Qureshi P, Lukies M, et al. Pulmonary plethora. Reference article, Radiopaedia.org (Accessed on 12 Dec 2024) https://doi.org/10.53347/rID-41136

Q45. Radiology NEET 2020 ID: MD0669

A patient complains of neck stiffness and severe headache. The CT image shown below is suggestive of which of the following conditions?

A Meningitis

B ✓ Subarachnoid hemorrhage

C Intraparenchymal hemorrhage

D Hemorrhagic stroke

Correct Answer: Subarachnoid hemorrhage

The clinical manifestations of neck stiffness and severe headache with CT revealing **blood** in the **basal cistern** points out to the diagnosis of **subarachnoid hemorrhage** (SAH).

Subarachnoid hemorrhage (SAH) is the bleeding into the subarachnoid space, which is the area between the arachnoid membrane and the pia mater surrounding the brain. The diagnosis is suspected when a **hyperattenuating material** is seen **filling the subarachnoid space**. Most commonly this is apparent **around the circle of Willis**, on account of the majority of berry aneurysms occurring in this region (65%), or in the Sylvian fissure (30%).

The **modified Fisher scale** is a method for grading SAH as seen on non-contrast CT (NCCT). The **Hunt and Hess scale** describes the severity of SAH and is used as a predictor of survival.

Xanthochromia on CSF is indicative of SAH in case of a negative CT scan.

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References: Harrison's Principles of Internal Medicine, 20th Edition, Page no: 2084
Harrison's Principles of Internal Medicine, 21st Edition, Page no: 3353

Q46. Radiology AIIMS 2019 (May & Nov) ID: MC7744

All lung carcinomas for diagnosis requires CECT thorax but MRI is useful in which of the following:

A Small cell lung carcinoma

B Adenocarcinoma

C Carcinoid

D ✓ Pancoast tumor

Correct Answer: Pancoast tumor

MRI is useful in Pancoast tumor because of the associated soft tissue infiltration, extent of which can be best visualised using an MRI

Pancoast syndrome/superior sulcus tumor results from a local extension of a tumor growing in the apex of the lung, with the involvement of **C8, T1, and T2** nerve roots.

It presents with pain that radiates to the ulnar distribution of the arm, often with the destruction of the 1st and 2nd ribs. Patients with the central or endobronchial growth of the primary tumor may present with a cough, hemoptysis, wheeze, stridor, dyspnea, or post-obstructive pneumonitis.

Regional spread can cause

- Tracheal obstruction,
- Esophageal compression with dysphagia
- Recurrent laryngeal paralysis with hoarseness
- Phrenic nerve palsy with the elevation of the hemidiaphragm and dyspnea
- Sympathetic nerve paralysis with Horner's syndrome (enophthalmos, ptosis, miosis, and anhydrosis).

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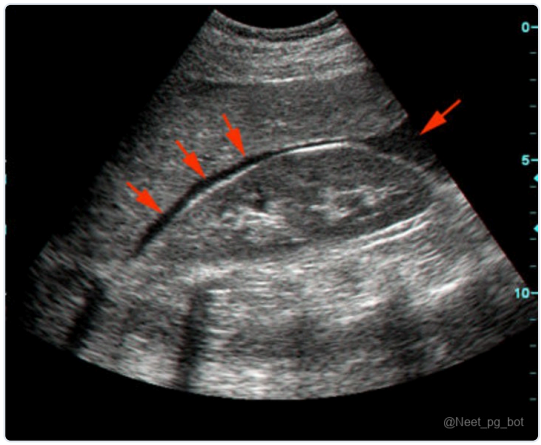
References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 331
Grainger & Allison's Diagnostic Radiology 7th edition, Page 197

Q47. Radiology AIIMS 2019 (May & Nov) ID: MC7743

The following image of a FAST scan in a trauma patient suggests:



A ✓ Positive fast

B Negative fast

C Equivocal fast

D Pneumoperitoneum

Correct Answer: Positive fast

The image of FAST in a trauma patient suggests **Positive FAST** as there is a fluid collection in the abdominal cavity. The **thin anechoic area** located between the liver(on the left) and the Kidney(on the right) is a collection of fluid visualized in this RUQ Sonogram image.

FAST stands for Focused assessment with sonography for trauma. It is a **rapid diagnostic examination** to assess patients with **potential thoracoabdominal injuries**. It is performed within 2-4 minutes.

Technique:

- Patient in the supine position
- 3.5-5.0 MHz convex transducer
- Following regions are scanned :
  - Subxiphoid: assess for pericardial effusion and left lobe liver injuries
  - Right upper quadrant: assess for right liver injuries, right kidney injury, and Morison pouch
  - Left upper quadrant: assess for splenic injury and left kidney injury
  - Suprapubic region: assess the bladder and pouch of Douglas

An "extended FAST or eFAST" scan may be performed: left and right thoracic views to assess for pneumothorax and haemothorax.

Sequentially order of FAST is:- **pericardial sac**, the Right Upper Quadrant (**RUQ**), Left Upper Quadrant (**LUQ**), and **pelvis.**

4 'Ps' are evaluated in sequence: Pericardial sac → Perihepatic region → Perisplenic region → Pelvis.

Note:-

In Pneumoperitoneum, air is collected in the peritoneal cavity which is usually hyperechoic not anechoic and would appear white in the ultrasound image.

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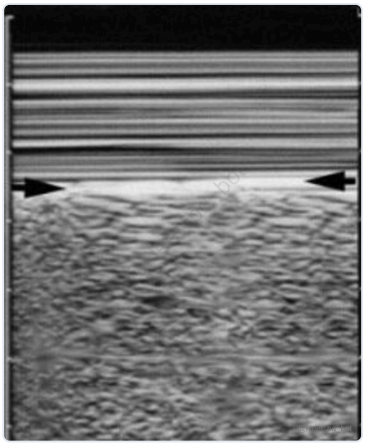
References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 378
Online Resource, https://radiopaedia.org/articles/focussed-assessment-with-sonography-for-trauma-fast-scan

Q48. Radiology AIIMS 2019 (May & Nov) ID: MD0247

The given USG of lung shows:



A Stratosphere sign

B ✓ Sea shore sign

C Troposphere sign

D Mirror image artifact

Correct Answer: Sea shore sign

The given image shows the **seashore sign** seen in M-mode imaging of the **normal lung**.

M-mode is defined as the **time-motion display** of the ultrasound wave along a chosen ultrasound line. With M-Mode, even rapid motions can be displayed, measured, and recorded.

In the M-mode of USG, the motionless portion of the chest above the pleural line creates horizontal 'waves', and the sliding below the pleural line creates a granular pattern, the 'sand'. The resultant picture is one that resembles waves crashing in onto the sand and is therefore called the 'seashore sign' and is present in a normal lung.

Option A: Stratosphere sign or **barcode sign** is a finding in the M-mode seen as parallel horizontal lines above and below the pleural line. This is seen in **pneumothorax**.

Option C: There is no named sign called troposphere sign.

Option D: Mirror image artifact is seen when there is a highly reflective surface (eg: Diaphragm) in the path of the primary beam.

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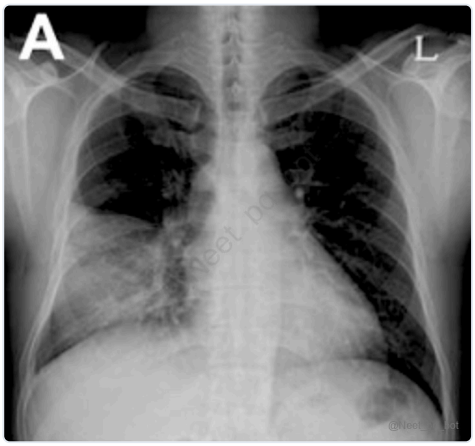
References: Online Resource, https://radiopaedia.org/encyclopaedia/signs/all
Online Resource, https://radiopaedia.org/articles/pneumothorax-ultrasound-1?lang=gb
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3299161/

Q49. Radiology AIIMS 2019 (May & Nov) ID: MD0248

An 8-year old child presents with the history of cough for 7 months. He also gives a history of haemoptysis for 1 month and fever for 4 days. The X-ray is given below. What is the diagnosis?



A Lung abscess

B ✓ Consolidation

C Hydatid cyst

D Thoracic duct malformation

Correct Answer: Consolidation

The given X-ray shows opacities in the **right middle lobe** and history of fever, cough and haemoptysis suggesting **consolidation due to lobar pneumonia**.

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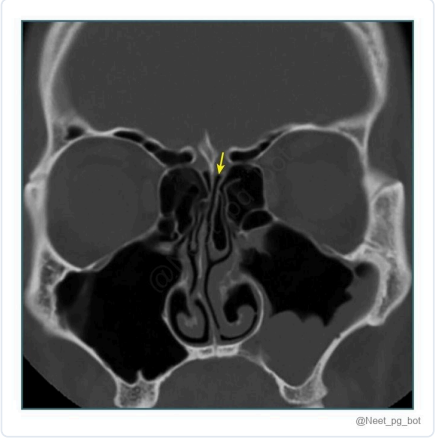
References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 247

Q50. Radiology INI-CET 2021 ID: MF0910

What is the structure marked in the CT scan ?



A Anterior ethmoidal canal

B Posterior ethmoidal canal

C Optic nerve

D ✓ Cribriform plate

Correct Answer: Cribriform plate

The structure marked in the image is the **cribriform plate** of the ethmoid bone. The image shown is a **coronal section of paranasal sinuses** at the level of anterior ethmoidal cells.

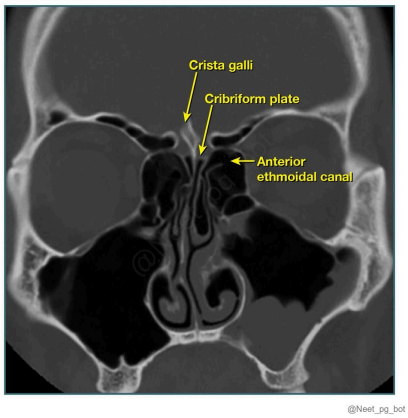
The cribriform plate is identified as a **bony structure** that separates the **nasal cavity from the anterior cranial fossa**. It is part of the **ethmoid bone** and is a sieve-like structure as it is pierced by the **olfactory nerve** fibers.

Crista Galli is also a part of ethmoid bone that projects **superiorl**y from the **cribriform plate**. It provides attachment for the falx cerebri.

The **anterior ethmoidal cana**l is identified by its location **below the roof of ethmoidal air cells.** The anterior ethmoidal artery passes through this canal and it is important to identify this to avoid bleeding from this vessel during sinus surgeries.

The **posterior ethmoid canal/forame**n is an opening in the ethmoid bone through which the posterior ethmoidal artery, vein, and nerve traverse. It is not visualized in this image as it is a coronal section at the level of anterior ethmoidal cells.

The **optic nerve** can be identified as an oval intraorbital structure in the coronal sections.



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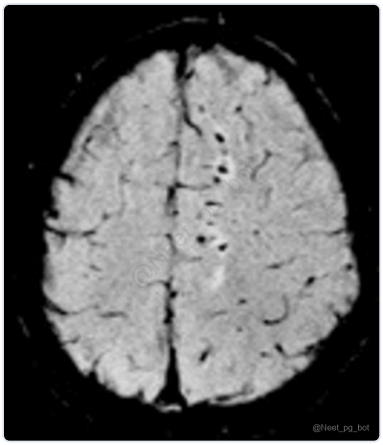
References: Online Resource, https://radiopaedia.org/articles/cribriform-plate
Online Resource, https://teachmeanatomy.info/head/osteology/ethmoid-bone/
Online Resource, https://radiopaedia.org/articles/posterior-ethmoidal-foramen

Online Resource, http://www.ajnr.org/content/ajnr/4/2/169.full.pdf
Online Resource, https://www.sciencedirect.com/science/article/pii/S1808869415308399#f4

Q51. Radiology INI-CET 2021 ID: MF0909

What type of MRI image is given below?



A Proton density image

B ✓ Susceptibility weighted image

C Gradient echo image

D Inversion recovery image

Correct Answer: Susceptibility weighted image

The image shown is a **susceptibility-weighted image(SWI)** as it clearly demonstrates **microhemorrhages as hypointense foci.**

SWI is a form of MRI sequence that is especially sensitive to substances that **distort the local magnetic field**. Compounds with **diamagnetic, paramagnetic, and ferromagnetic** properties will interact with the local magnetic field and produce distraction. This would lead to a loss of signal in these areas.

Deoxyhemoglobin, hemosiderin, and **ferritin** are paramagnetic. **Bone minerals** and **dystrophic calcifications** are diamagnetic in nature.

The main clinical use is the identification of microhemorrhages/blood products or calcium that are not detected in other MRI sequences.

Other options:

Proton density image:

It is based on the **number of nuclei** in the area that is imaged. In PD-weighted image tissues with a higher density of protons produce stronger signals and appear brighter. PD weighted image can be identified by comparing fluid with the fat signal. **Fluids** in these images **appear greyish white** similar or the appearance of fat in the body.

Gradient echo sequences(GRE):

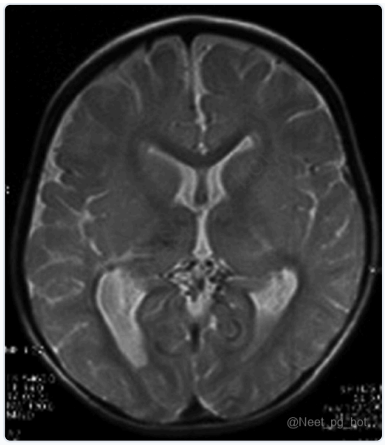
It uses gradient fields to generate transverse magnetization. In areas of hemorrhage, the iron that is present in the hemoglobin produces its own local magnetic field and dephases the spinning nuclei. This is useful to identify hemorrhagic contusions in the brain and pigmented villonodular synovitis.

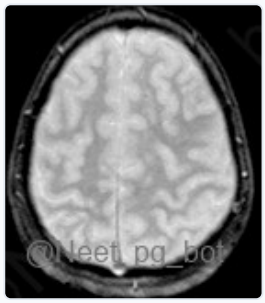
Inversion recovery sequences:

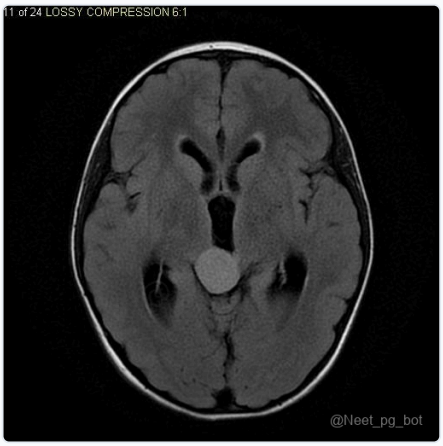
This MRI sequence is used to selectively null certain tissues. It can be:

- **Short tau inversion recovery(STIR)** where fat is nulled. Fluids appear bright and fat appears dark in these images.
- **Fluid attenuated inversion recovery(FLAIR)** or double inversion recovery(DIR where fluid is nulled.- The fluid(CSF) normally appears dark and lesions appear to be brighter on these images.

FLAIR image-







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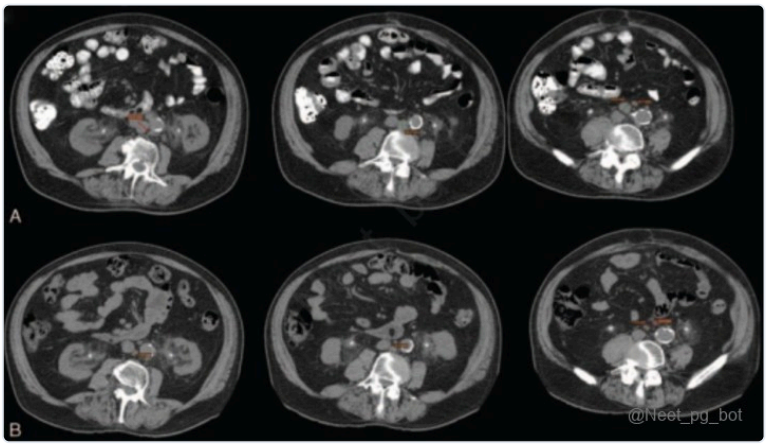
References: Online Resource, https://radiopaedia.org/articles/susceptibility-weighted-imaging-1
Online Resource, https://radiopaedia.org/articles/gradient-echo-sequences-1
Online Resource, https://radiopaedia.org/articles/pd-weighted-spin-echo-images
Online Resource, https://radiopaedia.org/articles/inversion-recovery-sequences
Online Resource, https://pubs.rsna.org/doi/10.1148/rg.262055063
Online Resource, https://mrimaster.com/characterise%20image%20stir.html

Q52. Radiology INI-CET 2021 ID: MF0915

During rounds, your senior was discussing the given image. The following is an image of which of the following investigations?



A Contrast Dye study

B ✓ CT scan

C X-ray

D Angiography

Correct Answer: CT scan

The image in the question is a **CT scan** as it shows the **bone as a high-intensity structure (white appearance)**.

In CT study X-rays are applied in a circular motion and body tissue slices are recreated mathematically. Different tissues are seen with different intensities depending on the proportion of attenuation of the X-rays that pass through them.

- **Air** and **water** are displayed as **low intensities (dark appearance)** because they produce only little attenuation.
- **Bone** is displayed as a **high-intensity structure (bright appearance**) because of its significant attenuation.
- **High density** is seen with **hyperemia, freshly clotted blood, and when contrast** is used.
- **Edema** and **necrosis** are seen as **low-intensity** lesions.

Other options:

Contrast dye study: In this imaging study, a **special dye or contrast** is given to evaluate structures that are not identified by conventional imaging. The contrast added would be visible in the image depending on the type of contrast and tissues targeted. Intravenous pyelogram, barium studies, angiography, etc are all examples of contrast studies.

X-ray: It shows different parts of the body in varying shades of black and white. The difference in the appearance of various tissues is due to the differences in the amount of radiation absorbed by them.

- **Bone appears white** because calcium has high x-ray absorption.
- **Fat** and **soft tissues appear grey** due to less x-ray absorption.
- The least absorption is seen with **air** which appears **black**.

Angiography: Here a special dye or contrast is injected to highlight the blood vessels are they are not clearly identified in normal imaging. Hence vessels would be highlighted in an angiogram.

Note: A contrast study is different from a contrast-enhanced CT scan, Intravenous pyelogram, barium studies, angiography, etc are all examples of contrast studies.

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References: Online Resource, https://case.edu/med/neurology/NR/CT%20Basics.htm
Online Resource, https://utswmed.org/conditions-treatments/contrast-radiography/
Online Resource, https://medlineplus.gov/xrays.html
Online Resource, https://www.nhs.uk/conditions/angiography/

Q53. Radiology INI-CET 2021 ID: MD6696

A 22-year-old male was brought to the emergency room following a road traffic accident. On examination, his GCS was E2M2V3. He had no fractures and no hydrocephalus or raised intracranial pressure. Non-contrast CT was performed and was found to be normal. The likely diagnosis is

- **A** ✓ Diffuse axonal injury
- **B** Post concussion syndrome
- **C** Extra dural hemorrhage
- **D** Herniation of brain

Correct Answer: Diffuse axonal injury

The given clinical scenario of an RTA with no imaging abnormality along with a disproportionate clinical manifestation suggests **diffuse axonal injury.**

It is caused due to **stretching of axons** in subcortical white matter due to sudden deceleration injuries. It is usually associated with **mild nonspecific changes** in **NCCT**. It is visualized on **T2 MRI** characterized by multiple lesions showing **microbleeds or linear hypointensities**. It may also show disruption of white matter tracts on tractography.

Other options

Post-concussion syndrome usually follows mild traumatic brain injury. It does not have a severe clinical status as seen in this patient.

Extradural hemorrhage is produced usually results secondary to trauma to head. The bleeding is usually arterial and blood collects between outer dura and inner table of the skull. It can very well be visualized on CT.

A cerebral herniation is defined as a shift of cerebral tissue in an adjacent space. Intracranial brain herniation can be classified into 3 types - subfalcine, transtentorial and tonsillar. A distinct clinical picture can be produced as a result of compression of neurovascular structures by the herniating part. CT is the imaging modality of choice because it can be done rapidly.

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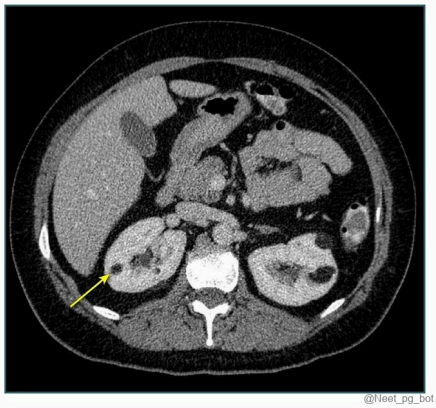
References: Harrison's Principles of Internal Medicine 20th Edition, Pg 3185,3186
Online Resource, https://radiopaedia.org/articles/diffuse-axonal-injury
Online Resource, https://www.ijri.org/article.asp?issn=0971-3026;year=2015;volume=25;issue=4;spage=404;epage=414;aulast=D

Q54. Radiology INI-CET 2021 ID: MF0882

A 45-year-old patient presented with vague abdominal pain. On USG, he was found to have a renal cyst of Bosniak class III. CECT was done, as shown below. What is the diagnosis?



A RCC

B Renal oncocytoma

C ✓ Renal angiomyolipoma

D Renal cyst

Correct Answer: Renal angiomyolipoma

The CT image showing renal lesions that has **hypoattenuating components consistent with macroscopic fat** is highly suggestive of **renal angiomyolipoma.**

Imaging in renal angiomyolipoma: Demonstration of macroscopic fat is the main part of making a diagnosis. **USG** may show **hyperechoic lesions** in the cortex with **posterior acoustic shadowing**. In patients with tuberous sclerosis, the entire kidney is affected, producing an echogenic appearance with loss of normal corticomedullary junction.

CT findings:

- Cortical lesions that demonstrate **macroscopic fat (less than -20 HU)**
- **Absence** of **ossification or calcification**
- The presence of macroscopic fat is strongly suggestive of this condition, but not pathognomonic as RCC may rarely demonstrate macroscopic fat.
- Approximately 5% may be fat poor.

MRI: It is an excellent tool to demonstrate intramural fat components. Two sequences are used:

- Fat-saturated techniques
- Phase-in and phase-out imaging that shows **India ink artifact** at the junction between fat and non-fat components.

Digital subtraction angiography: They are hypervascular lesions showing '**sunburst appearance**' in the arterial phase and '**onion peel appearance**' in the venous phase.

Angiomyolipomas are benign tumors that occur sporadically or in association with **tuberous sclerosis**. The mean age of presentation is **43 years** and there is a female predilection. It can present as spontaneous retroperitoneal hemorrhage, especially with larger tumors. The patient can present with flank pain, palpable mass, hematuria, urinary tract infections, hypertension, or renal failure. They usually do not require any intervention. mTOR inhibitors like **everolimus** can be used to decrease the size of the lesion.

Bosniak classification:

It classifies the renal cystic lesions mainly into four classes based on the CT characteristics.

Class I and II are **benign** and do not require follow-up imaging or surgery.

Class IIF requires follow-up for about 5 years to exclude malignancy.

Class III needs to be biopsied or surgically explored due to a **higher risk of malignancy**.

Class IV lesions are **clearly malignant** and need appropriate treatment.

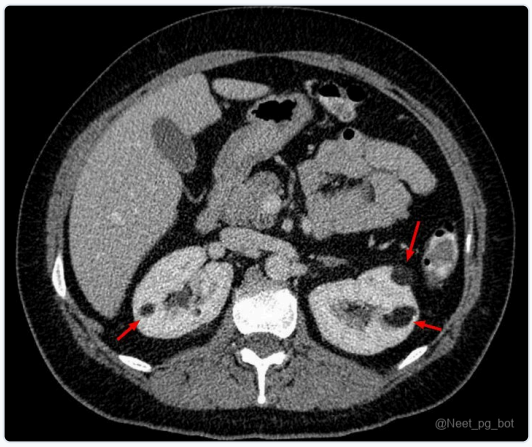
Other options:

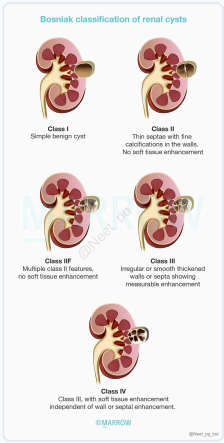
Renal cell carcinoma (RCC): It is the most common malignant tumor of the kidney. It usually occurs in the age group **50-70 years** and **macroscopic hematuria** is seen in about 60% of the cases. It is a differential for angiomyolipoma but it is important to note that **macroscopic fat** is almost always **associated with ossification or calcification.**

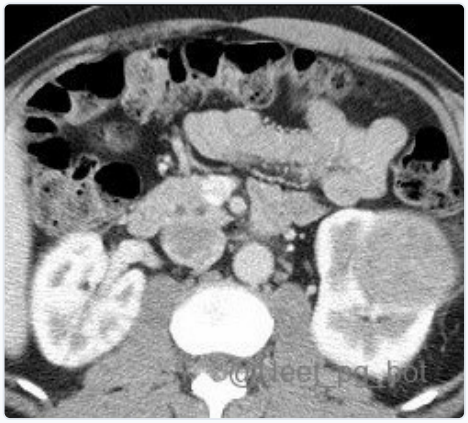
Renal oncocytoma: They are relatively benign in nature and usually present in the 6th to 7th decade. An important point regarding this tumor is its **similarity with RCC** and hence it is very difficult to differentiate between them preoperatively. A sharp **central stellate scar** may be seen in one-third of the cases. But RCC may also show this stellate scar appearance in some cases.

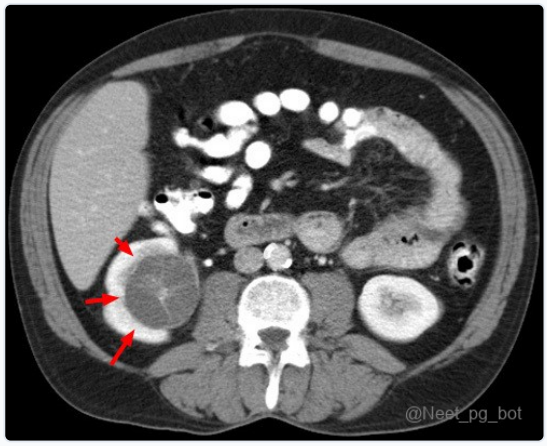
Renal cyst: It is a broader term that includes the cystic lesions of the kidney. They can be simple cysts that are benign. Complex or complicated cystic lesions need evaluation and can be malignant. Bosniak classification of renal cysts is given above.

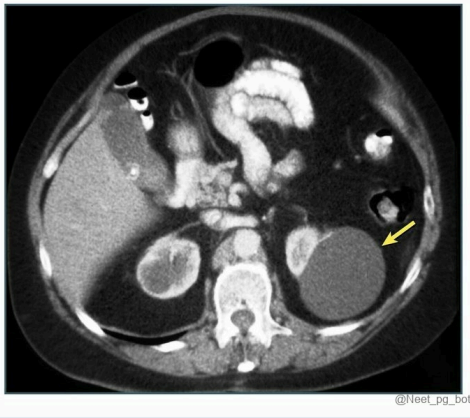
Image showing a simple renal cyst:











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References: Online Resource, https://radiopaedia.org/articles/renal-angiomyolipoma
Grainger & Allison's Diagnostic Radiology 6th edition, Page no 834,883
Online Resource, https://radiopaedia.org/articles/renal-oncocytoma
Online Resource, https://radiopaedia.org/articles/renal-cell-carcinoma-1?lang=us
Online Resource, https://radiopaedia.org/articles/renal-cyst-1

Q55. Radiology INI-CET 2021 ID: MF0893

A young child presented with mild intermittent upper abdominal pain. X-ray is given below. What is the diagnosis?

A Bochdalek hernia

B ✓ Morgagni hernia

C Gastric volvulus

D Eventration of diaphragm

Correct Answer: Morgagni hernia

The clinical features and the chest X-ray showing an opacity in the **right cardiophrenic angle** are suggestive of **Morgagni hernia**.

Congenital diaphragmatic hernia(CDH) of **anteromedial location** is the **Morgagni** hernia. Here the defect is anterior and retrosternal. It is **rarer** than bochdalek hernia and accounts for only 2-5% of CDH. About **90% of the cases** occur on the **right side**.

It is less symptomatic than bochdalek hernia and the diagnosis is usually delayed until childhood. The patient presents with respiratory or upper GI symptoms. Respiratory symptoms like distress, tachypnea, and recurrent pulmonary infections are common in children. Other symptoms are abdominal pain, dyspnea, flatulence, and indigestion.

Radiographic findings:

- **Radiolucent paracardiac shadow** may be seen **retrosternally** on lateral chest x-ray if there is bowel herniation.
- **Rounded opacity** in the **right cardiophrenic angle** may be seen if there is herniation of omentum or solid organs.
- A CT scan may be used to confirm the diagnosis.

Surgical repair is indicated in all cases either by a thoracic or abdominal approach.

Other options:

Bochdalek hernia: It is **more common** than Morgagni hernia. Here the defect is **posterolateral** in location. Respiratory distress is experienced by most infants. In the pediatric age group, symptoms are predominantly respiratory due to pulmonary hypoplasia. In adults, it is usually asymptomatic and is incidentally found. About **85% of the cases are left-sided.**

Gastric volvulus: This is an uncommon condition and presents with the characteristic **Borchardt triad** which includes:

- Sudden-onset severe upper abdominal pain
- Recurrent retching with the production of little vomitus
- Inability to pass a nasogastric (NG) tube

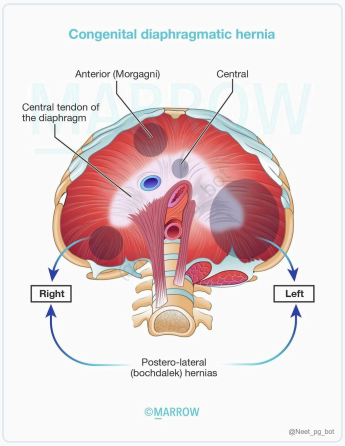
It can be **organoaxial or mesentroaxial**. Chest X-ray may show intrathoracic upside-down stomach and retrosternal or mediastinal air-fluid level.

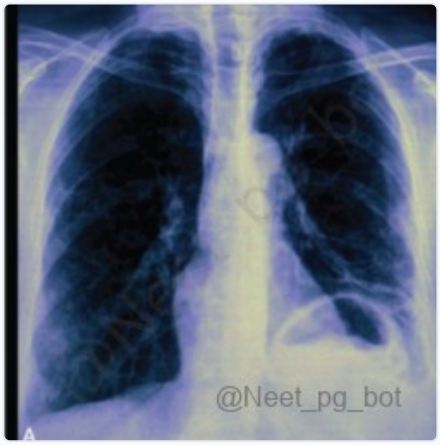
Image showing intrathoracic gastric volvulus with mediastinal gastric bubble-

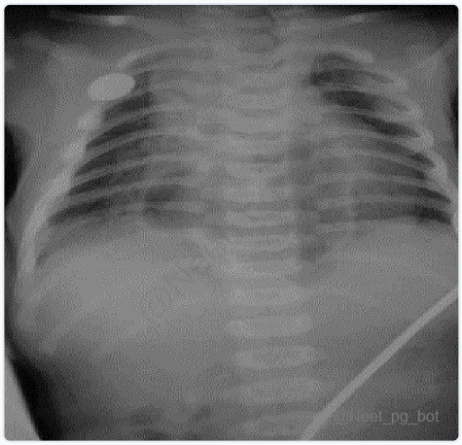
Eventration of the diaphragm: It refers to the **abnormal contour of the dome of the diaphragm** without any disruption to the diaphragmatic continuity. It usually involves a segment of the hemidiaphragm. It can cause respiratory distress in newborns and is usually asymptomatic in adults.

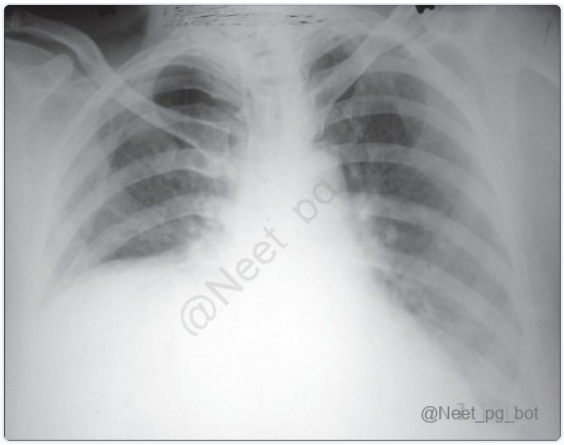
Radiographic findings: The affected portion of the hemidiaphragm is elevated and is seen as a smooth hump, while the normal contour is seen in the remaining part. '**Double diaphragmatic contour**' may be seen in frontal view and is confirmed with a lateral X-ray.

Image showing eventration of the right dome of the diaphragm-









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References: Sabiston Textbook of Surgery 20th Edition, Page no: 1232,1863,1864
Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK557501/
Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK539767/
Online Resource, https://radiopaedia.org/articles/gastric-volvulus
Online Resource, https://radiopaedia.org/articles/diaphragmatic-eventration

Q56. Radiology INI-CET 2021 ID: MF0873

A woman presented with confusion, constipation, and headache. Serum calcium was 13.1mg/dL and ALP was elevated. She was diagnosed with primary hyperparathyroidism. Which of the following is the imaging of choice?

A MRI

B FDG PET

C CT

D ✓ Sestamibi scan

Correct Answer: Sestamibi scan

Sestamibi scan is the imaging of choice to delineate **parathyroid abnormalities**.

Sestamibi is a monovalent lipophilic cation. It shows passive diffusion across cell membranes and **accumulates in mitochondria**. Because of its accumulation in the mitochondria, there is a difference in the rate at which it gets washed out depending upon the number of mitochondria in each tissue.

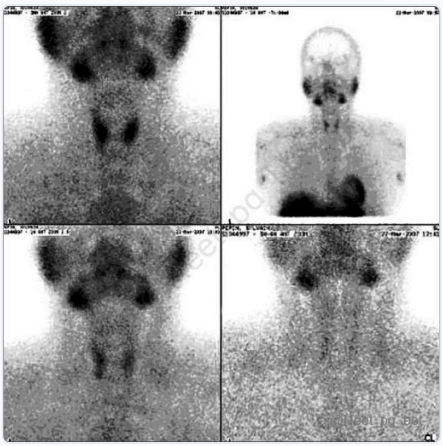
In **parathyroid adenomas**, there is a **high concentration** of **oxyphilic cells** with **high mitochondrial content**. The three protocols for sestamibi scanning are:

- Single-isotope dual-phase scan
- Dual-isotope subtraction imaging
- Single-photon emission computed tomography (SPECT).

SPECT is **commonly** used because of its improved spatial resolution. Sestamibi combined with CT can be used to detect smaller parathyroid tissues.

Other imaging modalities for parathyroid localization:

- **Ultrasonography:**
  - It is inexpensive and noninvasive
  - **Mediastinal** parathyroid tissue **cannot** be detected because USG cannot penetrate the sternum or clavicles.
- **Four-dimensional CT:**
  - It has a very high sensitivity
  - Increased radiation exposure than other routine modalities is a potential disadvantage
- **Selective venous sampling** for PTH can be used for the exact localization of adenomas



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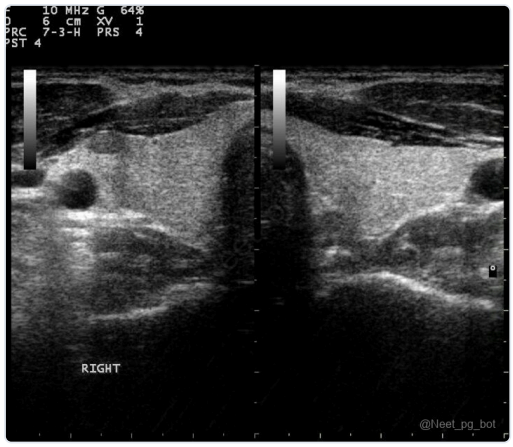
References: Bailey & Love's Short Practice of Surgery - 27th Edition, Page no:825-828
Sabiston Textbook of Surgery 20th Edition, Page no:930

Q57. Radiology INI-CET 2021 ID: MF2384

Identify the imaging modality given below.

USG- Ultrasonography
MRI- Magnetic resonance imaging



A ✓ USG

B Fluroscopy

C X-Ray

D MRI

Correct Answer: USG

The image shown in the question is the **USG** of the neck.

Sonographic examination of the neck shows the thyroid gland, muscles, blood vessels, bony structures or cartilages, and nerves. The **thyroid** of a healthy patient appears homogeneously echogenic with a uniform echotexture on USG. Lateral to the thyroid, the large blood vessels (the common carotid artery and more laterally, the internal jugular vein) can be identified. In the above image, **hypoechoic areas** are suggestive of the **lumen of vessels** seen in the neck i.e common carotid artery and internal jugular vein.

Neck ultrasonography is the preferred modality for routine radiographic evaluation of the thyroid gland, as it offers the most accurate visualization of thyroid nodular disease. It also allows for the concomitant performance of percutaneous fine-needle aspiration biopsy of thyroid lesions.

Neck ultrasound can also be used to localize parathyroid disease.

Other options:

Option B: **Fluoroscopy** is an imaging modality that allows real-time X-ray viewing of a patient with high resolution. During this procedure, radiologists usually use contrast material (dye) to highlight the area being examined.

Below image shows the fluoroscopy of neck.

Option C: The below image shows the **X-ray** appearance of neck in AP view.

Option D: Relative to the neck musculature, the thyroid is hyperintense on MRI images.

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References: Online Resource, https://radiopaedia.org/articles/fluoroscopy
Online Resource, https://www.google.co.in/books/edition/Ultrasonography_of_the_Head_and_Neck/HxegDwAAQBAJ?hl=en&gbpv=1&dq=Ultrasonography+of+the+Head+and+Neck&printsec=frontcover
Sabiston Textbook of Surgery 21st Edition, Pg 882

Q58. Radiology INI-CET 2021 ID: MF2319

Which of the following is true regarding CORADS?

CORADS- COVID-19 Reporting and Data System

A CORADS- 0 indicates normal lung

B ✓ CORADS-5 is typical of COVID-19

C Pleural thickening is typical of COVID-19

D Lobar consolidation is typical of COVID-19

Correct Answer: CORADS-5 is typical of COVID-19

Among the given options, **CORADS-5** is **typical** of **COVID-19** is the true statement.

COVID-19 Reporting and Data System (CO-RADS) is a categorical assessment for pulmonary involvement of COVID-19 on chest CT.

CO-RADS Category	Summary
CO-RADS 0	Scans are incomplete or of inferior quality
CO-RADS 1	Normal or non-infectious
CO-RADS 2	Typical features of other infections but not of COVID-19
CO-RADS 3	Features compatible with COVID-19 but also with other diseases
CO-RADS 4	Suspicious for COVID-19
CO-RADS 5	**Typical for COVID-19**
CO-RADS 6	Proven COVID-19 by positive RT-PCR test

CO-RADS categories:

- CO-RADS 0: **Not interpretable -** Technically insufficient to assign a score: Scans are incomplete or of inferior quality.
- CO-RADS 1: **Very low level of suspicion** for pulmonary involvement by COVID-19. Mild or severe emphysema, perifissural nodules, lung tumors, or fibrosis.
- CO-RADS 2: **Low level of suspicion** for pulmonary involvement by COVID-19. Includes infectious etiology that is not compatible with COVID-19 such as bronchitis, bronchiolitis, bronchopneumonia, lobar pneumonia, abscess.
- CO-RADS 3: **Equivocal findings for pulmonary involvement** by COVID-19. Findings overlap with other viral or infectious etiologies. Could include perihilar ground-glass, homogenous extensive ground-glass with or without sparing of secondary pulmonary lobules.
- CO-RADS 4: **High level of suspicion** for pulmonary involvement by COVID-19. Findings are similar to CO-RADS 5 but are not present in the typical locations. Findings may also be superimposed on severe pre-existing pulmonary abnormalities.
- CO-RADS 5 based on **typical CT findings**. It requires the presence of at least one confirmatory pattern (mentioned below).
- CO-RADS 6: Proven COVID-19 by **positive RT-PCR test.**

CT features typical of pulmonary involvement in COVID-19 include:

- Obligatory features
  - Ground glass opacities, **with or without consolidation** in areas close to the visceral pleural surface, including the fissures.
- Confirmatory features
  - Ground glass regions
  - Crazy paving pattern
  - Organizing pneumonia compatible patterns
  - **Thickened vessels** with parenchymal abnormalities

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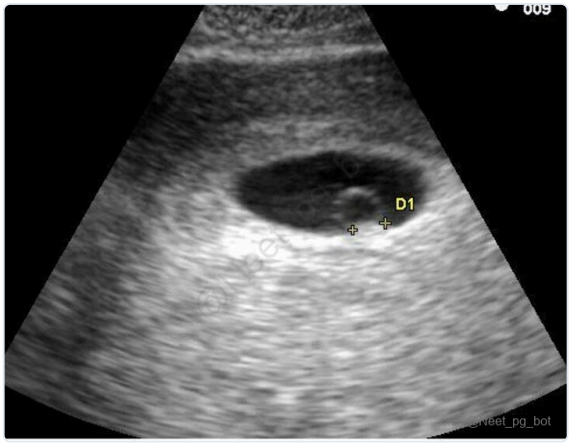
References: Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7233402

Q59. Radiology INI-CET 2021 ID: MF2314

Double bleb sign seen in early pregnancy is due to _____.



A Twin pregnancy

B Amnion and chorion

C ✓ Yolk sac and amniotic sac

D Heterotopic pregnancy

Correct Answer: Yolk sac and amniotic sac

The **double bleb sign** is due to the **yolk sac and amniotic sac.**

On ultrasonographic examination, a gestational sac may contain the yolk sac and amniotic sac, appearing like two small bubbles. This is known as the **double bleb sign.** The embryonic disc is located between the **yolk sac and the amniotic sac.** It is an important feature of an **intrauterine pregnancy**.

Other options:

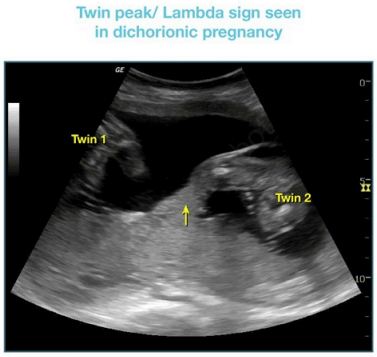
In a **twin pregnancy,** the chorionicity of the placenta is best diagnosed with the help of an ultrasound examination.

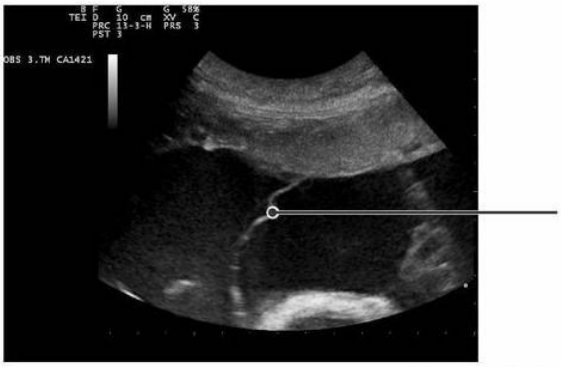
- In **dichorionic gestation**, the twins are separated by amniotic membranes and an **intervening chorion**. So, a **thicker twin peak sign or lambda (λ) sign** is formed at the point at which the amnions meet the placenta.

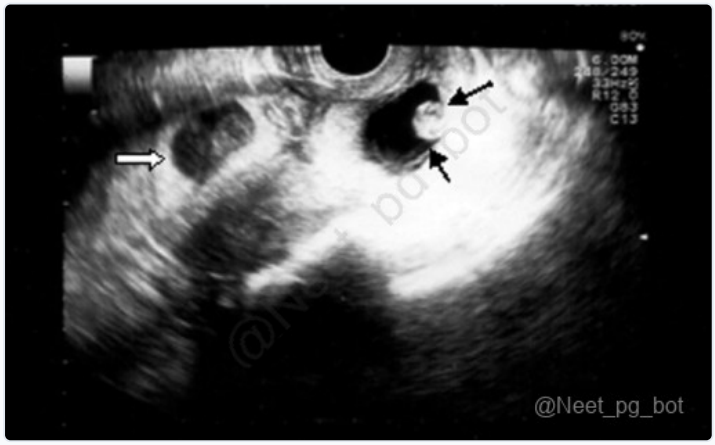
- In **monochorionic diamniotic** gestation, the twins are separated only by the two amniotic membranes with **no intervening chorion**. So, a **thin T sign** is formed at the point at which the amnions meet the placenta.

Heterotopic pregnancy is a condition in which a normal uterine pregnancy co-exists with an extrauterine (ectopic) pregnancy. Although heterotopic pregnancy can occur spontaneously, it is more likely after artificial reproductive technology **(ART)** and **ovulation induction.**

The below image shows heterotopic pregnancy with 2 sacs, one inside the uterus (white arrow) and the other in the cervix (black arrows).







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References: Online Resource, https://radiopaedia.org/articles/double-bleb-sign
Williams Gynecology 3rd edition, Pg 175
DC Dutta's Textbook of Obstetrics 8th Edition, Pg 237

Q60. Radiology INI-CET 2021 ID: MF2325

Which one of the following is the most radioresistant phase of the cell cycle?

A ✓ S phase

B M phase

C G1 phase

D G2 phase

Correct Answer: S phase

S phase is the **most radioresistant phase** of the cell cycle.

The replication of the DNA genome occurs only at a specified time period during the cell cycle. This period is referred to as the **synthetic or S phase.** The S phase is separated from the **mitotic**, **or M phase**, by nonsynthetic periods referred to as **gap 1 (G1)** and **gap 2 (G2) phases**, occurring before and after the S phase, respectively. The cell prepares for DNA synthesis in G1, and for mitosis in G2.

The most **radiosensitive** phase of the cell cycle is the **G2-M phase.** Cells in the G2 or M phase do not have time to repair DNA damage that occurs before they start to divide. This affects mitosis, leading to catastrophe.

Cells become gradually less sensitive to radiation in the G1 phase to the early S phase and are **least sensitive** in the **late S phase**.

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References: Basic Radiotherapy Physics and Biology, 2014 edition, Page no 257
Online Resource, https://pubmed.ncbi.nlm.nih.gov/15234026/
Harper's Illustrated Biochemistry 31st edition, Page no 368

Q61. Radiology AIIMS 2020 (May and Nov INI-CET) ID: MD4510

Which of the following statements is false regarding acute radiation syndrome?

A ✓ GI manifestations occurs at a lower radiation dose than bone marrow depression

B Bone marrow suppression appear at doses greater than 1 Gy

C Bone marrow suppression occurs earlier than neurovascular effects

D GI manifestations occur at a lower radiation dose than neurovascular effects

Correct Answer: GI manifestations occurs at a lower radiation dose than bone marrow depression

GI manifestations **do not** occur at a lower radiation dose than bone marrow depression.

Stages of Acute Radiation Syndrome (ARS)

The prodromal phase manifests with non-specific symptoms like nausea, vomiting, anorexia. Early and persistent symptoms predict severity.

Latent phase: The patient looks and feels generally healthy for a few hours or even up to a few weeks. In severe cases, this phase may be absent.

Manifest illness phase: Symptoms are seen as per the radiation dose. Tissues with greater rates of cellular division are most radiosensitive, and hence the hematopoietic and gastrointestinal systems are the most affected.

Recovery phase or Death: The recovery phase will be seen after the manifest stage when the patient recovers from the acute exposure. The time taken for recovery will depend on the severity of the injury and the organ system affected.

Radiation dose	Sub-syndrome	Features
~1-2 Gy	Hematopoietic	Bone marrow suppression
~6-10 Gy	Gastrointestinal	Nausea, vomiting, gastrointestinal bleeding, malabsorption,
~20Gy	Neurovascular	irritability, altered mental status, seizures, coma

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References: Rosen's Emergency Medicine: Concepts and Clinical Practice 9th Edition, Page no 1807
Online Resource, https://www.cdc.gov/nceh/radiation/emergencies/arsphysicianfactsheet.htm
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3863169/

Q62. Radiology AIIMS 2020 (May and Nov INI-CET) ID: MD4508

A 46-year-old male presented with chronic back pain and neck stiffness. X-ray cervical spine is given below. What is the diagnosis?

A Ankylosing spondylitis

B ✓ Diffuse idiopathic skeletal hyperostosis

C Klippel feil syndrome

D Sprengel deformity

Correct Answer: Diffuse idiopathic skeletal hyperostosis

The given clinical scenario and the radiograph showing **ossification of the anterior longitudinal ligament**, in front of the C2 to C7 cervical vertebral bodies, point towards a diagnosis of Diffuse Idiopathic Skeletal Hyperostosis (DISH). It is also referred to as **Forestier disease.**

DISH is a common disorder characterized by bone and calcium formation at the sites of ligamentous insertions. It occurs in the middle-aged or elderly. The radiograph shows calcification of anterior and sometimes posterior longitudinal ligaments appearing as **flowing or dripping-like candle wax**. It is usually asymptomatic but the calcification might cause compression of the spinal cord and can be better visualized on CT and MRI. In contrast to ankylosing spondylitis, the sacroiliac joints are usually spared.

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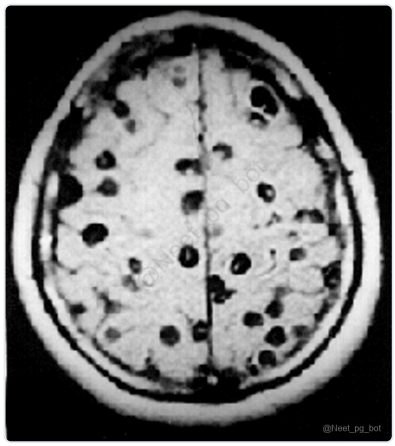
References: Learning Radiology: Recognizing the Basics, Page no 271
Harrison's Principles of Internal Medicine 20th Edition, Page no 2566

Q63. Radiology AIIMS 2020 (May and Nov INI-CET) ID: MD4505

An 18-year-old girl presented with seizures and her radiological image is as given below. Knowing that the MR spectroscopy of the lesion showed elevated choline, alanine, and lactate, what is the likely diagnosis?



A Cerebral toxoplasmosis

B ✓ Neurocysticercosis

C Cryptococcosis

D Multiple tuberculomas

Correct Answer: Neurocysticercosis

The given radiological image and history of seizures suggests a diagnosis of **neurocysticercosis**.

Neurocysticercosis presents with seizures, hydrocephalus, raised ICP, and may even cause signs of chronic meningitis. The given image shows multilobulated cystic lesions (with scolex) in the subarachnoid space, typical parenchymal brain calcifications are suggestive of **neurocysticercosis** caused by T. solium **(Cysticercosis cellulose).**

- **Cyst with a dot sign:** seen in both MRI and CT and the dot represents the eccentrically placed scolex of the parasite.

MR Spectroscopy features of neurocysticercosis include elevated lactate, alanine, succinate, and choline levels and reduced levels of N -acetyl aspartate and creatine. **Escobar's pathological stages** describe 4 stages - vesicular, colloidal vesicular, granular nodular, and nodular calcified.

Other options

- Solid or ring enhancement is typical in the post-contrast images of tuberculoma. A central focus of calcification with a ring of peripheral enhancement (the "target sign") is also described in tuberculosis but is not specific to it.
- Cerebral toxoplasmosis appears as multiple hypodense regions predominantly in the basal ganglia and at the corticomedullary junction. It can be associated with ring enhancing lesions as well.
- Cryptococcosis is usually associated with normal scans but may show cerebral atrophy, hydrocephalus or mass lesions as well.

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References: Harrison's Principles of Internal Medicine 20th Edition, Page nos 1642 and 1643
Online Resource, https://radiopaedia.org/articles/cns-cryptococcosis-2
Online Resource, https://radiopaedia.org/articles/intracranial-tuberculous-granuloma
Online Resource, https://radiopaedia.org/articles/neurotoxoplasmosis

Q64. Radiology AIIMS 2020 (May and Nov INI-CET) ID: MD4411

Which of the following is a finding of large bowel obstruction?

A Dilated loops of intestine with plica circularis

B ✓ Dilated loops with haustrations

C String of pearls sign

D Central dilated bowel loops

Correct Answer: Dilated loops with haustrations

Dilated bowel loops with haustrations is a feature of **large bowel** obstruction.

In large bowel obstruction, the colon is distended proximally with the distal portion collapsed. The colon is considered dilated if it is more than 6 cm in diameter while the cut off for caecum is 9cm.

Other options:

Dilated loops of the intestine with **plica circularis** (**valvulae conniventes**) and **predominantly central loops** are features of the **small bowel obstruction.** The small bowel is considered dilated if it is more than 3 cm in diameter.

The **string of pearls sign** is also a feature of **small bowel obstruction** seen in plain radiographs or CT. It represents a **row of gas bubbles trapped between the valvulae conniventes.**

Differences in small and large bowel obstruction on radiography are listed below:

	Small bowel	Large bowel
Valvulae conniventes	Present in jejunum	Absent
No of Loops	Many	Few
Distribution of loops	Central	**Peripheral**
Haustra	Absent	Present
Diameter	3-5 cm	>5 cm

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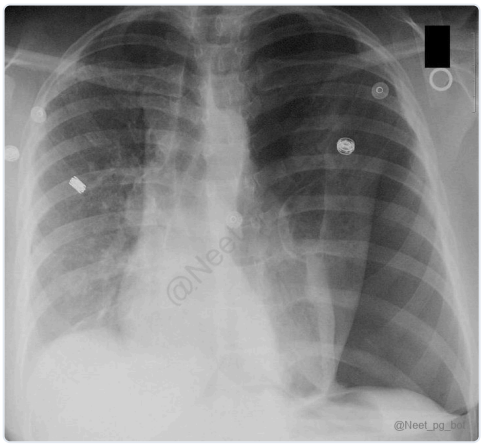
References: Bailey & Love's Short Practice of Surgery, 27th Edition, Page no 1288
Grainger & Allison's Diagnostic Radiology 6th edition, Page nos 601-603
Online Resource, https://radiopaedia.org/articles/small-bowel-obstruction, https://radiopaedia.org/articles/string-of-pearls-sign-gastrointestinal#:~:text=The%20string%20of%20pearls%20(or,slow%20resorption%20of%20intraluminal%20gas.

Q65. Radiology AIIMS 2020 (May and Nov INI-CET) ID: MA0730

A 25-year-old male with Marfanoid features presented with sudden onset breathlessness. The chest X-ray is given below. What is the likely diagnosis?



A ✓ Pneumothorax

B Pleural effusion

C Consolidation

D Fibrosis

Correct Answer: Pneumothorax

The chest X-ray demonstrating **tracheal deviation** and **displacement** of the **heart** away from the affected side along with the partial collapse of the left lung in the above condition points to a diagnosis of **left-sided pneumothorax**.

Marfan's syndrome is one of the causes for seconday spontaneous pneumothorax in adults.

Pneumothorax refers to the collection of air in the pleural space.

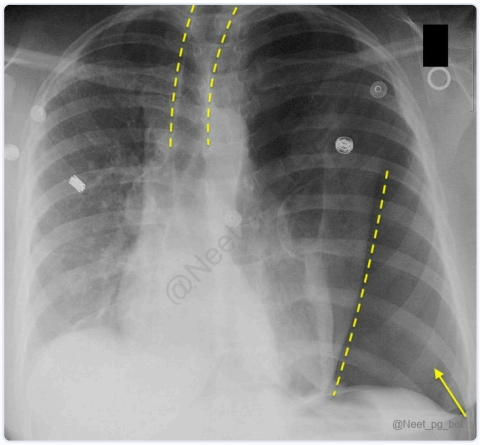
Chest radiograph in spontaneous pneumothorax demonstrates:

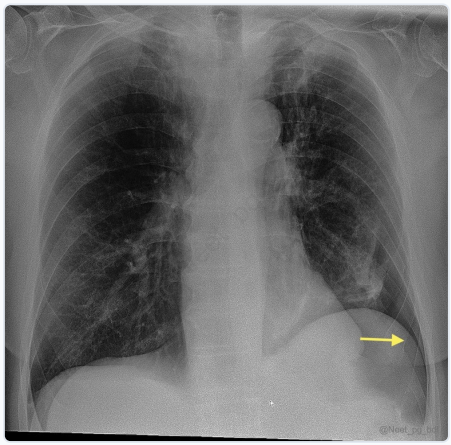
- **Thin, sharp white line**, demarcating the visceral pleural edge
- **Absence of lung markings** seen peripheral to this line
- **Peripheral space radiolucent** compared to adjacent lung
- **Complete lung collapse**, as well as **subcutaneous emphysema** and **pneumomediastinum**, may be seen
- **Deep sulcus sign** (on supine chest radiograph)

Deep sulcus sign: On a **supine position**(common in the ICU), air collects anteriorly and basally, within the non-dependent portions of the pleural space, **abnormally deepening** the costophrenic angle. This sign is indicative of **pneumothorax** in such patients and may be the only visible clue, especially when patient can not be imaged upright.

The image below is plain chest x-ray showing pneumothorax with deep sulcus sign (shown by arrow):

When there is a progressive accumulation of air intrapleurally in the thoracic cavity (caused by a valve effect during inspiration and expiration), it is called **Tension pneumothorax**. It can result in compression of mediastinal structures and compromise cardiopulmonary function. It is **life-threatening** and requires emergency intervention.





@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 7th edition, page no. 50,51,52
Harrison's Principles of Internal Medicine, 20th Edition, Page No. 2008, 2009
Online Resource, Viveiro C, Rocha P, Carvalho C, Zarcos MM. Spontaneous pneumothorax as manifestation of Marfan syndrome. BMJ Case Rep. 2013 Dec 5;2013:bcr2013201697. doi: 10.1136/bcr-2013-201697. PMID: 24311428; PMCID: PMC3863073.
Online Resource, Sharma R, Mahmud A, Tigges S, et al. Pneumothorax. Reference article, Radiopaedia.org (Accessed on 31 Dec 2024) https://doi.org/10.53347/rID-4578

Q66. Radiology AIIMS 2020 (May and Nov INI-CET) ID: MD6696

A 22-year-old male was brought to the emergency room following a road traffic accident. On examination, his GCS was E2M2V3. He had no fractures and no hydrocephalus or raised intracranial pressure. Non-contrast CT was performed and was found to be normal. The likely diagnosis is

A ✓ Diffuse axonal injury

B Post concussion syndrome

C Extra dural hemorrhage

D Herniation of brain

Correct Answer: Diffuse axonal injury

The given clinical scenario of an RTA with no imaging abnormality along with a disproportionate clinical manifestation suggests **diffuse axonal injury.**

It is caused due to **stretching of axons** in subcortical white matter due to sudden deceleration injuries. It is usually associated with **mild nonspecific changes** in **NCCT**. It is visualized on **T2 MRI** characterized by multiple lesions showing **microbleeds or linear hypointensities**. It may also show disruption of white matter tracts on tractography.

Other options

Post-concussion syndrome usually follows mild traumatic brain injury. It does not have a severe clinical status as seen in this patient.

Extradural hemorrhage is produced usually results secondary to trauma to head. The bleeding is usually arterial and blood collects between outer dura and inner table of the skull. It can very well be visualized on CT.

A cerebral herniation is defined as a shift of cerebral tissue in an adjacent space. Intracranial brain herniation can be classified into 3 types - subfalcine, transtentorial and tonsillar. A distinct clinical picture can be produced as a result of compression of neurovascular structures by the herniating part. CT is the imaging modality of choice because it can be done rapidly.

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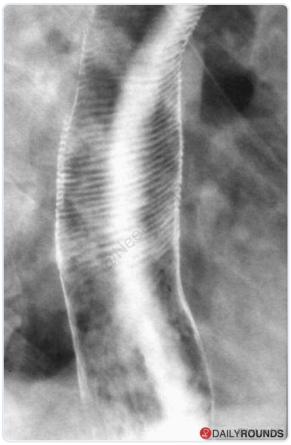
References: Harrison's Principles of Internal Medicine 20th Edition, Pg 3185,3186
Online Resource, https://radiopaedia.org/articles/diffuse-axonal-injury
Online Resource, https://www.ijri.org/article.asp?issn=0971-3026;year=2015;volume=25;issue=4;spage=404;epage=414;aulast=D

Q67. Radiology NEET 2018 ID: MC4003

In which of the following conditions is the given X-ray finding seen?



A Esophageal atresia

B Esophageal stenosis

C ✓ Reflux esophagitis

D Barrett's esophagus

Correct Answer: Reflux esophagitis

The given image is a barium swallow X-ray showing a **feline esophagus** suggestive of **reflux esophagitis**.

The **feline esophagus** is also known as **esophageal shiver**. These folds are 1-2 mm thick and run **horizontally** around the entire circumference of the esophageal lumen. It is thought to be due to the contraction of the muscularis mucosae of the esophagus. This results in the shortening of the esophagus and the bunching up of the mucosa in the lumen.

The appearance is almost always associated with active **gastroesophageal reflux**. The findings are **transient**, seen **following reflux** and not during swallowing. The appearance is confined to the **distal two-thirds** of the thoracic esophagus.

It can also be seen in **hiatus hernia** and **eosinophilic esophagitis**. Here, the folds are **permanent** and associated with **strictures**.

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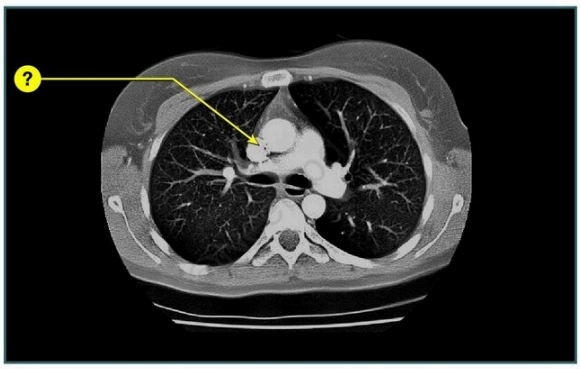
References: Harrison's Principles of Internal Medicine, 21st Edition, Page No. 2413, 2417
Online Resource, Dawes L, Hacking C, Sharma R, et al. Feline oesophagus. Reference article, Radiopaedia.org (Accessed on 09 Dec 2024) https://doi.org/10.53347/rID-1333 DOI:

Q68. Radiology NEET 2018 ID: MC4258

Identify the structure marked in the CT scan shown below:



A Descending aorta

B Ascending aorta

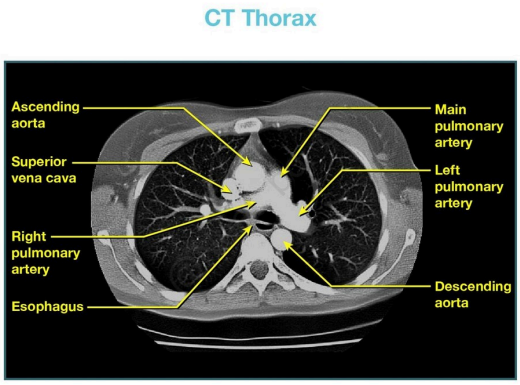
C ✓ Superior vena cava

D Right pulmonary artery

Correct Answer: Superior vena cava

The structure marked in the given CT thorax image is the **superior vena cava**.

The image below depicts all the major structures labelled in the CT thorax image:



@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no 4, 5, 6, 7, 8, 9, 10, 11

Online Resource, Hacking C CT chest arterial phase axial - labeling questions. Case study, Radiopaedia.org (Accessed on 02 Jan 2025) https://doi.org/10.53347/rID-62533

Q69. Radiology NEET 2018 ID: MC4000

What condition causes the skull appearance as in the given radiograph?

A ✓ Hyperparathyroidism

B Multiple myeloma

C Hyperthyroidism

D Paget's disease

Correct Answer: Hyperparathyroidism

The given X-ray of the skull shows **multiple radiolucent** lytic lesions, typical of the **salt** and **pepper** appearance (**pepperpot** appearance). This is seen in **hyperparathyroidism.**

In hyperparathyroidism, there is **resorption** of the **trabecular** bone of the skull (which causes the radiolucent areas) in combination with **cystic** areas of **osteopenia.** There is also a **loss** of **definition** between the **inner** and **outer tables** of the skull.

Other options:

Option B: In **multiple myeloma,** multiple well-defined **"punched out"** lytic lesions exist.

Option C: Untreated **hyperthyroidism** for several years can lead to **osteoporosis** of bones.

Option D: In **Paget's** disease, the pelvis, spine, skull, and proximal long bones are most frequently affected. Radiological skull changes include:

- Osteoporosis **circumscripta:** Large, well-defined lytic lesion
- **Cotton wool** appearance: Mixed lytic and sclerotic lesions of the skull
- **Diploic** widening: Both inner and outer calvarial tables are involved, with the former usually more extensively affected
- **Tam o'Shanter** sign: Frontal bone enlargement, with the appearance of the skull falling over the facial bones, like a Tam o'Shanter hat

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References: Online Resource, https://radiopaedia.org/articles/pepperpot-skull
Grainger & Allison's Diagnostic Radiology 7th edition, page 1713, 1114

Q70. Radiology NEET 2018 ID: MC3997

What dye is used for the diagnosis of esophageal perforation?

A Gastrografin

B ✓ Iohexol

C Gadolinium

D Barium sulphate

Correct Answer: Iohexol

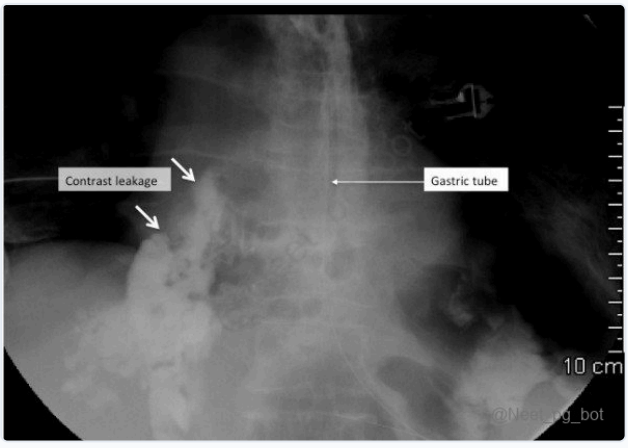
Iohexol is the contrast medium used for the diagnostic work-up for a suspected **esophageal perforation.**

Although **gastrografin** (water-soluble) was used previously, it has a high chance of causing aspiration pneumonitis and pulmonary edema consequently. Hence it is not preferred nowadays. (Option A)

Barium contrast is not used in esophageal perforation as it causes barium-related inflammation of the mediastinum. (Option D)

Contraindications to the use of barium contrast	Alternative dyes used
Tracheoesophageal fistula	Dianosil- water-soluble, non-ionic agent
Esophageal perforation	Iohexol
Perforation peritonitis	Gastrografin / Iohexol

Given below is a plain chest radiography with a water-soluble contrast swallow, showing contrast leakage in a patient with spontaneous rupture of the esophagus.

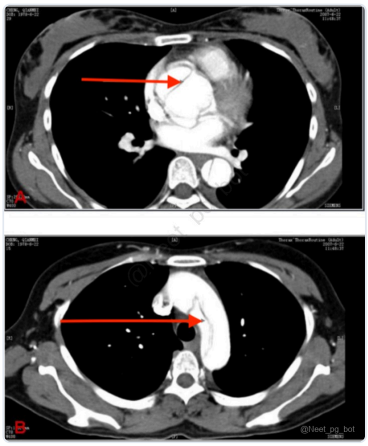


@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 7th edition, Page No. 483
Online Resource, Workman RB Jr. Imaging of the injured esophagus. Tex Heart Inst J. 2012;39(2):306-7; author reply 307. PMID: 22740765; PMCID: PMC3384036.

Q71. Radiology NEET 2018 ID: MC4008

A 50-year-old patient has come with sudden onset of chest pain and his CT scan shows the following picture. What is the diagnosis?



A Pulmonary embolism

B ✓ Aortic dissection

C Aortic aneurysm

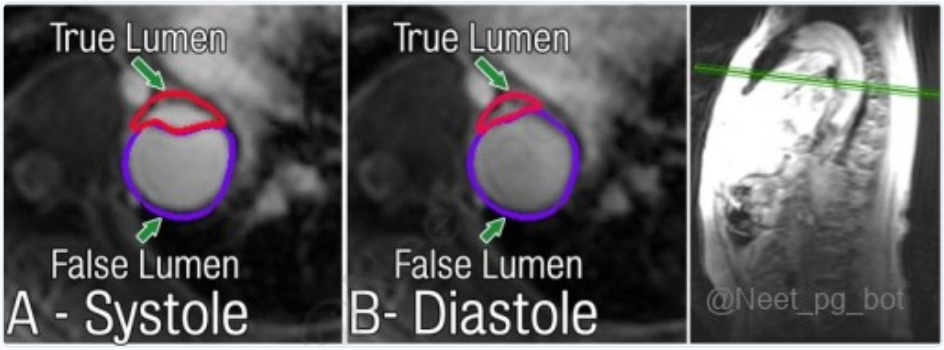
D Myocardial infarction

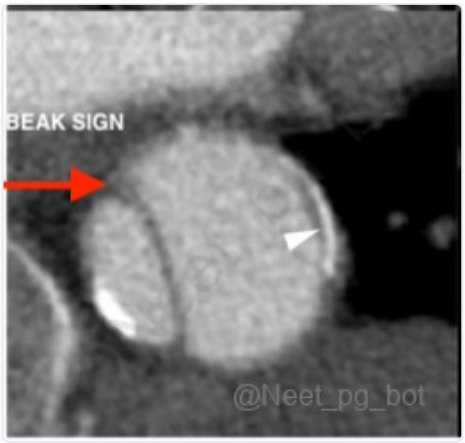
Correct Answer: Aortic dissection

The given CT scan showing a **double lumen** and dilatation of the aorta is suggestive of an **aortic dissection.**

In this condition, blood enters the space below the intima of the aorta. This results in the formation of a **true lumen** and **false lumen**. The true lumen is often the smaller of the two because of being compressed by the false lumen, as shown below.

The **beak sign** in aortic dissection is a reliable sign to identify the **false lumen**. It refers to the acute angle formed at the edge of the false lumen in aortic dissection in axial cross-section. It is formed by the borders of the outer aortic wall and the intimal flap, and maybe partially thrombosed (blunted beak).



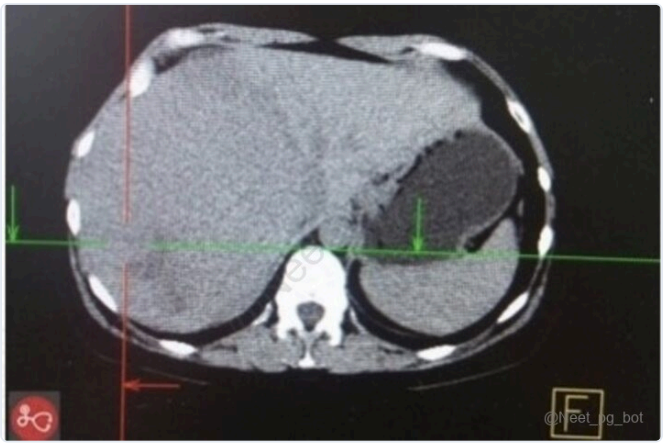


@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 7th edition, Page No. 424, 425, 426,427,428
Online Resource, Gaillard F, Silverstone L, Gajera J, et al. Aortic dissection. Reference article, Radiopaedia.org (Accessed on 12 Dec 2024) https://doi.org/10.53347/rID-918

Q72. Radiology NEET 2018 ID: MC4010

The following scan is of a young woman on oral contraceptives. The most likely diagnosis is:



A Hepatocellular carcinoma

B Metastatic infiltrates

C ✓ Hepatic adenoma

D Hemangioma

Correct Answer: Hepatic adenoma

The image given is of **hepatic adenoma.** It is the most frequent hepatic tumour in young women on oral contraceptive pills.

Hepatic adenomas, also referred to as **hepatocellular adenomas**, are **benign** liver tumours. They are well-defined vascular lesions, which are generally solitary. They are almost always seen in **women** between the age of 25 and 50 years and are associated with the use of the **oral contraceptive pill.**

On CT, uncomplicated adenomas are usually **homogeneous**, with a well-defined margin. They **enhance markedly** and uniformly during the **arterial phase** and then **merge** with the surrounding liver during the **portal phase.**

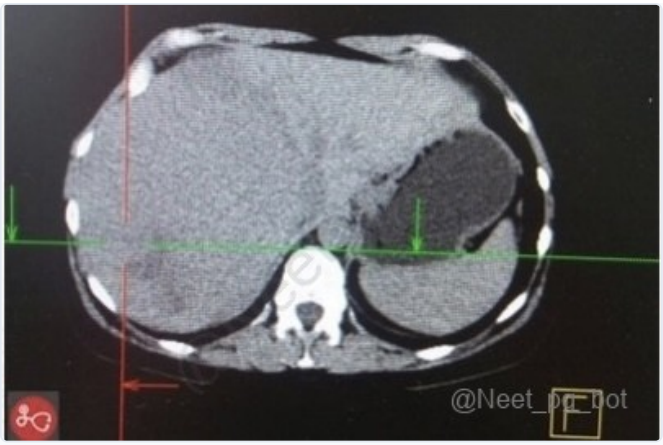
Hepatic adenoma on the CT scan is shown below.

Other options:

Option D: On contrast-enhanced CT scan**, hemangiomas** show **asymmetrical, nodular peripheral enhancement** that is isodense with large vessels and progressive **centripetal enhancement.** Liver hemangioma on CT scan is shown below.

Option B: Most **metastatic deposits** appear as areas of low attenuation on unenhanced CT and portal-phase CT. Hypervascular tumours are often visible as **low-attenuation** lesions on unenhanced images and may enhance transiently in the arterial phase, some becoming invisible in the portal phase. Metastatic infiltrates in the liver in the portal phase on CT scan are shown below.

Option A: **Hepatocellular carcinoma** (HCC) typically shows a decreased signal on T1-weighted images and a moderately increased signal on T2-weighted images, with internal heterogenecity. **MRI** is regarded as the most sensitive imaging modality for detecting HCC. Given below is the MRI image of HCC: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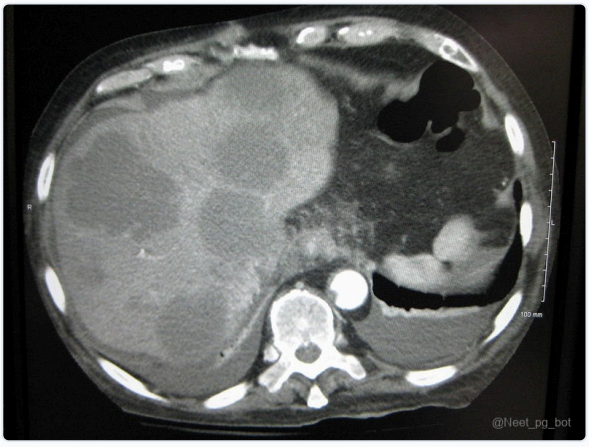
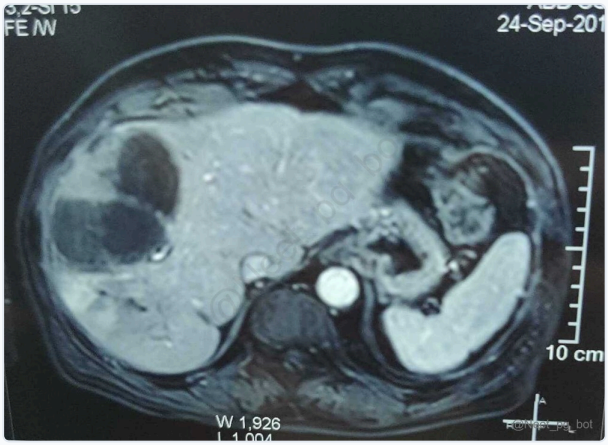
Spin:
Tilt:
I.V. & Oral Con.


@Neet_pg_bot


FE /W
24-Sep-201
10 cm
W 1,926

@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no 613, 616, 618, 624, 628, 629
Harrison's Principles of Internal Medicine, 21st Edition, Page No. 657

Q73. Radiology NEET 2018 ID: MC4004

A child presents with a left-to-right shunt and high-output cardiac failure. The following is his MRA. What is the most likely diagnosis?

- **A** ✓ Vein of Galen malformation
- **B** Dandy Walker syndrome
- **C** Pneumocephalus
- **D** Venous sinus thrombosis

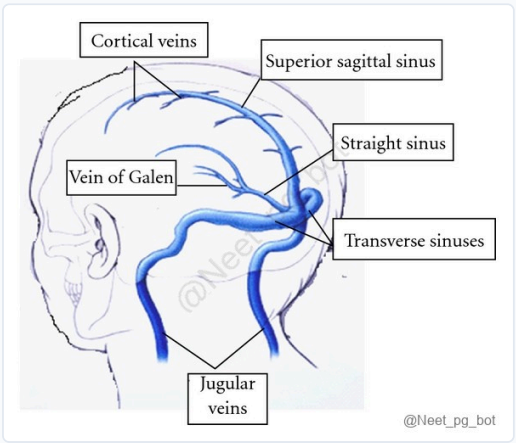
Correct Answer: Vein of Galen malformation

The MR-Angiography shown is of **Vein of Galen Malformation.**

Vein of Galen aneurysmal malformations (VGAM's), also known as **median prosencephalic arteriovenous fistula**, are uncommon intracranial anomalies that tend to present in early childhood with features of a left-to-right shunt and high-output cardiac failure.

The vein of Galen, also known as the great cerebral vein or great vein of Galen, is a short trunk formed by the union of the two internal cerebral veins and basal veins of Rosenthal. It lies in the quadrigeminal cistern. It curves backwards and upward around the posterior border of the splenium of the corpus callosum and drains the anterior and central regions of the brain into the sinuses of the posterior cerebral fossa.

The following image shows the Vein of Galen:

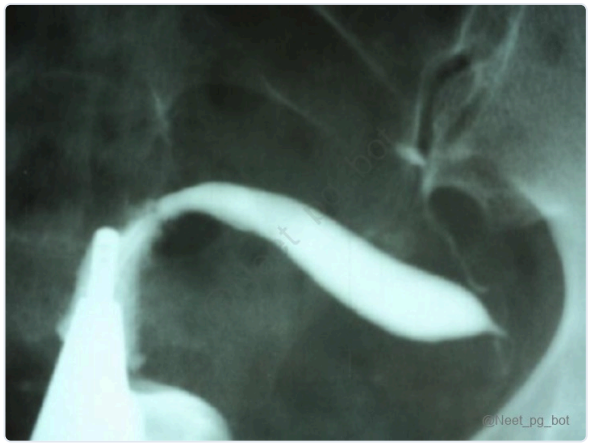


@Neet_pg_bot

References: Gray's Anatomy: The Anatomical Basis of Clinical Practice, 41st Edition, Page no 288
Grainger & Allison's Diagnostic Radiology 7th edition, Page No. 2031
Online Resource, D'Souza D, Campos A, Neto A, et al. Vein of Galen aneurysmal malformation. Reference article, Radiopaedia.org (Accessed on 17 Dec 2024) https://doi.org/10.53347/rID-2263

Q74. Radiology NEET 2018 ID: MC4005

What is the diagnosis based on the given image?



A Uterus didelphys

B Bicornuate uterus

C ✓ Unicornuate uterus

D Septate uterus

Correct Answer: Unicornuate uterus

The given image shows the **hysterosalpingogram** of a **unicornuate uterus.**

The endometrial cavity usually assumes a **fusiform (banana type)** shape, tapering at the apex and draining into a **single fallopian tube**. The uterus is generally shifted off the midline.

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References: Williams Gynecology - 4th edition, page no. 419-424

Q75. Radiology NEET 2018 ID: MC3998

The Bragg peak effect is pronounced in:

A X ray

B ✓ Proton

C Neutron

D Electron

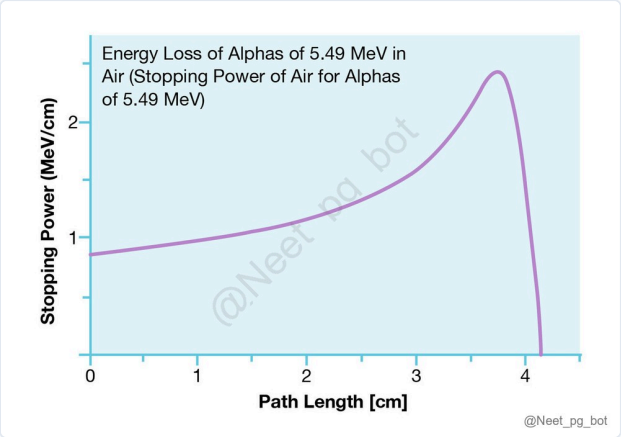
Correct Answer: Proton

The **Bragg peak effect** is pronounced in **protons**.

The proton is a heavy and charged particle that gradually loses its speed as it interacts with human tissue. It is easily controlled and delivers its maximum dose at a precise depth and can go as far as 32 cm. The absorbed dose increases very gradually with greater depth and lower speed, **suddenly rising** to a **peak** when the proton is ultimately stopped. This is known as the Bragg peak.

The behavior of the proton can be precisely determined, and the beam can be directed, allowing the Bragg peak to occur within the tumor site. Immediately following this burst of energy, the protons cease radiating and come to a complete stop. This property of **proton therapy** enables **targeting tumors** within the body, delivering radiation directly to the tumor while **sparing** the **surrounding healthy cells**, offering a much less invasive alternative to treat cancer.

The Bragg curve shown below plots the energy loss of ionizing radiation during its travel through matter.

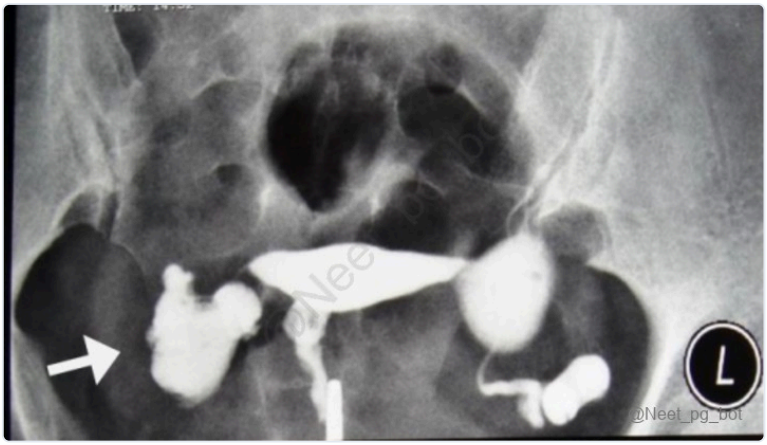


@Neet_pg_bot

References: Basic Radiotherapy Physics and Biology, 2014 edition, Page no. 49-51, 295
Grainger & Allison's Diagnostic Radiology 7th edition, Page no. 1740
Online Resource, Lane SA, Slater JM, Yang GY. Image-Guided Proton Therapy: A Comprehensive Review. Cancers (Basel). 2023 Apr 29;15(9):2555. doi: 10.3390/cancers15092555. PMID: 37174022; PMCID: PMC10177085.
Online Resource, https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/proton-radiation

Q76. Radiology NEET 2018 ID: MC4002

The image seen below shows_____.



A Magnetic Resonance Imaging

B Positron Emission Tomography

C ✓ Hysterosalpingography

D Laparoscopic chromopertubation

Correct Answer: Hysterosalpingography

The image given is of the findings seen in **hysterosalpingography**. The given image shows bilateral retort-shaped fallopian tubes, indicating **bilateral hydrosalpinx.**

Hysterosalpingography (HSG) is a technique used for visualization of the uterine cavity and fallopian tubes by the injection of a **radioopaque dye**, and visualization of the dye by **fluoroscopy** as it passes through the uterine cavity and fallopian tubes.

It is best performed between the end of the menstrual period and ovulation (i.e. **9th or 10th day of the cycle**). **10-15ml** of the dye is injected using a **Leech-Wilkinson cannula, Rubin's cannula**, or Foley's catheter. If the **tubes** are **patent**, the **dye** will **spill out** of the tubal Ostia and smear the adjacent bowel.

It is used to **detect**:

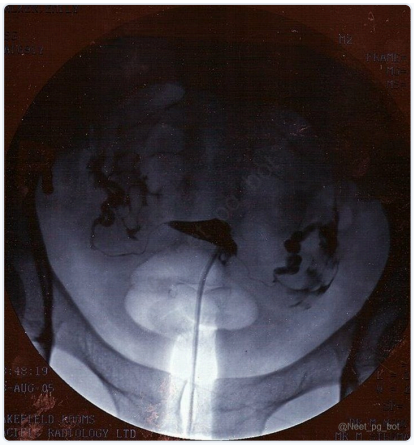
- **Tubal block:** This can be due to strictures (fibrotic block), inspissated amorphous material, or spasms. The tubal spasm can occur in an apprehensive woman, and this is prevented by premedication with atropine and analgesia.
- Uterine anomalies
- Uterine polyps
- Uterine synechiae
- The feasibility of tuboplasty, by studying the location and extent of tubal pathology.

Contradictions:

- Presence of genital tract infection and bleeding
- Genital tuberculosis, due to the risk of spreading infection.
- Suspected pregnancy
- Allergy to the dye
- Premenstrual phase (there is a risk of embolism and endometriosis if HSG is done during this phase).

The image given below shows the appearance of a normal hysterosalpingogram.

Laparoscopic chromopertubation is a laparoscopic procedure for establishing the **patency of fallopian tubes** or verifying the findings of blocked tubes on HSG. **Methylene blue** is injected through the cervix to visualize the free spill. The absence of spill indicates blocked tubes.



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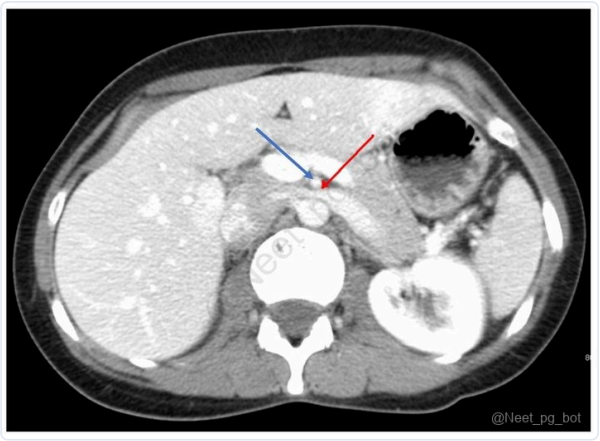
References: Williams Gynecology - 4th edition, Page no 44, 45
Shaw's Textbook of Gynaecology, 17th edition, Page no: 216, 507, 519, 520

Q77. Radiology NEET 2018 ID: MC4011

Identify the artery marked by blue arrow.



A ✓ Superior mesenteric artery

B Inferior mesenteric

C Celiac artery

D Intestinal artery

Correct Answer: Superior mesenteric artery

The artery marked by the blue arrow is the **superior mesenteric artery (SMA),** which gives rise to various branches that supply the small intestines, cecum, ascending, and part of the transverse colon.

The CT also depicts **nutcracker syndrome,** a left renal vein **entrapment syndrome** characterized by extrinsic **compression** of the **left renal vein** (marked by a red arrow), most commonly between the abdominal aorta and superior mesenteric artery. This syndrome is one of the postulated reasons for **left-sided varicoceles** being more common than the right side.

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References: Online Resource, D'Souza D, Bell D, Saber M, et al. Superior mesenteric artery. Reference article, Radiopaedia.org (Accessed on 12 Dec 2024) https://doi.org/10.53347/rID-5206
Online Resource, D'Souza D, Walizai T, Guan H, et al. Nutcracker syndrome. Reference article, Radiopaedia.org (Accessed on 12 Dec 2024) https://doi.org/10.53347/rID-1745

Q78. Radiology NEET 2018 ID: MC4255

What is the probable diagnosis based on the image of the scan given below?

A Pituitary apoplexy

B Pituitary macroadenoma

C Rathke cleft cyst

D ✓ Craniopharyngioma

Correct Answer: Craniopharyngioma

The given MRI of the brain demonstrates a **craniopharyngioma,** which is a suprasellar tumour that arises from **remnants of Rathke's pouch.**

The two variants of craniopharyngioma are:

- **Adamantinomatous type** (common in children): Predominantly solid with cystic components. Diagnostic features include compact lamellar keratin ("**wet keratin**") and areas of dystrophic **calcification**.
- **Papillary type** (common in adults)**:** Contains both solid sheets and papillae lined by well-differentiated non-keratinized squamous epithelium.

These tumours are usually large and heterogeneous, displaying **solid and cystic** components. The cysts contain a cholesterol-rich, thick brownish-yellow fluid that is compared to '**machine oi**l'.

Clinical features include **headaches** (most common), visual abnormalities - decreased acuity and **bitemporal hemianopia, GH deficiency,** growth failure, delayed sexual maturation due to pituitary hypofunction, and polyuria (central diabetes insipidus).

Radiologically:

- CT may show **calcifications** associated with the solid and cystic wall components
- MRI is the imaging modality of choice which shows:

  Cysts

  T1 Iso to hyperintense to the brain (due to high protein content "machinery oil cysts")

  T2 Mostly hyperintense

  Solid component

  T1 Vivid enhancement

  T2 Variable or mixed

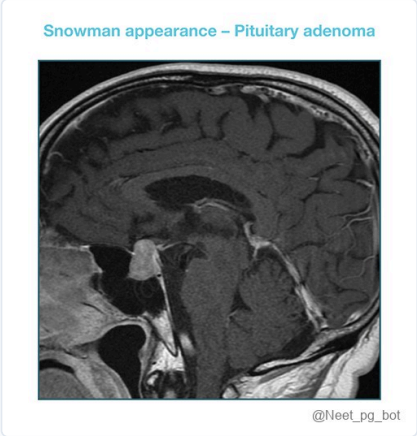
Craniopharyngiomas tend not to expand the pituitary fossa unless they become huge, which is a differentiating feature from pituitary macroadenomas.

Other options:

Option A: **Pituitary apoplexy** is an acute clinical condition associated with the rapid expansion of the pituitary gland, which is evident on an MRI as enhancement around a non-enhancing infarcted centre.

Option B: **Pituitary macroadenoma** (with cystic degeneration or necrosis) shows indentation at the diaphragma sellae, giving rise to a **snowman/figure of eight appearance/cottage loaf**. Calcification is often absent. The image below shows the snowman appearance:

Option C: **Rathke cleft cyst** shows no solid or enhancing component. Calcification is rare. It is unilocular and the majority are completely or mostly intrasellar.



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References: Nelson textbook of pediatrics 21st edition, Page No. 2676
Grainger & Allison's Diagnostic Radiology 7th edition, Page No. 1436
Harrison's Principles of Internal Medicine, 21st Edition, Page No. 2906
Online Resource, Ruth Prieto, Tareq A Juratli, Evan D Bander, Sandro Santagata, Laura Barrios, Priscilla K Brastianos, Theodore H Schwartz, José M Pascual, Papillary Craniopharyngioma: An Integrative and Comprehensive Review, Endocrine Reviews, 2024;, bnae028, https://doi.org/10.1210/endrev/bnae028

Q79. Radiology INI-CET 2024 ID: MG5998

Which of the following statements is correct about CT scan?

A ✓ CT score of water is 0

B CT consists of unfiltered x-ray beams

C It is the investigation of choice for gallstones

D 20mSV for NCCT head

Correct Answer: CT score of water is 0

The correct statement is that the CT score of water is 0.

On CT, each image consists of a matrix of data points (pixels) that are assigned a number that represents the X-ray attenuation(μ) at that point in the body. CT number is a scaled version of μ that is **normalised** to the **X-ray attenuation of water** (μW).

$CT = 1000 \times (\mu/\mu W - 1)$

The resulting numbers are called **'Hounsfield units'(HU)**. Due to this normalisation, water is assigned 0 HU, and air is assigned a CT number of −1000 HU.

CT numbers:

- **Water = 0 HU**
- Solid soft tissues are centred around 0 HU
- Lungs = -1000 HU
- Air = −1000 HU
- Dense cortical bone = +1000 HU
- Acute Blood Clot = 60-90 HU
- Fat = -100 HU

Other options:

Option B: Computed tomography (CT) imaging involves the use of a narrow beam of X-rays that are emitted from a rotating X-ray tube.CT scanners typically use a combination of X-ray beams at different energies and filters to reduce scatter and improve image quality. Therefore, the X-ray beams used in CT imaging **are often filtered** to **enhance image contrast** and **reduce radiation** dose to the patient.

Option C: The investigation of choice for gallstones is an **Abdominal ultrasound** and not a CT Abdomen. An ultrasound can **assess for complications** of gallstones, such as inflammation of the gallbladder (cholecystitis) or obstruction of the bile ducts. In cases where ultrasound findings are inconclusive or further characterisation is needed, other imaging modalities such as CT scan or magnetic resonance cholangiopancreatography (MRCP) may be used.

Option D: The mSv (millisievert) value for a non-contrast computed tomography (NCCT) scan of the head typically ranges from around **1 to 2 mSv**. This radiation dose can vary depending on factors such as the specific CT scanner used, the scanning protocol, and the individual patient characteristics. However, in general, NCCT head scans are associated with relatively **low radiation** doses compared to other medical imaging procedures.

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References: Online Resource, https://radiopaedia.org/cases/hounsfield-scale-diagram
Grainger & Allison's Diagnostic Radiology 6th edition, Page no: 77

Q80. Radiology INI-CET 2024 ID: MG5876

'Mount Fuji sign' is seen in _____

A Tension pneumomediastinum

B Tension pneumothorax

C Tension pneumoperitonium

D ✓ Tension pneumocephalus

Correct Answer: Tension pneumocephalus

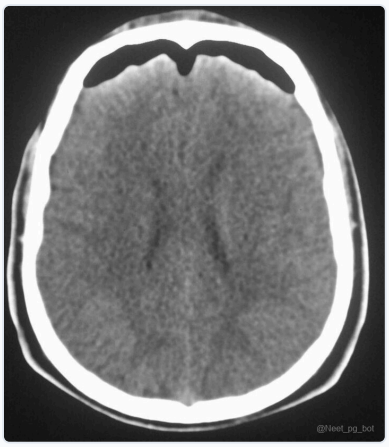
'Mount Fuji sign' is seen in **tension pneumocephalus.**

Mount Fuji sign refers to the presence of air **(pneumocephalus) between the tips of the frontal lobes** giving the appearance of Mount Fuji.

Tension pneumocephalus occurs with **subdural air** causing a mass effect over the underlying brain parenchyma, often from a **ball valve mechanism** causing **one-way entry of air into the subdural space.**

The appearance of this sign often warrants **immediate surgery** to prevent permanent damage.

The CT image below shows the Mount Fuji sign.

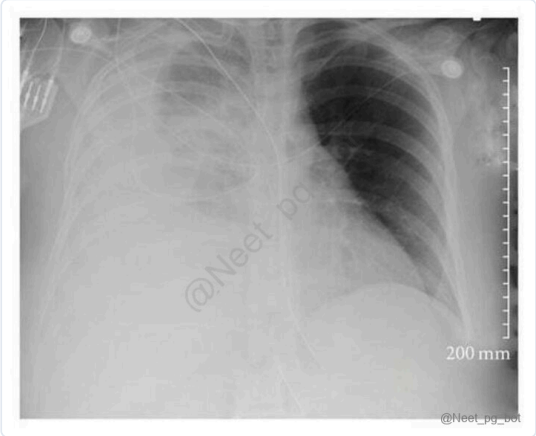


@Neet_pg_bot

References: Online Resource, https://radiopaedia.org/articles/mount-fuji-sign-1

Q81. Radiology INI-CET 2024 ID: MG6005

A patient presented with a stab wound over the right chest. The supine X-ray image of the patient is given below. What is the diagnosis?



A Right pneumothorax

B Pulmonary contusion

C ✓ Right hemothorax

D Right diaphragmatic rupture

Correct Answer: Right hemothorax

The history of stab injury to the right chest and the X-ray showing **opacity** of the **right hemithorax**, with a slight **tracheal deviation** towards the opposite side, indicates a diagnosis of **right hemothorax.**

In a plain chest radiograph, it's difficult to differentiate an acute hemothorax from other collections of pleural fluid as they are indistinguishable. As blood clots, it often leads to localisation, occasionally forming a fibrin body. Pleural thickening and calcification are acknowledged as potential consequences.

The following are the findings commonly seen in a case of hemothorax:

- **Blunting of Costophrenic Angle:** Hemothorax results in the accumulation of blood in the pleural cavity, causing the lung to collapse partially. This can lead to the blunting or obliteration of the costophrenic angle on the affected side.
- **Increased Opacity:** The presence of blood in the pleural space leads to increased opacity or density in the affected hemithorax. This can manifest as a homogenous shadowing that obscures the lung markings.
- **Mediastinal Shift:** Depending on the volume of blood in the pleural cavity, there may be a mediastinal shift towards the opposite side. However, in mild to moderate hemothorax, this may not be readily apparent on supine X-rays.
- **Air-Fluid Level:** In cases where there is concurrent pneumothorax (air in the pleural cavity), an air-fluid level may be visible, with the blood layering dependently and the air layer superiorly.
- **Unilateral Findings:** The abnormalities will typically be limited to one side of the chest, corresponding to the side of the hemothorax.

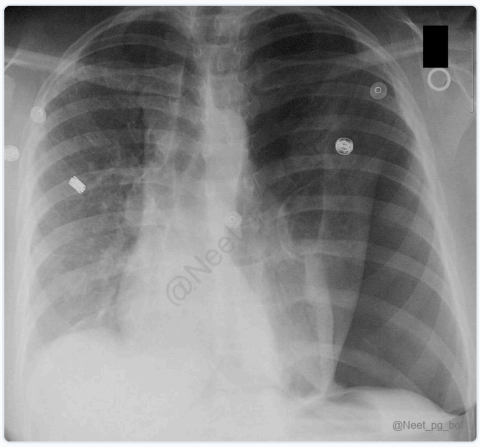
In cases of suspected hemothorax, a **lateral decubitus** or **chest X-ray in erect position** may be more sensitive for detection. Additionally, other imaging modalities such as ultrasound or computed tomography (CT) scans can provide more detailed information about the presence and extent of a hemothorax.

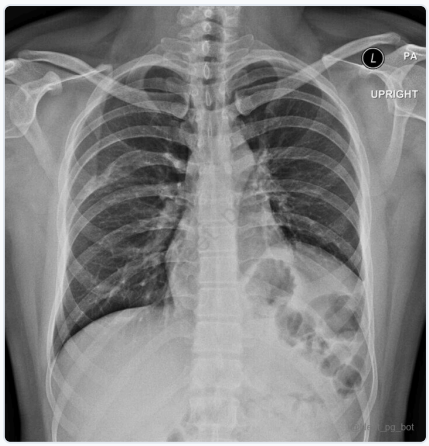
Other options:

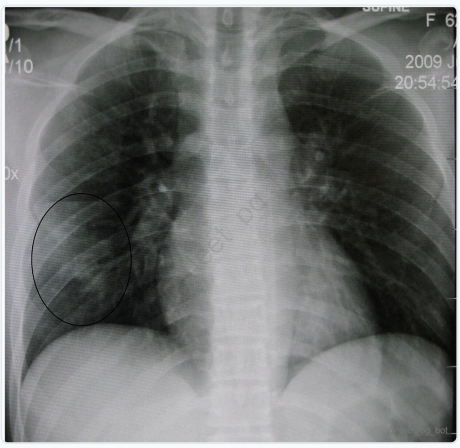
The following image is characteristic of a **pneumothorax** (left-sided) with **radiolucency** over the left lung, lung collapse, and mediastinal shift on x-ray.

The given radiograph is a case of **diaphragmatic hernia** that shows elevation of the left hemidiaphragmatic line along with the bowel loops and stomach seen in the pleural cavity.

The following is an image of a **right-sided pulmonary contusion** post-blunt chest trauma. Note the increased opacity just below the horizontal fissure.







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References: Grainger & Allison's Diagnostic Radiology 6th edition, Page No: 50,51

Q82. Radiology INI-CET 2024 ID: MG6002

Which of the following contrast agents is safe to use in renal failure patients?

A Gadolinum contrast used in MRI

B Iodinated ionic contrast agent

C Iodinated non-ionic contrast agent

D ✓ Ultrasound contrast agent

Correct Answer: Ultrasound contrast agent

Ultrasound contrast agents are gas-filled microbubbles that are administered intravenously to the systemic circulation. Hence, considered as low risk in patients with renal impairment.

Ultrasound contrast agents are used to enhance the visibility of blood vessels, organs, and tissues during ultrasound imaging. These agents typically consist of tiny **gas-filled microbubbles** encapsulated within a shell. When injected intravenously, they flow through the bloodstream and resonate in response to ultrasound waves, producing strong echoes that enhance the ultrasound image.

Perflutren/Optison is an ultrasound contrast agent. **Levovist** (consisting of galactose microparticles, air bubbles, and palmitic acid) is another example of an ultrasound contrast agent. Sulfur hexafluoride with a phospholipid shell is also used as an ultrasound contrast agent.

Other options:

Option A: **Gadolinium contrast** used in MRI is predominantly excreted through the kidney. In renal failure, it gets accumulated in the body and releases gadolinium ions. Since the ion is toxic, the body tries to contain it by fibrosis resulting in multisystem painful fibrosis called **nephrogenic systemic fibrosis.**

Option B: **Iodinated ionic contrast agents** increase the risk of **Contrast-induced nephropathy(CIN)**. Hence alternative imaging methods, without iodinated contrast media, should be chosen in patients with considerably impaired renal function (CKD 4 or 5).

Option C: **Non-ionic contrast agents** have a lower risk of causing adverse reactions compared to ionic contrast agents. They are generally **better tolerated** by patients, with fewer incidences of allergic reactions and nephrotoxicity. However, USG contrast agents are among the safest in renal failure patients.

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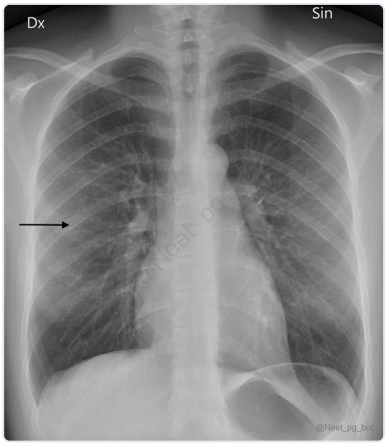
References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no: 31 , 36
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5206822/

Q83. Radiology INI-CET 2024 ID: MG6007

Which of the following structures is the arrow pointing to?



A Anterior end of rib

B Costal cartilage

C ✓ Posterior end of rib

D Right pulmonary artery

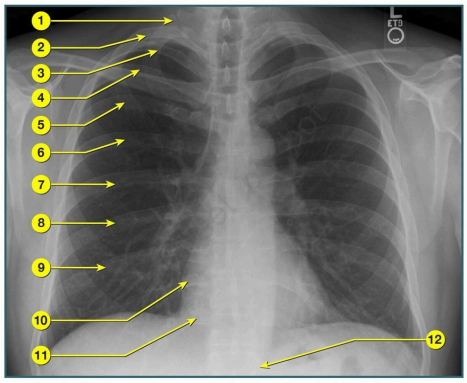
Correct Answer: Posterior end of rib

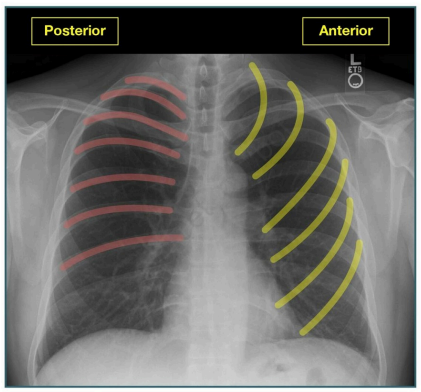
The structure pointed by the arrow mark in the given chest X-ray PA view is the **posterior end of the rib**.

In the chest X-ray PA view, ribs appear as **light grey strips**. The **anterior** part of the rib is **more oblique** and the medial part cannot be demonstrable as it ends with costal cartilage. The **posterior part** of the rib is **more horizontal**, hyperdense and the attachment of the rib with the vertebra can be easily made out.

Normal Chest X-ray Rib Identification

The anterior(yellow) and posterior(red) ribs are highlighted in the image given below.





@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 6th edition, pg :186

Q84. Radiology INI-CET 2024 ID: MG5844

Which of the following radiological interventions are used in the management of cerebral aneurysm?

A ✓ Coiling

B Sclerotherapy

C Stenting

D Clipping

Correct Answer: Coiling

Coiling is the **radiological intervention** used in the management of **cerebral aneurysm**.

Aneurysms occur at the **bifurcations** of the large to medium-sized intracranial arteries. Rupture is into the subarachnoid space in the basal cisterns. Approximately 85% of the aneurysms occur in the anterior circulation, mainly on the circle of Willis.

Most unruptured intracranial aneurysms are completely asymptomatic. When the aneurysm ruptures, 45% of cases present with **sudden onset severe headache**, described as the **worst headache** of the patient's life. Other features include neck rigidity and transient loss of consciousness.

Diagnosis -

CT shows **hyperattenuating** material filling the **subarachnoid space**. CT angiography has a high sensitivity for aneurysms and AVMs, but **digital subtraction angiography** remains the **gold standard**.

Treatment -

Early aneurysm repair prevents re-rupture. At many centers, definitive repair is carried out within 24 h of the bleed in all patients who are stable enough to tolerate the procedure. An aneurysm can be "**clipped**" by a **neurosurgeon** or "**coiled**" by an **endovascular approach.** **Surgical repair** involves placing a **metal clip** across the aneurysm neck, thereby immediately eliminating the risk of rebleeding. This approach requires **craniotomy** (option D) and brain retraction, which is associated with neurologic morbidity.

Endovascular techniques involve placing **platinum coils** within the aneurysm via a catheter that is passed from the femoral artery. The aneurysm is packed tightly to enhance thrombosis and over time is walled off from the circulation.

Intracranial pressure should be measured and the cerebral perfusion pressure targeted at **60–70 mmHg**.

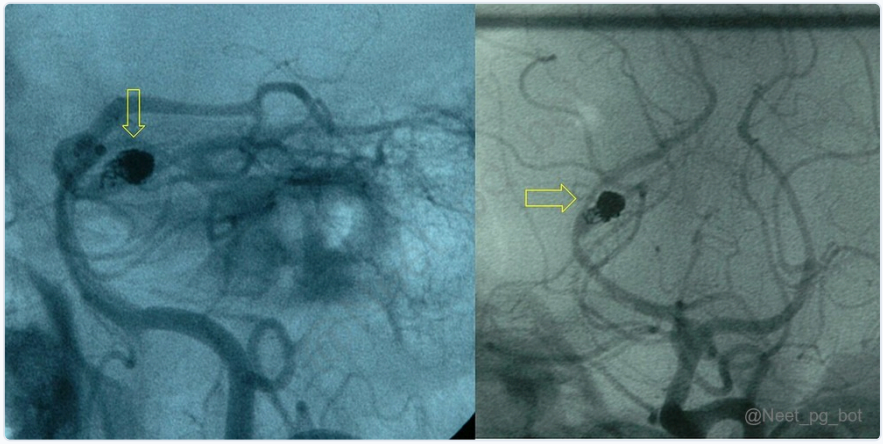
Delayed cerebral ischemia remains the leading cause of morbidity and mortality following aneurysmal SAH. Treatment with the calcium channel antagonist **nimodipine** (60 mg PO every 4 h) improves outcome, perhaps by preventing ischemic injury rather than reducing the risk of vasospasm.

The image below is an arteriogram showing a coiled aneurysm (indicated by yellow arrows) of the posterior cerebral artery with a residual aneurysmal sac.

Other options :

Option B - Sclerotherapy is used in the management of venous insufficiency.

Option D - Stenting is employed mainly for stenotic vascular lesions.

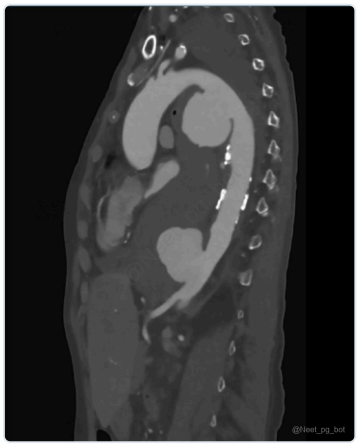


@Neet_pg_bot

References: Harrison's Principles of Internal Medicine, 21st Edition, Page no : 3355

Q85. Radiology NEET 2024 ID: MG6846

A patient presented with chest pain and hoarseness for one month, and on a CT angiogram, the following image is seen, what is the possible diagnosis?



A Stenosis of aortic arch

B Aortic dissection

C ✓ Saccular aneurysm of distal aorta

D Coarcatation of aorta

Correct Answer: Saccular aneurysm of distal aorta

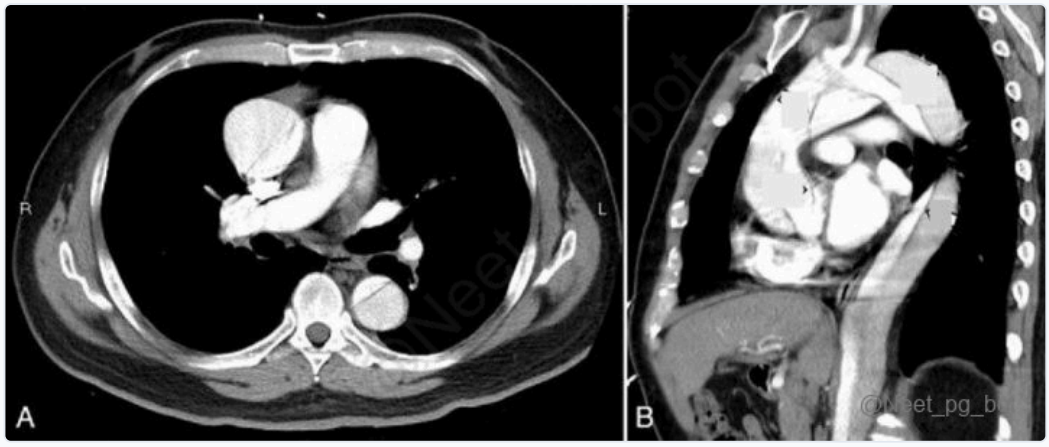
The given CT shows **saccular aneurysm of the distal aorta**. The clinical presentation of **chest pain** and **hoarseness** supports the diagnosis.

Other options:

Option A: **Stenosis of the aortic arch** would show a **narrowing** rather than an aneurysmal dilation, and it is more likely to present with symptoms related to decreased blood flow to the upper body.

Option B: **Aortic dissection** would usually present with a **tearing chest pain** that radiates to the back and would show a **double lumen or false lumen** on imaging. The below image shows **Stanford type A** aortic dissection:

Option D: **Coarctation of the aorta** typically presents with **hypertension** in the **upper limbs** and **weak** or **absent pulses** in the lower limbs.

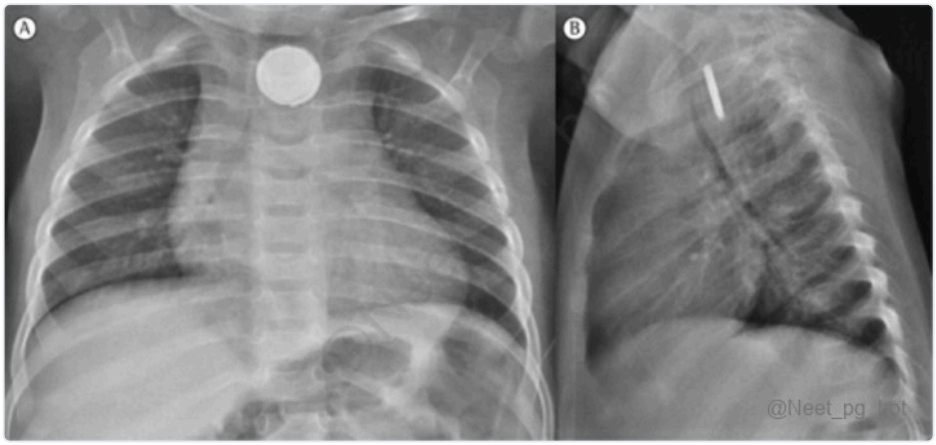


@Neet_pg_bot

References: Schwartz's Principles Of Surgery - 10th Edition, Page no 672

Q86. Radiology NEET 2024 ID: MG6556

This is the radiograph of a child with a foreign body in the oesophagus. What is the sign seen in this?



A Thumb sign

B Steeple sign

C ✓ Double ring sign

D String sign

Correct Answer: Double ring sign

The presence of a **halo sign** or **double ring sign** is indicative of the foreign body being a **button battery**.

Button batteries and sharp objects in the oesophagus and foreign bodies associated with **respiratory symptoms** all mandate urgent endoscopic removal.

Button batteries, in particular, are very dangerous foreign bodies, which must be expediently removed because they can cause:

- **Mucosal injury** in 1-hour contact time
- Involve **all layers** of the **oesophagus** in 4 hours

Injury is thought to be secondary to electrochemical burns from electrical discharge. **Chemical burns** from electrolyte alkali leakage and pressure necrosis also may contribute to mucosal injury.

Asymptomatic blunt objects and coins lodged in the oesophagus can be observed for up to 24 hr in anticipation of passage into the stomach.

Another AP and lateral X-ray demonstrating the presence of a button battery in the oesophagus is as shown below.

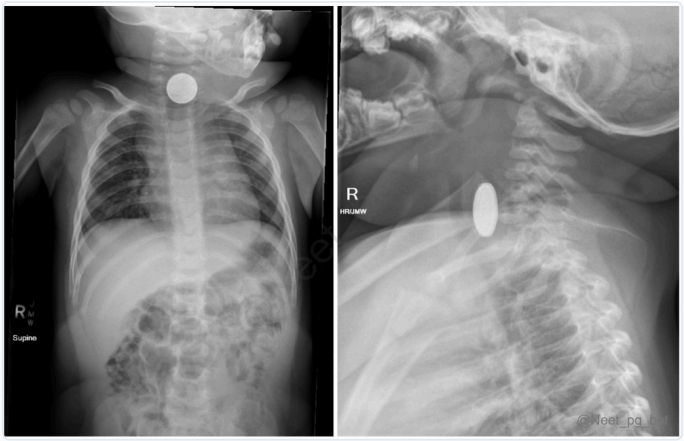
Other options

Option A: **Thumb sign** is seen in **acute epiglottitis**.

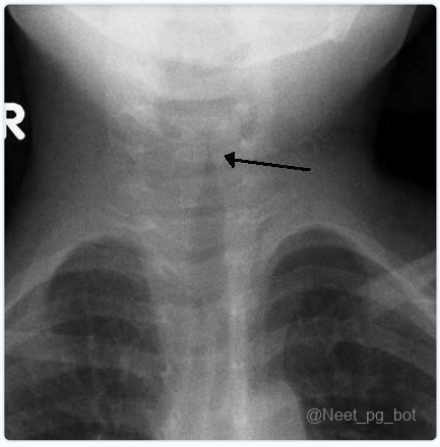
Option B: A **steeple sign** is seen in **croup** or **laryngotracheobronchitis**.

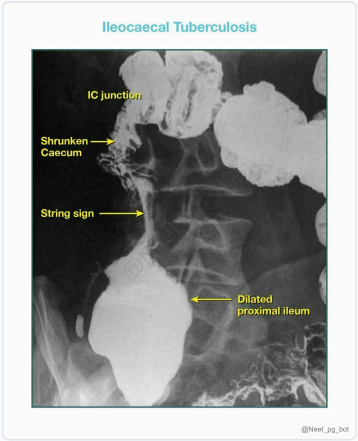
Option D: **String sign** on the barium enema is in **ileocecal tuberculosis**.

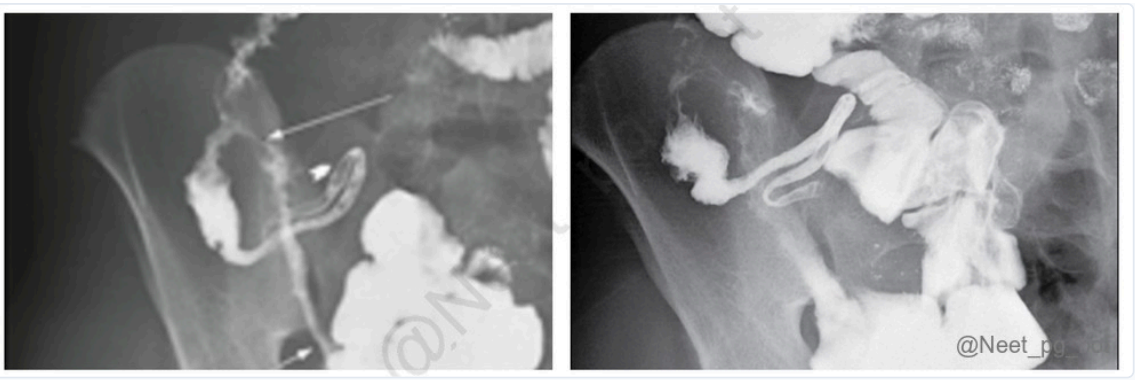
The image below shows the **string sign of Kantor** associated with **Crohn's disease**.


R
Supine
R
HRUMW

R


R
@Neet_pg_bot


Ileocaecal Tuberculosis
IC junction
Shrunken Caecum
String sign
Dilated proximal ileum
@Neet_pg_bot

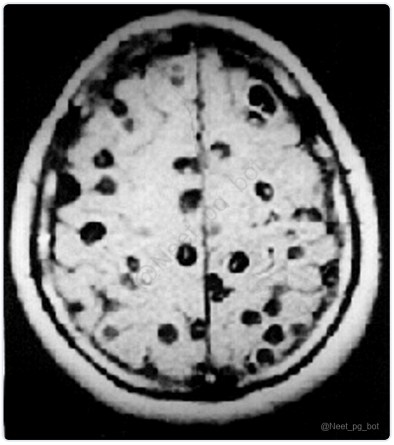


@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 7th edition, Page No. 1790
Nelson textbook of pediatrics 21st edition, Page No.2203
Online Resource, https://radiopaedia.org/cases/button-battery-oesophageal-impaction-1

Q87. Radiology NEET 2024 ID: MG6843

A patient presents with a one-month history of lethargy, headache, fever, and vomiting. Shortly after being admitted to the hospital, the patient tragically passes away. Radiological imaging reveals multiple cystic lesions in the cerebrum with invasion into the ventricles and subarachnoid space. What is the likely cause?



A ✓ Neurocysticercosis

B Astrocytoma

C AV malformation

D Brain metastasis

Correct Answer: Neurocysticercosis

The given clinical scenario along with CT showing **cystic lesions** in the brain, particularly within the cerebrum, ventricles, and subarachnoid space is clearly suggestive of **neurocysticercosis**.

Neurocysticercosis is a form of cysticercosis caused by **Taenia solium.**

It usually presents with **seizures** and symptoms of **increased intracranial pressure,** including headache, nausea, vomiting, vision changes, dizziness, ataxia, or confusion. **Papilledema** may develop secondary to hydrocephalus.

Diagnosis is done by demonstrating **eggs** of Taenia in **faecal examination**.

A biopsy of the brain or spinal cord lesion or fundoscopic visualization of subretinal cysticercus can also be done.

Imaging shows a **scolex** within a **cystic lesion**, cystic lesions with or without enhancement (e.g., ring enhancement), nodular calcifications, focal enhancing lesions, or multilobulated cystic lesions in the subarachnoid space.

Treatment of intestinal T. solium infection is by a **single dose** of **praziquantel**. Neurocysticercosis is treated with **anti-epileptic** drugs, **anticysticidal** drugs like **albendazole** or **praziquantel**, and steroids given before anticysticidal therapy.

Other options:

Option B: **Astrocytoma** is a type of **glial cell tumour** in the brain, ranging from low to high grade. They do not typically present as multiple cystic lesions and are more likely to appear as a **single mass with variable enhancement** on imaging.

Option C: **Arteriovenous malformation (AVM)** is a congenital vascular anomaly characterized by a tangle of abnormal blood vessels connecting arteries and veins. AVMs are more likely to cause hemorrhage or seizures and typically appear as **highly vascular lesions on imaging,** not as cystic lesions.

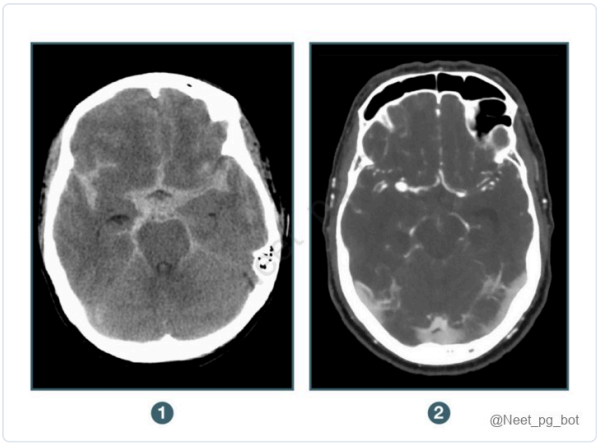
Option D: While **metastases** can present as multiple lesions on imaging, they usually **appear solid** rather than cystic masses and often enhance with contrast.

@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 1510-1511
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1176337/
Online Resource, https://radiopaedia.org/articles/cyst-with-dot-sign-neurocysticercosis

Q88. Radiology NEET 2024 ID: MG6840

A patient complains of severe headache. NCCT and CT angiography images are as below. What is the likely etiology?



A AV malformation

B ✓ Subarachnoid hemorrhage

C Intraparenchymal hemorrhage

D Subdural hemorrhage

Correct Answer: Subarachnoid hemorrhage

The clinical manifestations of severe headache with CT revealing **blood** in the **basal cistern** point to a diagnosis of **subarachnoid haemorrhage** (SAH).

Subarachnoid haemorrhage (SAH) is bleeding into the subarachnoid space, which is the area between the arachnoid membrane and the pia mater surrounding the brain. The diagnosis is suspected when a **hyperattenuating material** is seen **filling the subarachnoid space**. Most commonly this is apparent **around the circle of Willis**, on account of the majority of berry aneurysms occurring in this region (65%), or in the Sylvian fissure (30%).

The **modified Fisher scale** is a method for grading SAH as seen on non-contrast CT (NCCT). The **Hunt and Hess scale** describes the severity of SAH and is used as a predictor of survival. **Xanthochromia on CSF** is indicative of SAH in case of a negative CT scan.

Other options:

Option A: **Arteriovenous malformation** (AVM) is a congenital vascular anomaly characterized by a **tangle** of **abnormal blood vessels** connecting arteries and veins. AVMs are more likely to cause **haemorrhage** or **seizures** and typically appear as highly vascular lesions on imaging, not as cystic lesions.

Option C: **Intraparenchymal hemorrhage** results from damage to **lenticulostriate arteries**. **Hyperdensity** within the brain parenchyma is seen on CT.

Option D: Acute **subdural haemorrhage** results from the accumulation of blood between the **arachnoid membrane** and the **dura**. The tearing of a **bridging vein** from the cerebral cortex to the dural sinuses is the most common cause of acute SDH. On head CT scan, acute SDH appears as a hyperdense, crescent-shaped (concavo-convex).

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References: Harrison's Principles of Internal Medicine, 20th Edition, Page no: 2084
Harrison's Principles of Internal Medicine, 21st Edition, Page no: 3353
Grainger & Allison's Diagnostic Radiology 7th edition, Page no 1510-1511

Q89. Radiology NEET 2024 ID: MG6845

An elderly woman with a known history of Myasthenia Gravis presents with symptoms including fatigue, facial congestion, dilated neck veins, and breathlessness, which are indicative of superior vena cava (SVC) obstruction. What is the likely diagnosis?

A Lipoma

B Ectopic thyroid

C ✓ Thymoma

D Germ cell tumor

Correct Answer: Thymoma

The diagnosis of **myasthenia gravis** along with the CT scan image showing an **anterior mediastinal mass,** is suggestive of **thymoma**.

Thymoma is the **most common mediastinal tumor** and is derived from the thymus gland. Thymomas vary in behaviour from benign to aggressively invasive. **Superior vena cava obstruction** is a possible complication of mediastinal mass.

Masoaka classification system is used to stage thymomas.

Myasthenia gravis is often related to thymomas. Symptoms of myasthenia gravis can improve after excision of the thymus gland even in the absence of a thymoma.

Complete thymectomy is the treatment for thymoma. It can be performed by two approaches:

- A **median sternotomy** approach
- A **transcervical** approach with or without an additional **VATS procedure** (Video-Assisted Thoracoscopic Surgery) for smaller thymomas

Other options:

Option A: **Lipomas** are **benign** and **slow-growing**, less likely to cause significant compression of the SVC and produce the acute symptoms described.

Option B: **Ectopic thyroid tissue** can be found in the mediastinum, but it is typically **asymptomatic** and does not usually cause SVC obstruction.

Option D: **Germ cell tumours** can occur in the mediastinum and cause SVC obstruction, but they are **less commonly associated with Myasthenia Gravis** compared to thymomas.

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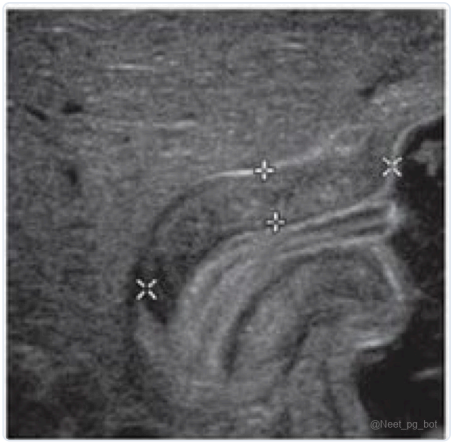
References: Schwartz's Principles Of Surgery - 10th Edition, Page no 672

Q90. Radiology NEET 2024 ID: MG6557

A 4-week-old infant presented to the ER with projectile vomiting. USG abdomen findings of the infant are given below. Probable diagnosis is _____

USG-Ultrasonography



A Sigmoid volvulus

B Duodenal atresia

C ✓ Congenital hypertrophic pyloric stenosis

D Intussusception

Correct Answer: Congenital hypertrophic pyloric stenosis

The probable diagnosis is **Congenital Hypertrophic Pyloric stenosis** since the **cervix sign** and **nipple sign** are signs of Congenital Hypertrophic Pyloric stenosis on USG.

- **Double-layered hypertrophic mucosa protrudes** into the **stomach**, creating the so-called **Nipple sign.**
- The sonographic appearance of the **hypertrophied pylorus** resembles that of the **uterine cervix** and is sometimes referred to as the **cervix sign.**

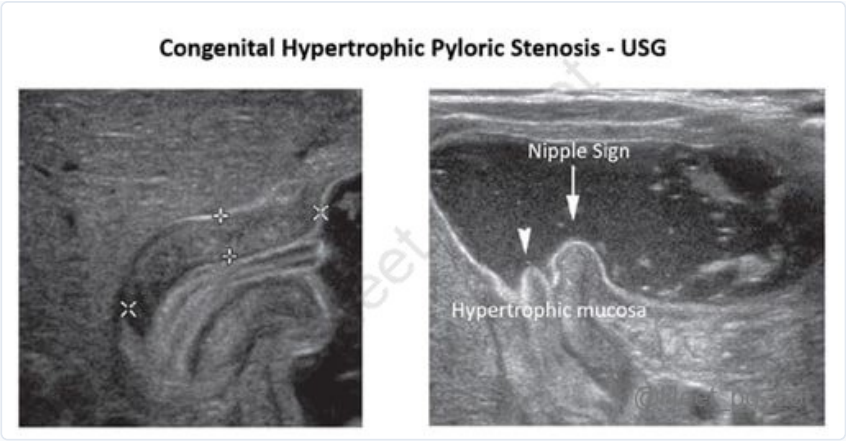
Hypertrophic pyloric stenosis most presents with **non-bilious projectile vomiting**. It is common between **2** and **8 weeks** of age. Hypertrophy of the **circular muscle** of the pylorus results in **constriction** and **obstruction** of the gastric outlet. **Male infants** are at a higher risk.

Clinically, an **olive-shaped mass** is palpable in the **right upper quadrant** or **epigastrium**. When the infant is given a test feed, **visible peristalsis** is usually observed moving from **left** to **right** across the abdomen.

Infants with hypertrophic pyloric stenosis develop a **characteristic metabolic abnormality**- **hypochloraemic hypokalaemic metabolic alkalosis**. Persistent vomiting results in loss of hydrochloric acid leading to the development of metabolic alkalosis (due to loss of H+) and hypochloremia (due to loss of Cl-).

Ultrasound examination is the **investigation of choice** for this condition. The diagnostic criteria for pyloric stenosis are a **muscle thickness** of **≥4 mm** and a **pyloric length** of **≥16 mm**. The cervix sign and nipple sign are signs seen on USG.

As for the management, the child should be **rehydrated** and the metabolic abnormality has to be corrected before proceeding to surgery. The **Ramstedt pyloromyotomy** remains the **standard surgical procedure** for its treatment.



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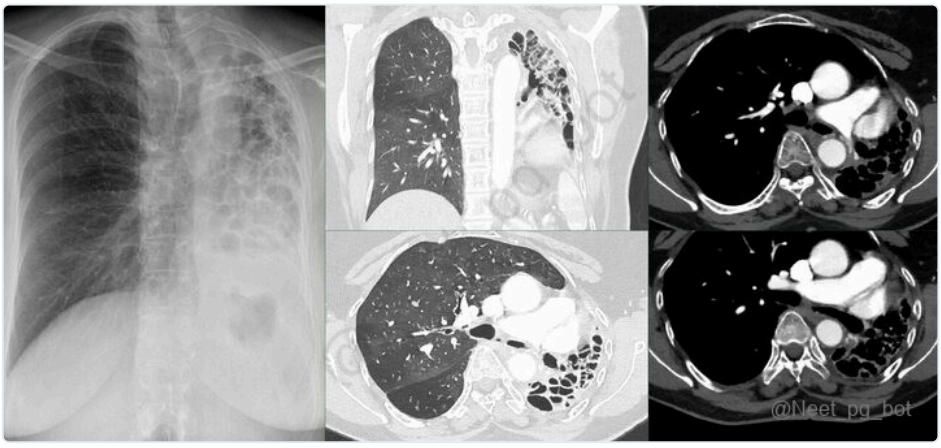
References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no: 1829
Bailey & Love's Short Practice of Surgery, 27th Edition, Page no: 128

Q91. Radiology NEET 2024 ID: MG6841

An infant presents with respiratory distress. He has a previous history of recurrent respiratory infections. What is the diagnosis?



A Lung abscess

B ✓ Congenital malformations of the airway

C Congenital diaphragmatic hernia

D Pyopneumothorax

Correct Answer: Congenital malformations of the airway

The given clinical scenario along with the image is suggestive of **congenital malformations of the airway**.

Congenital pulmonary airway malformation (CPAM) is a condition where more **normal lung tissue** is mixed with **hamartomatous** or **dysplastic lung tissue**, usually confined to one lobe. The lesion most likely originates from a maldevelopment of **terminal bronchiolar structures** caused by an embryologic injury that occurred prior to the **35th day** of gestation.

Pneumothorax, **recurrent respiratory infections**, and **respiratory distress** can all manifest in patients during the **newborn** or early infancy period. One could **mistake** the lesion for a **diaphragmatic hernia**. Until mid-childhood, when episodes of recurrent or persistent pulmonary infection or chest pain occur, patients with **smaller lesions** typically are asymptomatic.

Treatment includes **aspiration** of macrocystic lesions, **excision** of the **affected lobe** for microcystic lesions, and infrequently, **open foetal surgery** for severely affected infants. For patients with symptoms in the postnatal period, surgery is recommended.

Other options:

Option A: **Lung abscess** represents **necrosis** and **cavitation** of the lung following **microbial infection**.

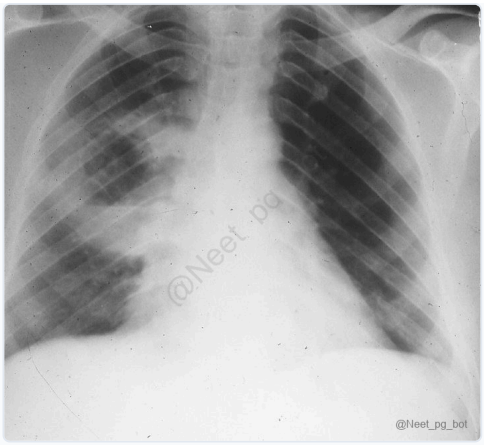
Chest X-ray features of lung abscess include a **lung cavity** with an **air-fluid level** and a thick and irregular cavity wall with surrounding pulmonary infiltrate. The image showing the lung abscess is given below:

Option C: **Congenital diaphragmatic hernia** (CDH) is a **birth defect** of the diaphragm. Malformation of the diaphragm allows the abdominal organs to push into the chest cavity, hindering proper lung formation and resulting in pulmonary hypoplasia.

The most common type of CDH is a **Bochdalek hernia,** which in most cases involves the left side. CDH is a life-threatening pathology in infants and a major cause of morbidity and mortality due to two complications: **pulmonary hypoplasia** and **pulmonary hypertension**. The image of Bochdalek hernia is given below:

Option D: **Pyopneumothorax** is a **pleural collection** of **pus** and **gas**, also referred to as **empyemic hydropneumothorax** or **infected hydropneumothorax**. It could be compared to a thoracic empyema variation that contains gas. Chest pain and fever are typically the patient's initial complaints.

X-ray shows **gas-fluid level**. It could have **loculations**. There appears to be a **loss of silhouette** around the **diaphragm** dome. The thick pleural lining is strongly indicative of the pyopneumothorax.



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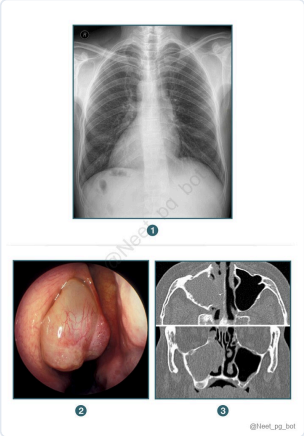
References: Nelson textbook of pediatrics 21st edition, Page no 2230-2231,911,944,945
Online Resource,
https://radiopaedia.org/articles/pyopneumothorax#:~:text=Pyopneumothorax%20(also%20known%20as%20infected,the%20etiology%20may%20be%20different.

Q92. Radiology NEET 2024 ID: MG6842

A young patient presents with nasal polyps and rhinosinusitis. X-ray and clinical images are as below. What is the diagnosis?



A Young's syndrome

B ✓ Kartagener's syndrome

C Churg Strauss syndrome

D Cystic fibrosis

Correct Answer: Kartagener's syndrome

The given clinical scenario along with an **X-ray** showing **dextrocardia**, **CT** showing **maxillary polyp,** and **endoscopy** showing **nasal polyp** is suggestive of **Kartagener's syndrome**.

Primary ciliary dyskinesia (also called immotile cilia syndrome) is an **autosomal recessive** disorder affecting **ciliary motility**. When a patient has bronchiectasis, chronic sinusitis, and situs inversus triad, he is known to have **Kartagener syndrome**. It occurs due to the absence of **protein dynein** in the **A tubules** of the respiratory mucosa.

The disease is usually diagnosed based on the clinical picture and should be evaluated in later stages when a child presents with **recurrent lower** and **upper respiratory tract infections**. Other features of primary ciliary dyskinesia include congenital heart diseases and neonatal hydrocephalus. Male patients with this condition tend to be **infertile**, due to **asthenospermia** (immotile spermatozoa due to impaired sperm flagella function).

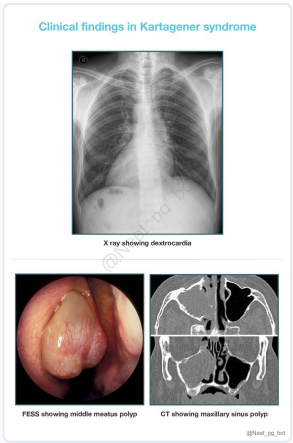
Transmission electron microscopy is the **gold standard investigation** to analyze cilia. There is no specific treatment for this condition and is managed according to the symptoms.

Other options:

Option A: **Churg Strauss syndrome** clinically presents with **asthma, fever** and **eosinophilia.**

Option C: **Young's syndrome** clinically presents with i**nfertility** (azoospermia) and **bronchiectasis.**

Option D: **Cystic fibrosis** is an **autosomal recessive** condition due to a **mutation** of the **CFTR gene** on **chromosome 7**. Clinical features include **recurrent sinopulmonary infection**, **intestinal obstruction**, **pancreatic insufficiency**, and **male infertility**. Diagnosis is made by **elevated sweat chloride** levels, abnormal nasal potential difference, and genetic testing for **CFTR mutation**.



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References: P L Dhingra Diseases of Ear Nose and Throat and Head and neck surgery - 8th Edition, Page no 201,524,
Nelson textbook of pediatrics 21st edition, Page 2299, 2230

Q93. Radiology NEET 2024 ID: MG6844

Given below is a USG taken at approximately 8 weeks of gestation. Which anomaly can be observed here?

USG - ultrasound

A ✓ Anencephaly

B Hydranencephaly

C Meningocele

D Encephalocele

Correct Answer: Anencephaly

The given ultrasound image shows an open defect in the calvaria and skin. This is consistent with the diagnosis of **anencephaly.**

Anencephaly results from **incomplete closure** of the **neural tube.** It is one of the earliest detectable congenital defects that can be **detected** as early as **10 to 11 weeks** of gestation. Screening can also be done by serum estimation of maternal serum alpha-fetoprotein **(MSAFP).** The following are the radiological signs seen in anencephaly.

- **Frog eye sign** - Absence of cranial calvaria, prominent orbits thereby resembling the face of a frog.
- **Shower cap sign -** The absence of the cranium could lead to the protrusion of a disorganized mass of brain tissue. This resembles a shower cap giving rise to a characteristic triangular facial appearance.

Neural-tube defects can be prevented with **pre-conceptional** supplementation of **folic acid.** A dose of 4 mg daily is recommended in subsequent pregnancies.

The frog eye and showercap signs are shown below:

Other options:

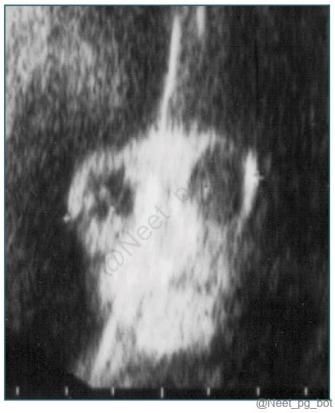
Option B: **Hydranencephaly:** This condition involves the **absence** of cerebral hemispheres, replaced by fluid-filled sacs.

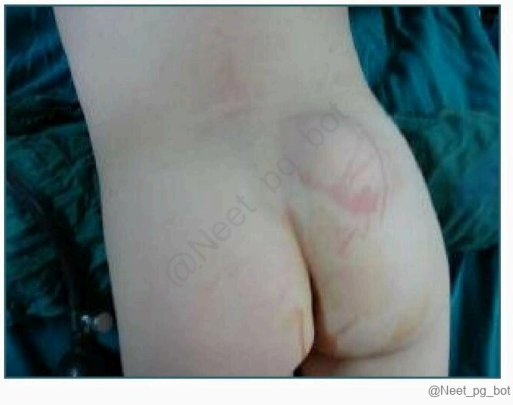
Option C: **Meningocele:** It involves herniation of the meninges through a defect in the skull or spinal column, forming a sac filled with cerebrospinal fluid.

The image below shows meningocele:

Option D: Encephalocele is caused due to **defects in the skull** through which the **meninges** and/or **neural tissue herniate**. The squamous part of the occipital bone is the most frequently affected bone.

The image shows a newborn with **encephalocele**.





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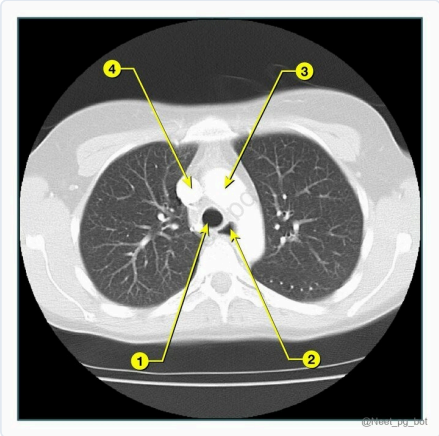
References: Williams Obstetrics 26th Edition, Page no: 186-187, 192-193, 197-198, 205-206, 229, 240, 270, 282, 288, 411, 540, 1216
DC Dutta's Textbook of Obstetrics 10th Edition, Page no: 383,602
Grainger & Allison's Diagnostic Radiology 7th edition, Page no: 231, 1805, 1986
Online Resource, https://pubmed.ncbi.nlm.nih.gov/24627678/
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7447666/

Q94. Radiology NEET 2024 ID: MG7208

Match the following using the radiological image given below:

1	A. Arch of Aorta
2	B. Trachea
3	C. Esophagus
4	D. Superior vena cava



A 3-A, 2-B, 1-C, 4-D

B 2-A, 3-B, 4-C, 1-D

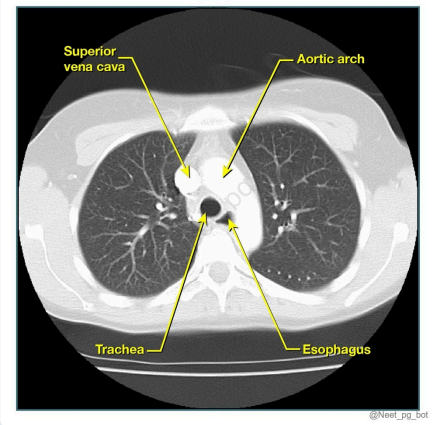
C ✓ 1-B, 2-C, 3-A, 4-D

D 2-A, 1-B, 3-C, 4-D

Correct Answer: 1-B, 2-C, 3-A, 4-D

Here are the correctly matched structures:

1	B. Trachea
2	C. Esophagus
3	A. Arch of Aorta
4	D. Superior vena cava

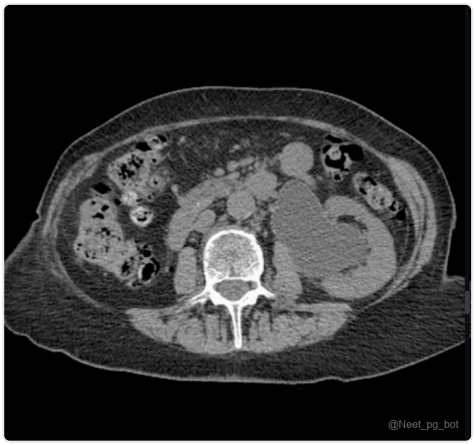


@Neet_pg_bot

References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 404

Q95. Radiology NEET 2024 ID: MG7207

Identify the diagnosis from the given CT scan.



A ✓ PUJ obstruction

B Renal cyst

C Renal tumour

D Multicystic kidney

Correct Answer: PUJ obstruction

The CT scan showing **ballooning** of the **left renal pelvis** with moderate **hydronephrosis** of the affected kidney indicating a likely diagnosis of **ureteropelvic junction obstruction (UPJ)**of the **left side.**

PUJ obstruction refers to a blockage or narrowing at the junction where the renal pelvis transitions into the ureter, leading to impaired drainage of urine from the kidney.

Indications for surgery in patients with UPJ obstruction include:

- On the diuretic renogram, UPJO with less than 40% in the split function of the afflicted kidney.
- Renal parenchymal atrophy in the presence of severe bilateral UPJO.
- **Despite preventive medicines, recurring urinary tract infections.**
- Obstructive UPJO that is **symptomatic** or **linked with an abdominal mass**

Hydronephrosis may be detected on antenatal ultrasound; it can, however, be transient or mild and resolves on its own. Re-evaluation should be done after birth and in cases presenting with **symptoms of recurrent UTI, palpable flank mass or pain, urinary obstruction, and in cases of failure to thrive**.

If the hydronephrosis is persistent, its **severity** is **graded** by the extent of the dilatation of the pelvicalyceal system and parenchymal thickness. It needs to be ascertained whether the other kidney is involved, the lower ureter is dilated, the bladder wall is thickened, and whether emptying of the bladder is incomplete. Based on these findings, a diagnosis can be made.

In cases of hydronephrosis, **long-standing urine stasis** and pressure may **damage the kidney**s.

The **causes** of UPJ obstruction are

- **Intrinsic** (due to an **adynamic** or stenotic segment of the proximal ureter) or
- **Extrinsic** (e.g. **compression by accessory renal arteries**). This needs to be corrected primarily.

The renal anatomy and function can be assessed using **radioscintigraphy** studies. The **investigation of choice** for UPJ obstruction is ^{99m}Tc-MercaptoAcetyl-triGlycine **(MAG3) scan. Micturating cystourethrogram** is the investigation of choice in **vesicoureteric reflux.** ^{99m}Tc-dimercaptosuccinic acid **(DMSA) scan** is used for assessing **renal scarring**.

Treatment of intrinsic UPJ obstruction is done **surgically if renal function is compromised** or if there is bilateral involvement of the kidneys. In other cases, **serial follow-up with ultrasonography** is done to assess the spontaneous resolution of hydronephrosis.

Surgical options include **pyeloplasty** which entails the surgical division of the obstruction and the creation of a funnel-shaped anastomosis. Additionally, a temporary stent and a drain may be inserted. It is necessary to do further serial ultrasounds and **MAG-3 imaging** in order to ensure proper monitoring and evaluation. This procedure is now commonly being performed via **laparoscopy** and also with **robotic assistance. Congenital PUJ obstruction** is treated surgically by **Anderson Hyen's pyeloplasty**.

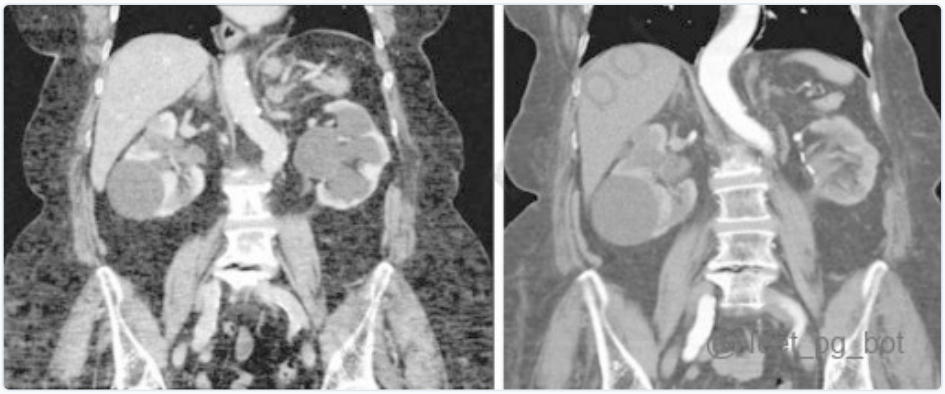
Below are CT scans of a left PUJ obstruction **before**(**left**) and **after**(**right**) surgery.

Other options:

Option B: A **renal cyst** would appear as a fluid-filled, well-demarcated structure within the kidney but does not involve the PUJ or cause obstruction.

Option C: A **renal tumor** would present as a solid, abnormal mass within the kidney, not typically involving the PUJ unless there is direct extension.

Option D: **Multicystic kidney** refers to the presence of multiple cysts within the kidney, often congenital, but is not specific to the PUJ and would not result in obstruction at the PUJ.

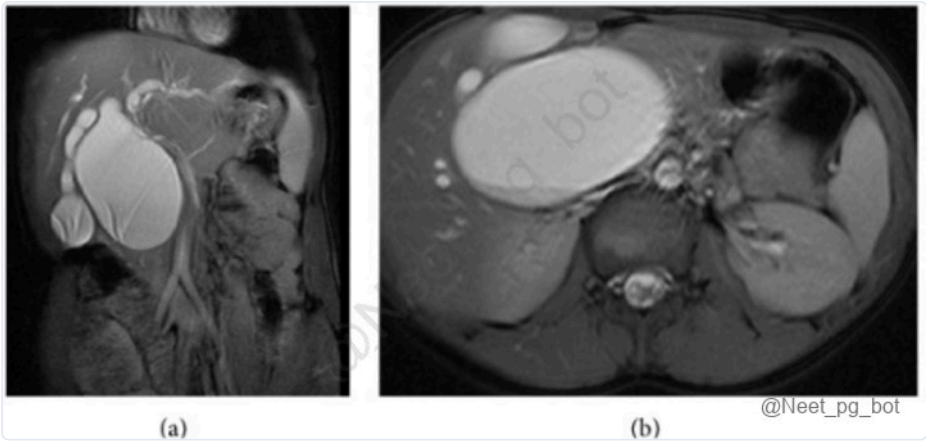


@Neet_pg_bot

References: Bailey & Love's Short Practice of Surgery - 28th Edition, Page no 281
Online Resource, https://www.ncbi.nlm.nih.gov/books/NBK560740/

Q96. Radiology NEET 2024 ID: MG7192

What does this image below indicate?



A ✓ Choledochal cyst

B Gall bladder carcinoma

C Cholangiocarcinoma

D Choledocholithiasis

Correct Answer: Choledochal cyst

The given MRCP image showing **saccular dilatation** of the common bile duct suggests a diagnosis of type I **choledochal cyst**.

Todani classification:

- Type I- saccular dilatation of the whole (Ia) or a segment (Ib) of the common bile duct (CBD)
- Type II- diverticulum from the CBD
- Type III- choledochocele that opens into the duodenum
- Type IV- multiple cysts (both intrahepatic and extrahepatic)
- Type V- multiple cysts that are only intrahepatic (Caroli's disease)

The **diagnosis** is made initially with a **right upper quadrant ultrasound**, following which a confirmatory MRCP can be carried out to help in pre-operative planning.

Treatment is usually surgical with a **Roux-en-Y hepaticojejunostomy** with radical resection of all the biliary tissue. There is a **risk** of **malignant transformation** in **long-standing disease,** so surgical removal is usually recommended.

Other options:

Option B: **Gall bladder carcinoma** would show abnormal growths or masses in the gall bladder rather than cystic dilations of the bile ducts.

Option C: **Cholangiocarcinoma** is a malignancy of the bile ducts and would present as irregular masses or bile duct strictures rather than cystic structures.

Option D: **Choledocholithiasis** refers to the presence of stones in the bile duct, which would appear as filling defects within the bile ducts rather than cystic dilations.

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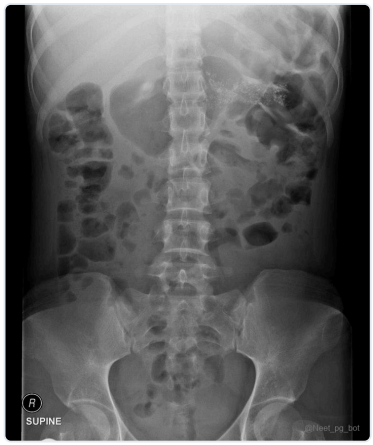
References: Nelson textbook of pediatrics 21st edition, Pg 2137
Grainger & Allison's Diagnostic Radiology 7th edition, Pg 1839
Bailey & Love's Short Practice of Surgery, 27th Edition, Pg 1197

Q97. Radiology NEET 2024 ID: MG7189

Identify the condition shown in the image:



A ✓ Chronic calicific pancreatitis

B Mesenteric calcification

C Horseshoe kidney

D Jejunal fecolith

Correct Answer: Chronic calicific pancreatitis

As the given image shows, there are **multiple calcifications** within the region of the pancreas, which is characteristic of **chronic calcific pancreatitis.**

This condition results from long-standing inflammation of the pancreas, leading to the **deposition of calcium.** Chronic pancreatitis is often associated with **alcohol abuse, genetic factors, or recurrent acute pancreatitis**. The calcifications are a hallmark feature and help differentiate it from other abdominal pathologies.

It presents with symptoms like **chronic abdominal pain** with a history of **alcoholism** and **bulky stools** which indicates a lack of **pancreatic lipases** seen in chronic disease leading to impaired fat digestion and absorption.

Chronic pancreatitis is a progressive inflammatory disease in which there is an **irreversible destruction** of pancreatic tissue. It is **more common in males**, with a male: female ratio of **4:1**.

The etiology of chronic pancreatitis and pancreatic exocrine insufficiency can be classified based on the TIGAR-O classification system:

- **Toxic-metabolic -alcohol,** smoking, hypercalcemia, hyperlipidemia
- Idiopathic
- **Genetic- PRSS1,** CFTR, and SPINK1 genes
- **Autoimmune-** autoimmune chronic pancreatitis, IgG4-related disease
- **Recurrent** and **severe acute pancreatitis**
- **Obstructive-** pancreas divisum, duct obstruction, malignancy

The diagnosis can be made using **Rosemont criteria.** It is based on endoscopic ultrasonographic (EUS) findings. It is explained in the pearl attached below.

ERCP is the **most sensitive** radiologic test for the diagnosis of chronic pancreatitis.

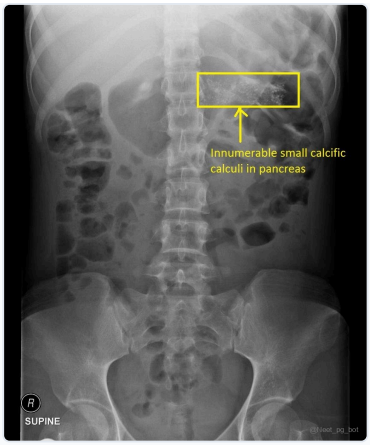
The below image shows the X-ray findings:

Other options:

Option B: **Mesenteric calcification** would present in a different anatomical region and is associated with conditions like ischemia.

Option C: **Horseshoe kidney** involves the kidneys and does not cause pancreatic calcifications.

Option D: **Jejunal fecolith** refers to hardened stool within the small intestine.



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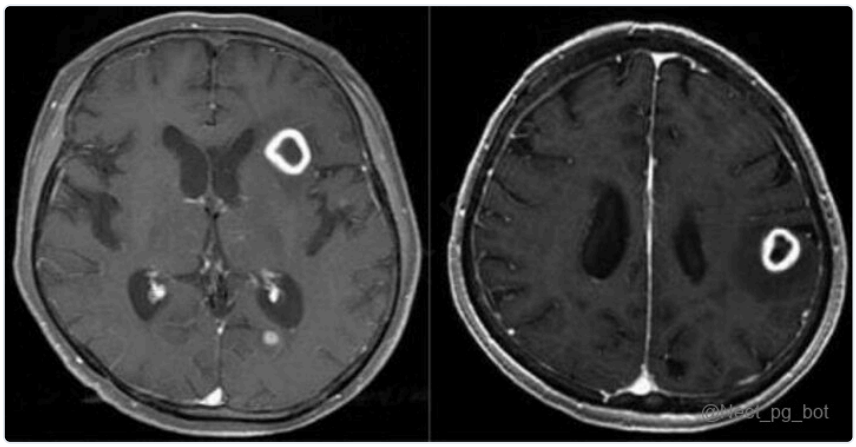
References: Harrison's Principles of Internal Medicine, 21st Edition, page 2666, 2667

Q98. Radiology NEET 2024 ID: MG7044

A 6-year-old child presented with seizures which subsided with injection Midazolam. Blood investigations and lumbar puncture studies were normal. MRI is done and the image is given below. What is the next best investigation of choice for further assessment?



A ✓ MR spectroscopy

B Ocular B scan

C CBNAAT of CSF

D PET scan

Correct Answer: MR spectroscopy

The given clinical history of a child along with an **MRI** showing a **ring-enhancing intraaxial lesion** of the brain could likely be a **tuberculoma** or **neurocysticercosis** and **MR spectroscopy** would be the next best investigation for further assessment.

Magnetic Resonance (MR) spectroscopy is a non-invasive test based on the principle of magnetic resonance used to analyse the metabolite distribution within the brain tissue. It contains **N-acetyl aspartate** (NAA), a neuronal viability signal; **choline** (a component of cell membranes and a marker of cellular turnover); **creatine** (present in metabolically active tissues); **lipids** (released with cell membrane disruption); and **lactate** (a hallmark of anaerobic metabolism).

The image below shows a normal MR spectroscopy.

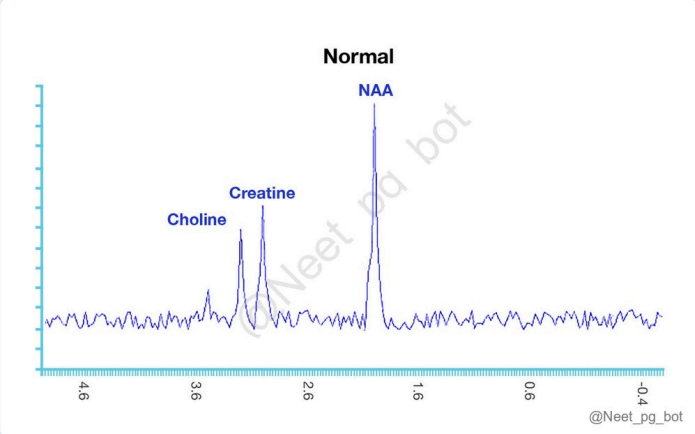
In **tuberculoma**, MR spectroscopy has **reduced NAA** owing to the destruction of the normal brain parenchyma and **increased** levels of **lactate** or **lipids**. In **neurocysticercosis**, there is **reduced NAA** and **creatine** and **elevated lactate**, **alanine**, **succinate**, and **choline** levels.

Other options:

Option B- **Ocular B- scan** is done to rule out diseases in the **posterior segment** in the presence of media opacities which is not significant for this case.

Option C- **CBNAAT** of CSF will not be the next investigation as the lumbar puncture analysis was normal.

Option D- **PET**-**CT** is done to assess the stage of cancer and is hence not required in this case.



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References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no: 1376, 1413

Q99. Radiology AIIMS 2018 ID: MC3482

A child presents with history of having ingested a button battery. What is your next course of action?

A Give oral fluids

B ✓ Remove endoscopically

C Reassure and to check whether passed in stools

D Induce vomiting

Correct Answer: Remove endoscopically

This case should be managed by **removing the button battery endoscopically** after confirming its position on X-Ray.

Button battery ingestion causes **alkaline caustic injury,** more in the areas surrounding the negative pole.

Management :

- The child is to be kept NPO until the oesophageal position of the battery is ruled out by X-ray scan. Vomiting should never be induced since the battery, which has passed through the esophagus, might lodge back again into it. Reassurance can be done only if the position of the battery is confirmed and if there are no symptoms.
- Since the question does not mention that the child is asymptomatic, the management of choice would be removing the battery endoscopically.
- Management also depends on battery life. Fresh batteries need to be removed as soon as possible as they have a higher concentration of injurious substances.

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References: Online Resource, https://www.ncbi.nlm.nih.gov/pubmed/29262219
Online Resource, https://www.poison.org/battery/guideline

Q100. Radiology AIIMS 2018 ID: MC6171

Which of the following statements are false regarding X Rays?

A Loss of soft tissue planes is the earliest X-ray change of infection

B Glass pieces can be seen in X-ray images.

C Joint space is a misnomer as there is no real space in the joint.

D ✓ Live bone is more radio-opaque than dead bone on X- Ray images.

Correct Answer: Live bone is more radio-opaque than dead bone on X- Ray images.

Live bone is **not more radio-opaque** than dead bone on X-ray images.

In fact, **dead bone looks more white (more radio-opaque)** than live bone on X-Ray as there's no bone resorption activity.

Loss of soft tissue planes is the earliest X-ray change in infections and is seen 24-48 hours after the onset of disease.

Joint space in X-rays is the radiolucency of cartilage being apparently perceived as space.

Glass pieces can be seen on X-ray images as they contain lead.

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References: Grainger & Allison's Diagnostic Radiology 6th edition, Pg No. 3-9

Q101. Radiology AIIMS 2018 ID: MC5356

A child presented with a history of road traffic accident. The axial CT image of the child, shown below, is suggestive of _____

- **A** ✓ Extradural Hemorrhage
- **B** Subdural Hemorrhage
- **C** Subarachnoid Hemorrhage
- **D** Intraventricular Hemorrhage

Correct Answer: Extradural Hemorrhage

The given CT image shows the typical **bi-convex hyperdense** lesion **not crossing** cranial sutures, characteristic of an **extradural hemorrhage.**

As the dura mater tends to adhere to the skull, the hematoma collecting between the skull and this dura is seen on CT sections as a biconvex dense area immediately beneath the skull vault, convex towards both the brain and the vault. They **tend not to cross the suture lines.**

The **temporoparietal** convexity is the **most common site**. The hematoma usually lies **beneath a fracture of the squamous part of the temporal bone**. The underlying brain is displaced but often appears intrinsically normal.

Monitoring an extradural hemorrhage and immediate intervention if necessary is an important step in management since the brain can herniate with increased bleeding that is causing raised ICP.

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References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 1420

Q102. Radiology AIIMS 2018 ID: MC5359

A 39-year-old male patient presented with complaints of fever and cough. His chest radiograph shows air bronchogram sign. The probable diagnosis is :

A Pleural effusion

B ✓ Consolidation

C Hilar lymphadenopathy

D Bronchiectasis

Correct Answer: Consolidation

The radiograph shows **air bronchogram sign** in a **lung with consolidation,** seen in pneumonia, which is supported by the clinical history of this patient.

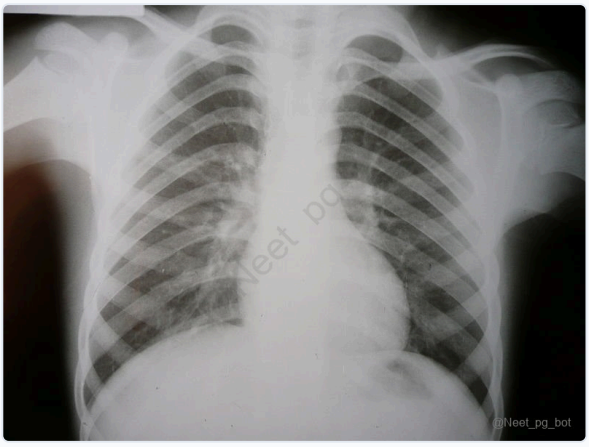
An **air bronchogram** is a tubular outline of an airway made visible when there is filling of the surrounding alveoli by fluid or inflammatory exudates.

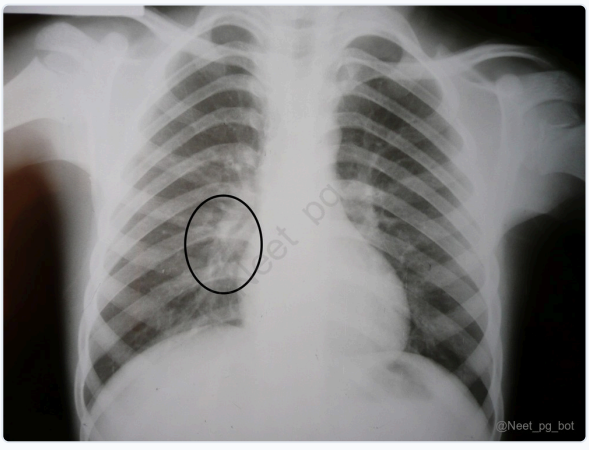
The causes of air bronchogram are as follows:

- Lung consolidation
- Pulmonary edema
- Nonobstructive pulmonary atelectasis
- Severe interstitial disease
- Neoplasm
- Normal expiration

Pleural effusion is seen as a homogenous opacity on chest radiograph.

The given radiograph shows peribronchial thickening and **air bronchogram sign:**





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References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 248

Q103. Radiology AIIMS 2018 ID: MC5357

The Doppler waveform of the venous blood flow in lower limb DVT is _____.

- **A** ✓ Monophasic
- **B** Biphasic
- **C** Triphasic
- **D** Multiphasic

Correct Answer: Monophasic

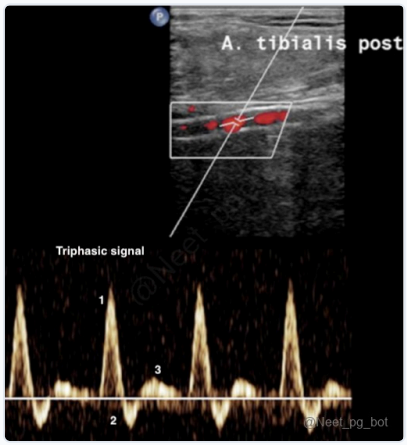
The **Doppler** waveform of the **venous blood flow** in lower limb DVT is **monophasic** proximal to the level of obstruction.

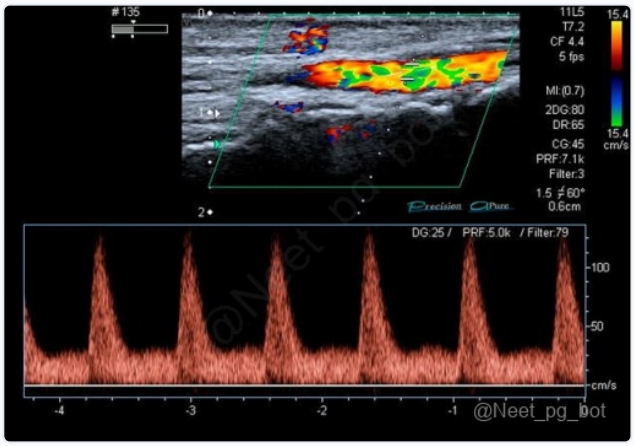
The **normal venous** waveform shows a variation (phasicity) secondary to respiration referred to as **respirophasic waveform**. Loss of this phasic variation results in a **monophasic waveform** suggestive of **venous obstruction** proximal to the level of interrogation.

A **normal artery** has a **triphasic** signal. The waveform becomes **biphasic** in mild-moderate stenosis and monophasic in severe stenosis.

The image given below shows the normal triphasic signal from an artery.

Monophasic waveform:





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References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 64
Online Resource, https://www.ahajournals.org/doi/pdf/10.1161/circulationaha.113.002966

Q104. Radiology AIIMS 2018 ID: MC5358

A middle-aged female patient with a history of cholelithiasis, presented with vomiting and abdominal pain. The abdominal radiograph showed Rigler's triad. Which of these is not a component of Rigler's triad?

A Pneumobilia

B Small bowel obstruction

C Ectopic gall stone

D ✓ Large bowel obstruction

Correct Answer: Large bowel obstruction

Rigler's triad is seen in gallstone ileus and consists of

- Pneumobilia
- Small bowel obstruction
- Ectopic gall stone (usually in the right iliac fossa)

This should **not be confused with Rigler's sign and Rigler's notch sign.**

Rigler's Sign:

Seen in pneumoperitoneum. Here, the air is seen on both sides of the bowel wall unlike normally, where the air is seen only in the intraluminal side of the bowel.

Rigler's notch sign:

Refers to indentation seen in the border of a solid lung mass, which is thought to represents a feeding vessel, thus suggesting the presence of bronchial carcinoma. But, this is not specific and can also be seen in granulomatous infections.

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References: Bailey & Love's Short Practice of Surgery - 27th Edition, Page no 1282

Q105. Radiology AIIMS 2018 ID: MA8281

All among the following are ionizing radiations except:

A Alpha rays

B Beta rays

C X-rays

D ✓ MRI

Correct Answer: MRI

MRI is a technique that produces topographic images by means of **magnetic fields** and **radiofrequency waves.**

The ionizing radiations

- Alpha
- Beta
- Gamma
- X-Rays

The non-Ionizing radiations are:

- Radiofrequency waves
- UV rays
- Microwaves

Alpha and beta rays are **particulate radiation** (particles) whereas gamma and x-rays are **electromagnetic rays**. All electromagnetic waves have a **constant velocity** of **3×10^8 m/s** and **no mass**. The electromagnetic spectrum, from **low frequency** to **high frequency** consists of:

Radiowaves < Microwaves < Infrared < Visible light < Ultraviolet < X-rays < **Gamma rays**

Note: **W. C. Roentgen** is the father of X-ray and the first medical X-ray by Roentgen was of his wife Anna Bertha's left hand. Physicists **Peter Mansfield** and **Paul Lauterbur** developed MRI-related imaging techniques in medicine and they were awarded the Nobel prize in 2003.

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References: Grainger & Allison's Diagnostic Radiology 7th edition, page 17
Grainger & Allison's Diagnostic Radiology 6th edition, page 74

Q106. Radiology AIIMS 2018 ID: MC7028

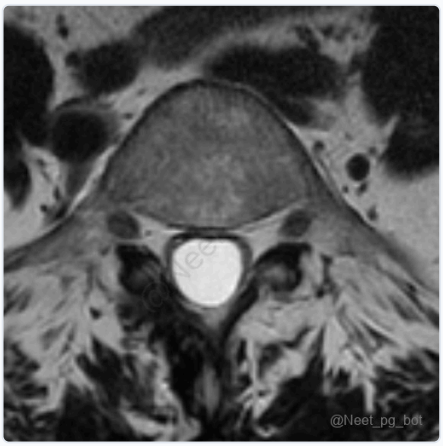
Empty thecal sac sign on MRI is seen in:

- **A** ✓ Arachnoiditis
- **B** Discitis
- **C** Vertebral osteomyelitis
- **D** Tethered cord syndrome

Correct Answer: Arachnoiditis

An **empty thecal sac** sign is seen in **arachnoiditis.**

Adhesive arachnoiditis can be diagnosed if the empty thecal sac indication is observed. This occurs due to **inflammation,** which leads to **clumping** of nerve roots. This leads to an empty sac appearance. It is best observed on **T2-weighted MRI** scans of the lumbar spine when the thecal sac appears empty.



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References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no 1330
Online Resource, https://radiopaedia.org/articles/empty-thecal-sac-sign#:~:text=Groups%201%20and%202%20were%20consistent%20with%20adhesions%20developing%20between,of%20an%20empty%20thecal%20sac.

Q107. Radiology AIIMS 2018 ID: MC7029

From the X-ray given, pick out the probable diagnosis:

A Subdiaphragmmatic abscess

B Pneumothorax

C ✓ Pneumoperitoneum

D Intestinal obstruction

Correct Answer: Pneumoperitoneum

The above X-ray clearly shows the presence of **subdiaphragmatic free gas** in the right upper quadrant. This indicates **pneumoperitoneum.**

Pneumoperitoneum is the presence of free gas within the peritoneal cavity but outside the viscera. The most common cause of pneumoperitoneum is **bowel perforation.**

- Gastric and posterior duodenal perforations are seen in long-standing cases of **peptic ulcers.**
- Intestinal perforations are seen with ischemia, inflammation, diverticula, or obstruction.
- Other causes also include iatrogenic, penetrating injuries, prolonged mechanical ventilation, peritoneal dialysis, etc.

Signs of pneumoperitoneum:

- Bowel-related signs
  - Double wall sign/**Rigler's sign**/gas-relief sign- Air is seen on both sides of the bowel wall.
  - Telltale triangle sign/triangle sign - Triangle of gas formed between three loops of the bowel or between the abdominal wall and two loops of the bowel.
- Peritoneal ligament-related signs
  - Football sign- Air outlines the confines of the peritoneal cavity
  - Falciform ligament sign- Air outlines the falciform ligament
  - Lateral umbilical ligament sign/**inverted "V"** sign- Air outlines the lateral umbilical ligament
  - Urachus sign - Air outlines the urachus
- Right upper quadrant signs
  - **Cupola sign**- Air outlines the median subphrenic space
  - Fissure for ligamentum teres sign- Air outlines the ligamentum teres
  - Hepatic edge sign- Cigar-shaped free-air pocket in the subhepatic region. It tracks superomedial following the contours of the liver
  - Lucent liver sign - Reduction of hepatic radiodensity due to accumulation of free gas anterior to the liver
  - Morison's pouch sign / **Doge cap sign** - Triangular/ crescentic gas lucency bound by the 11th rib
  - Periportal free gas sign.

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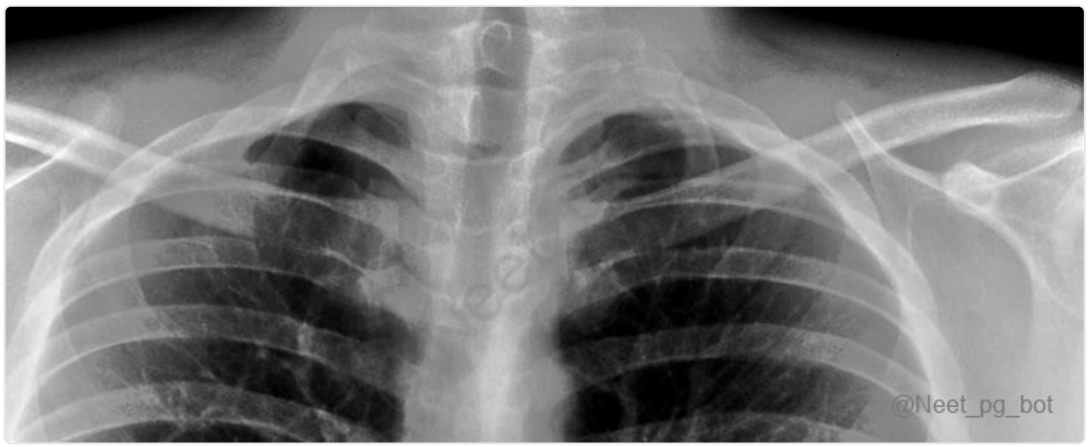
References: Online Resource, Jones J, González Herrera G, Campos A, et al. Pneumoperitoneum. Reference article, Radiopaedia.org (Accessed on 10 Dec 2024) https://doi.org/10.53347/rID-6165
Online Resource, Sureka B, Bansal K, Arora A. Pneumoperitoneum: What to look for in a radiograph? J Family Med Prim Care. 2015 Jul-Sep;4(3):477-8. doi:

10.4103/2249-4863.161369. PMID: 26288798; PMCID: PMC4535122.
Grainger & Allison's Diagnostic Radiology 7th edition, Page No. 463-466, 528

Q108. Radiology AIIMS 2018 ID: MC7030

What is the diagnosis shown in the image below?



A Costochondritis

B Spondylolisthesis

C ✓ Cervical rib

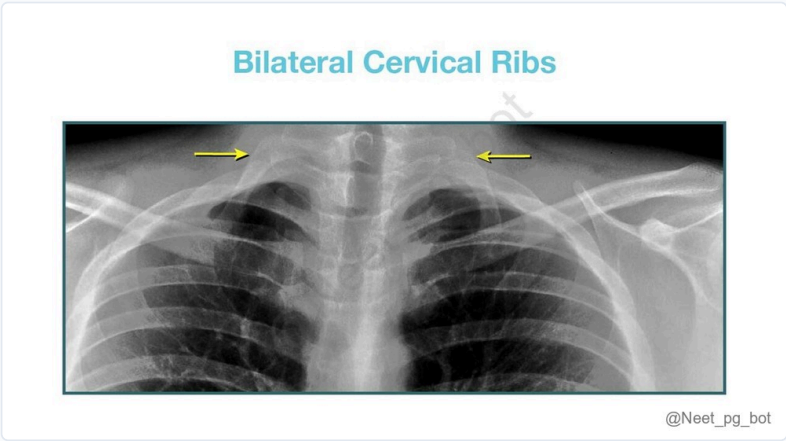
D Rib fracture

Correct Answer: Cervical rib

The given X-ray shows **cervical rib**.

Patients with a cervical rib usually show a lump and tenderness in the supraclavicular fossa. The pallor and paresthesias are due to vascular and neural compressions respectively. A radiograph of the neck is useful in detecting a cervical rib.

However, the absence of a rib on the **x-ray does not rule out** the possibility of a cervical rib**. CT** and **MRI** are more **sensitive.**



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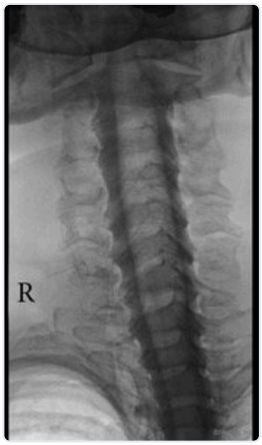
References: Textbook of Orthopedics, Page no 365
Grainger & Allison's Diagnostic Radiology 7th edition, page no 39

Q109. Radiology AIIMS 2018 ID: MC7031

Identify the procedure.



A ✓ Myelography

B Angiography

C DEXA

D Projectional radiography

Correct Answer: Myelography

The procedure depicted in the image represents **myelography.**

Myelography is a procedure used to evaluate **spinal** conditions. X-ray myelography can be used to **evaluate** the **spinal canal** in patients with **spondylolisthesis**. While MRI has largely replaced X-ray myelography it is done when there is a contraindication to MRI (Eg; non-compatible pacemaker, severe claustrophobia).

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References: Grainger & Allison's Diagnostic Radiology 7th edition, page no 1232

Q110. Radiology AIIMS 2017 ID: MA1502

A 30-year-old female came to OPD with complaints of backache. On examination, Schober test was positive and her eyes and nose were hyperpigmented. Probable diagnosis is

A Hypoparathyroidism

B Fluorosis

C ✓ Ochronosis

D Ankylosing spondylitis

Correct Answer: Ochronosis

Given X-ray shows intervertebral disc calcification and hyperpigmentation of nose and eyes are present. Both features are indicative of ochronosis as diagnosis.

- Ochronosis also known as **alkaptonuria** is an **autosomal recessive disorder**.
- It is due to **deficiency of homogentisic acid oxidase** leading to accumulation of **homogentisic acid.**
- Clinical features: Triad of
  - Homogentisic aciduria (urine turns dark color on standing due to oxidation or alkalinisation)
  - Eumelanin like pigmentation of skin, sclera, cartilages etc.
  - Degenerative chronic arthropathies especially of spine, hip, knees.

Option a: Hypoparathyroidism doesn't explain hyperpigmentation and other clinical features of hypoparathyroidism aren't mentioned.

Option b: Fluorosis - dental symptoms are not mentioned and osteoporosis will be marked in case of fluorosis. Also, hyperpigmentation is not a typical feature of fluorosis.

Option d: Ankylosing spondylitis – females are unlikely to be affected and it doesn't explain hyperpigmentation.

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References: Rook's Textbook of Dermatology 9th edition, Page no. 88.51

Q111. Radiology AIIMS 2017 ID: MA1162

A 30-year-old woman presented with complaints of bone pain and abdominal cramps. Her family says she has a history of abnormal behavior. The consultant doctor arrived at a provisional diagnosis based on the clinical features. Which of the following would be the best investigation to arrive at a definitive diagnosis?

A MRI

B Ultrasonogram

C ✓ Sestamibi scan

D CT scan

Correct Answer: Sestamibi scan

The given clinical features are suggestive of **primary hyperparathyroidism**. **Sestamibi scan** is the best investigation to arrive at a definitive diagnosis. **Parathyroid adenoma** is the most common cause of primary hyperparathyroidism.

Clinical features of hyperparathyroidism:

- Bones- bone pain, pathological fractures
- Stones- nephrocalcinosis
- Groans- abdominal pain
- Moans- psychotic symptoms
- Fatigue overtones- fatigue, malaise

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References: Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1514518/

Q112. Radiology AIIMS 2017 ID: MB9234

Based on the given image, what is the most likely diagnosis?

A Carotico-cavernous fistula

B ✓ Pseudoaneurysm of internal carotid artery

C Tumour blush of angiofibroma

D Extensive head and neck hemangioma

Correct Answer: Pseudoaneurysm of internal carotid artery

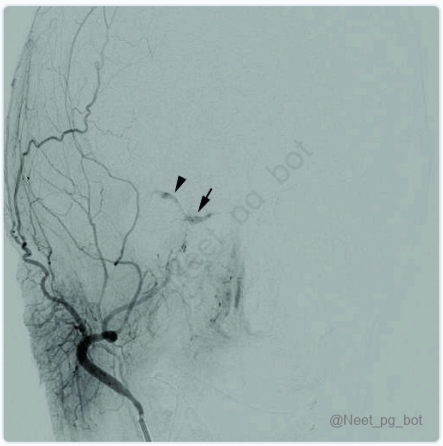
The given image is a **digital subtraction angiogram(DSA)** and it shows a **pseudoaneurysm** of the **internal carotid artery**.

The right ICA injection anteroposterior view shows the neck of the pseudoaneurysm originating from the **transverse segment** of the **cavernous portion** of the ICA.

Option A: Carotico-cavernous fistula is characterized by the premature filling of the veins as shown in the image below:

Option C: Angiofibroma is characterised by tumour blush. It is the radiological enhancement of a tumour that reveals opacification of the well-vascularized tumours.

Option D: Hemangioma can be differentiated from other vascular malformations by the presence of well-circumscribed mass demonstrating intense tissue staining, usually in a lobular pattern.



@Neet_pg_bot

References: Online Resource, Cruciata G, Parikh R, Pradhan M, Shah J, Greif E, Stein EG. Internal carotid artery dissection and pseudoaneurysm formation with resultant ipsilateral hypoglossal nerve palsy. Radiol Case Rep. 2017 Mar 6;12(2):371-375. doi: 10.1016/j.radcr.2017.01.016. PMID: 28491191; PMCID: PMC5417729.

Q113. Radiology AIIMS 2017 ID: MB9138

Which of the following will have reduced uptake on FDG PET scan?

- **A** ✓ Typical carcinoid
- **B** Small cell carcinoma
- **C** Atypical carcinoid
- **D** Large cell neuroendocrine carcinoma

Correct Answer: Typical carcinoid

Typical carcinoid is **more likely** to have reduced uptake on FDG PET scan than an atypical carcinoid.

The low metabolic rate of tumors makes the tumor **negative (show reduced uptake)** on an FDG PET scan.

Typical examples are:

- **Bronchoalveolar carcinoma** (Adenocarcinoma in situ)
- **Carcinoid tumour** (**Typical** > Atypical)

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References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 352

Q114. Radiology AIIMS 2017 ID: MB9214

A thin, 34-year-old male, presents with sharp pain and sudden onset dyspnea. Chest X-ray was performed and is shown below. What is the most probable diagnosis?

A Pulmonary embolism

B ✓ Pneumothorax

C Hydropneumothorax

D Pleural effusion

Correct Answer: Pneumothorax

The given chest radiograph clearly depicts the presence of **pneumothorax,** which represents air in the pleural space.

- The right hemithorax shows **blackening** (depicts area of low density) due to the presence of air.
- The **visceral pleural line** has **separated from the chest wall** by a **transradiant zone** devoid of vessels.
- The **collapsed lung** can be seen with a clear border.
- There is a **mediastinal shift** to the **left side** (contralateral), which is typically seen with cases of tension pneumothorax.

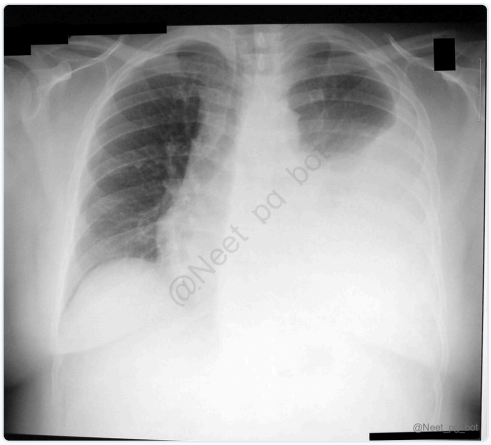
All the above features are suggestive of **pneumothorax**.

Note: **Pleural effusions** signify the presence of fluid accumulation in the pleural space. The fluid may have varying degrees of mobility depending on the nature of effusion (free-flowing or loculated).

While a small amount of free fluid is usually hard to detect on an erect PA chest x-ray, a **significant pleural effusion (200-500 mL)** is represented by obscured lateral and posterior costophrenic angles and the hemidiaphragm affected. Subsequently, **homogenous opacification** of the lower lung field develops, associated with **concave superior margin** relative to the lung. This margin is higher laterally than medially.

A massive pleural effusion cause dense opacification usually associated with a contralteral mediastinal shift.

Here is an image of a left-sided pleural effusion representing the homogenous opacification of the left lower and middle lung fields. There is associated blunting of the left sided costophrenic angle. The superior border of the effusion is concave and is higher laterally than medially.



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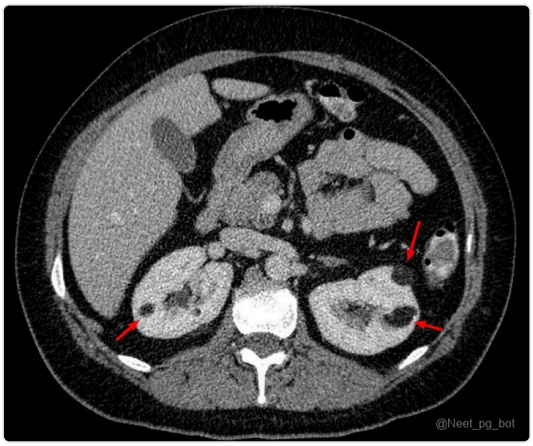
References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no 188-189, 193-195

Q115. Radiology AIIMS 2017 ID: MC9590

A 45-year-old woman comes to the emergency department with severe back pain, vomiting, and hypotension. CT findings are given below. What is the diagnosis?



A Renal cyst

B Renal cell carcinoma

C Para pelvic cyst

D ✓ Renal angiomyolipoma

Correct Answer: Renal angiomyolipoma

The given CT image shows well-circumscribed lesions with **low attenuation** (black in colour) that indicate **fatty** tissue. This is suggestive of **renal angiomyolipomas**.

Angiomyolipomas (AMLs) are benign tumours that occur sporadically or in association with **tuberous sclerosis**. It is commonly seen in adults between **30 and 50 years** of age, with a **female** predilection. AMLs predominantly affect the kidneys, though they can occur in other sites such as the liver (a common extra-renal location) and less frequently in the ovary, fallopian tube, spermatic cord, palate, and colon.

Histologically, angiomyolipomas consist of **perivascular epitheloid** cells with varying proportions of **vascular**, immature **smooth muscle** cells, and **fat cells**. Patients with AMLs present with flank pain, a palpable mass, hematuria, recurrent urinary tract infections, hypertension, or renal failure. Larger tumours increase the risk of complications, such as **spontaneous retroperitoneal hemorrhage**, which can manifest as sudden-onset abdominal or flank pain, along with symptoms like nausea, **vomiting**, and **hypotension**.

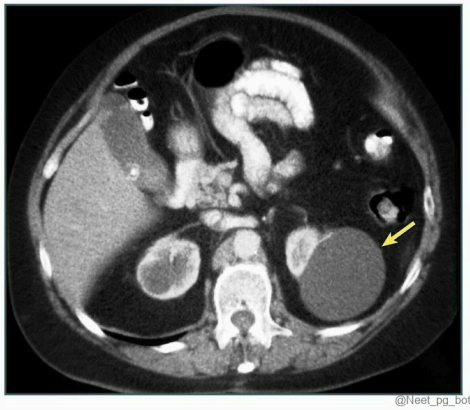
Everolimus is used for the treatment of angiomyolipomas. Treatment should be considered for **asymptomatic**, growing AMLs measuring **>3 cm** in diameter. Angiomyolipomas do not normally require surgery unless life-threatening bleeding is present.

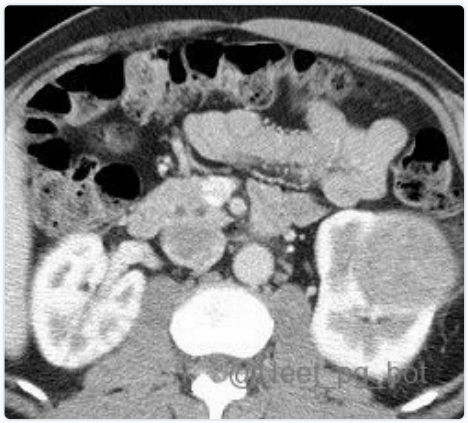
Other options:

Renal cyst is a broader term that includes the **cystic lesions** of the kidney. They can be **simple** cysts that are benign. **Complex** or complicated cystic lesions need evaluation and can be malignant. **Bosniak classification** is used for renal cysts. Cysts appear as hypodense lesions (greyish) with a thin wall, with or without septations.

CT image showing a simple renal cyst:

Renal cell carcinoma (RCC) is the most common **malignant** tumour of the kidney. It usually occurs in the age group **50-70 years** and macroscopic **hematuria** is seen in about 60% of the cases. It is a differential for angiomyolipoma but it is important to note that **macroscopic fat** is almost always associated with **ossification or calcification.**





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References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no 781, 796, 797, 1877, 1878
Textbook of Radiology and Imaging: Volume 2, Page no: 952 and 953
Harrison's Principles of Internal Medicine, 21st Edition, Page no: 2354
Online Resource, Amini B, Walizai T, Niknejad M, et al. Renal angiomyolipoma. Reference article, Radiopaedia.org (Accessed on 17 Dec 2024) https://doi.org/10.53347/rID-899
Online Resource, Vos N, Oyen R. Renal Angiomyolipoma: The Good, the Bad, and the Ugly. J Belg Soc Radiol. 2018 Apr 20;102(1):41. doi: 10.5334/jbsr.1536. PMID: 30039053; PMCID: PMC6032655.

Q116. Radiology AIIMS 2017 ID: MC9591

A 25-year-old male came with complaints of acute abdominal pain. A plain X-ray abdomen erect was taken which is shown below. What is your diagnosis?

A Subdiaphragmatic abscess

B ✓ Pneumoperitoneum

C Pneumopericardium

D Intestinal obstruction

Correct Answer: Pneumoperitoneum

The above X-ray clearly shows the **presence of subdiaphragmatic free gas** in the right upper quadrant. This indicates **pneumoperitoneum.**

Causes of pneumoperitoneum include:

1. Bowel perforation
    1. Peptic ulcer
    2. Ischemic bowel
    3. Necrotizing enterocolitis.
2. Small bowel perforation
    1. Trauma
    2. Penetrating injuries
3. During surgery / laparoscopy

Signs of pneumoperitoneum:

- Bowel-related signs
    - Double wall sign (also known as **Rigler's sign** or gas-relief sign)
    - Telltale triangle sign (also known as triangle sign)
- Peritoneal ligament-related signs
    - Football sign
    - Falciform ligament sign
    - Lateral umbilical ligament sign (also known as **inverted "V"** sign)
    - Urachus sign
- Right upper quadrant signs
    - **Cupola sign**
    - Fissure for ligamentum teres sign
    - Hepatic edge sign
    - Lucent liver sign
    - **Morison's pouch sign (doge cap sign)**
    - Periportal free gas sign

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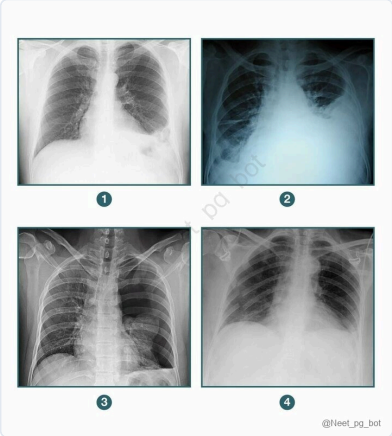
References: Online Resource, https://radiopaedia.org/articles/pneumoperitoneum

Q117. Radiology INI-CET 2022 ID: MF5178

A patient presents to the casualty following blunt trauma to the chest. A chest X-ray was done. Among the following radiographs, in which case would you further evaluate the patient before putting a chest tube?

1. Diaphragmatic hernia
2. Hemothorax
3. Pneumothorax
4. Flail chest



A 4

B ✓1

C 2

D 3

Correct Answer: 1

Among the following radiographs, image 1 shows dilated bowel loops in the hemithorax, and is suggestive of **diaphragmatic hernia** and it **needs to be evaluated further** before putting a chest tube since it can cause injury to the intestinal contents in the thorax.

Blunt injury to the diaphragm can lead to **rupture** and therefore the **abdominal contents** can **herniate** into the **chest**. **Nasogastric tube placement** followed by **chest radiography** can be useful in case of herniation of the stomach into the chest. The most accurate investigation is **video-assisted thoracoscopy** or **laparoscopy**.

Operative repair should be done. Repair should be done via the abdomen in case of a diaphragmatic rupture.

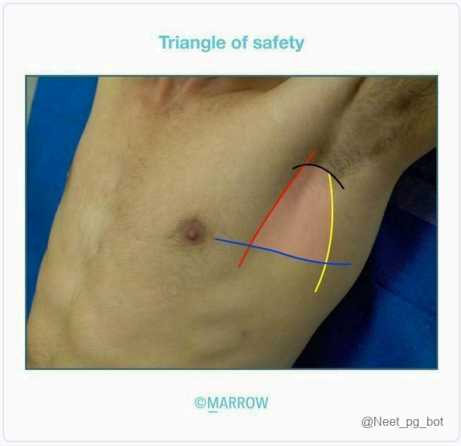
Radiographs 2,3,4 are **hemothorax**, **pneumothorax,** and **flail chest** respectively. These conditions are life-threatening. Hence, the diagnosis and management have to be immediate. Rapid insertion of **intercostal chest drains** should be done. They are inserted into the **fifth intercostal space.** The safe landmark is identified as a triangle that lies anterior to the midaxillary line above the nipple, below, and lateral to the pectoralis major muscle.

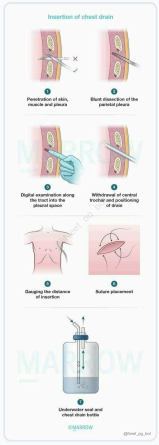
The below images show the **triangle of safety** and technique of chest drain insertion.

Hemothorax can occur following **blunt trauma**. It occurs due to bleeding from the intercostal vessels or internal mammary artery secondary to rib fracture. Features include hemorrhagic shock, flat neck veins, unilateral absence of breath sounds, and dullness on percussion. Correction of hypovolemic shock, **intercostal drain insertion,** and intubation are useful in management.

Pneumothorax can be open or tension pneumothorax. In tension pneumothorax, the common etiology is penetrating or blunt trauma, and features such as tachypnea, dyspnea, distended neck veins, tracheal deviation, hyper resonance and decreased breath sounds are seen. **Immediate decompression** can be attained by large bore cannula insertion into the second intercostal space on the affected side in the midclavicular line after which a **chest tube** can be placed in the **5th intercostal space** along the anterior axillary line. Chest radiograph shows **mediastinal shift** to the **opposite side** and diaphragmatic depression on the same side.

Flail chest occurs as a result of blunt trauma and is associated with rib fractures. Clinically, paradoxical movement of the chest can be present. Management includes oxygen administration, analgesia, chest drain, and surgery to stabilize the flail segment. Rib fractures can be seen on chest radiograph.





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References: Bailey & Love's Short Practice of Surgery, 27th Edition, Page no 323, 367-368, 370, 920
Grainger and Allison, textbook of medical imaging, 7th edition, Page no 52
Online Resource, https://emedicine.medscape.com/article/822999-overview

Q118. Radiology INI-CET 2022 ID: MF4969

A 30-year-old patient presents with progressive difficulty in swallowing both solids and liquids for 8 months. A barium swallow was done and is shown in the image below. What other investigations should be done to confirm the diagnosis in this patient?

UGIE- Upper gastrointestinal endoscopy

PET- Positron emission tomography

A ✓ UGIE + manometry

B UGIE + contrast enhanced CT

C UGIE + PET

D UGIE + 24 hr pH monitoring

Correct Answer: UGIE + manometry

The given clinical scenario and the barium swallow showing **bird beak sign** are suggestive of **achalasia cardia. Upper gastrointestinal endoscopy** (UGIE) and **manometry** should be done to confirm the diagnosis.

Achalasia is a motility disorder and is less common. It typically presents as dysphagia for **solids** and **liquids**. Other features such as **regurgitation** and **chest pain** are also present.

On fluoroscopy, a grossly dilated esophagus with tapering end towards the gastroesophageal junction is seen. This is called the **bird beak** sign.

Barium is retained in the esophagus for more than 5 minutes and is a significant finding.

Diagnosis is confirmed with **endoscopy** and **manometry**.

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References: Grainger and Allison, textbook of medical imaging, 7th edition, Page no 495
Online Resource, https://radiopaedia.org/articles/achalasia

Q119. Radiology INI-CET 2022 ID: ME0550

A 28-year-old woman with a severe headache is scheduled to undergo a CT scan. During history taking, which of the following will you definitely ask for before the commencement of the scan?

LMP - Last menstrual period

A Use of cochlear implant

B ✓ LMP

C History of fall

D History of hypertension

Correct Answer: LMP

The most important information that should be ascertained during history taking in a **28-year-old woman** of **reproductive age group** posted for a CT scan is the **last menstrual period** (LMP). **CT scan** is **avoided** during **pregnancy** due to the potential adverse effects of radiation on the fetus; hence, LMP is ascertained to rule out pregnancy.

CT is a high-resolution technique using x-rays to generate cross-sectional images of any area of the body. It is associated with high radiation doses.

Other options:

Option A: **Use of cochlear implant** is a **contraindication** for **MRI** and not CT. CT scan can be performed on a patient with a cochlear implant.

Option B: **History of fall** in a patient with a headache is asked for **suspected head trauma.** A **CT scan** in this patient can help in the diagnosis and assess extent of damage.

Option C: **History of hypertension** points to a possible **intracranial bleed.** A CT scan can be useful in ruling out the bleed.

Since it is imperative to rule out pregnancy before taking a CT scan, last menstrual period is the most appropriate answer to this question.

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References: Grainger & Allison's Diagnostic Radiology 7th edition, Page no: 479, 736, 817
Online Resource, https://emedicine.medscape.com/article/2110836-overview
Online Resource, https://www.fda.gov/medical-devices/cochlear-implants/cochlear-implants-and-mri-safety#:~:text=Some%20MR%20Conditional%20cochlear%20implants,manufacturer%20for%20use%20during%20MRI.
Online Resource, https://www.thieme-connect.com/products/ejournals/pdf/10.1055/s-0035-1553413.pdf

Q120. Radiology INI-CET 2022 ID: MF7317

Which of the following findings are seen in a high-resolution CT scan of fungal pneumonia?

1. Interlobular septations
2. Peripheral wedge-shaped consolidation
3. Pleural effusion
4. Cavitatory lesions with surrounding ground glass opacities

A 1, 2, 4

B 1, 3, 4

C ✓ 1, 2, 3

D 2, 3, 4

Correct Answer: 1, 2, 3

High-resolution computed tomography (HRCT) of **fungal pneumonia** shows interlobular septations, peripheral wedge-shaped consolidation, and pleural effusion. **HRCT** of fungal pneumonia shows a **nodule** (not a cavity) with **surrounding ground-glass opacities**.

A **nodule** surrounded by **ground-glass attenuation** on HRCT imaging is seen in **aspergillosis,** candidiasis, *Pneumocystis jirovecii* pneumonia, coccidioidomycosis, and paracoccidioidomycosis. **Cavitation** on **HRCT** is seen in cases of community-acquired pneumonia due to ***Mycobacterium tuberculosis, S. aureus*****,** and gram-negative bacilli.

Aspergillosis is a fungal disease caused by *Aspergillus* sp. Pulmonary aspergillosis has been categorized into saprophytic, allergic, and invasive forms.

HRCT findings of various types of aspergillosis are as follows:

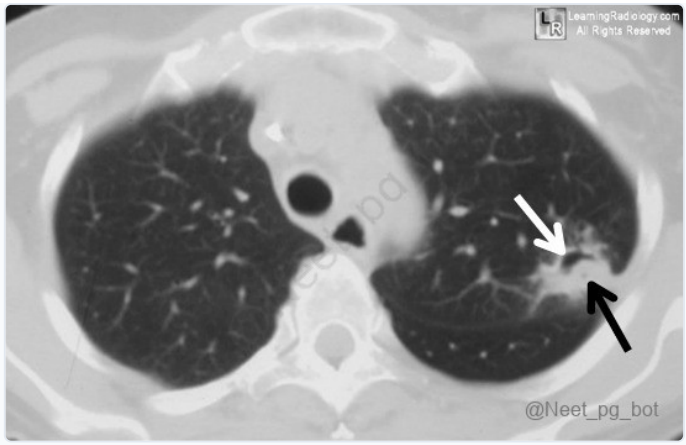
- **Aspergilloma** (saprophytic) shows a mass with soft-tissue attenuation, which is separated from the thoracic wall by airspace called **the air crescent sign.** Multiple nodules and bilateral involvement may be seen.
- **Allergic bronchopulmonary aspergillosis (ABPA-allergic)** shows hypodense mucous plugs.
- **Angioinvasive aspergillosis** shows multiple nodules surrounded by a **halo of ground-glass attenuation** ('halo sign') or **pleural-based wedge-shaped areas of consolidation.**

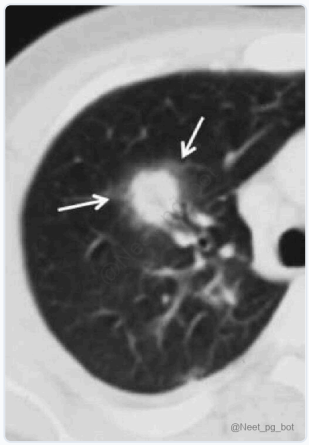
The following is an HRCT image of aspergilloma showing the **air crescent sign:**

The following CT image of aspergilloma shows the **halo sign:**

Candidiasis pneumonia caused by *Candida* sp. in immunocompromised patients shows multiple bilateral nodular opacities, areas of consolidation, and ground-glass opacities on HRCT. **Pleural effusion**, thickening of the bronchial walls, and cavitation are less commonly seen.

Paracoccidioidomycosis (PCM), is a disease caused by *Paracoccidioides brasiliensis.* On HRCT, ground-glass opacities, nodules, the 'halo' and 'reversed halo' signs, **interlobular septal thickening**, airspace consolidation, cavitation, and fibrosis.





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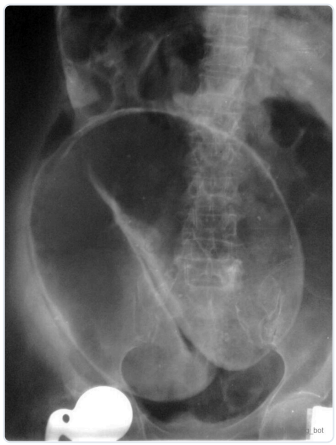
References: Grainger and Allison, textbook of medical imaging, 7th edition, Page no 105, 117-120

Q121. Radiology INI-CET 2022 ID: ME0696

An elderly lady presents with abdominal pain and distension. She has a history of scoliosis. Given below is the erect X-ray abdomen radiograph of the lady. What is the most likely diagnosis?



A ✓ Large bowel obstruction due to sigmoid volvulus

B Large bowel obstruction due to fecolith

C Cecal volvulus

D Small bowel obstruction due to bowel perforation

Correct Answer: Large bowel obstruction due to sigmoid volvulus

The given clinical scenario with an old woman and the radiograph showing characteristic **coffee bean appearance** are highly suggestive of **sigmoid volvulus causing a large bowel obstruction**.

Sigmoid volvulus is a cause of **large bowel obstruction** and occurs when the sigmoid colon **twists** on the sigmoid mesocolon. Symptoms include constipation, abdominal bloating, nausea, and/or vomiting. The onset may be acute or chronic. It occurs more commonly in patients with a history of mental illness and chronic constipation.

X-ray abdomen shows a large, dilated loop of the colon, often with a few air-fluid levels. Specific signs include the **coffee bean sign, Frimann-Dahl sign** (three dense lines converge towards the site of obstruction), and the **absence of rectal gas. CT** may reveal a large gas-filled loop **without haustral** markings, forming a **closed-loop** obstruction.

Here are some pointers that help in differentiating the two volulus:

Sigmoid volvulus	Cecal volvulus
More common incidence	Less common incidence
Coffee bean sign is typical.	Coffee bean sign is occassionally seen
Seen in old age, psychiatric patients	Usually seen in those aged 30-60 years old
The top of the sigmoid volvulus usually lies very high in the abdomen with its apex on the left side.	• In half the patients, the caecum twists and inverts so that the pole of the caecum and the appendix occupy the left upper quadrant. • In other patients, the twist occurs in an axial plane without inversion and the caecum still occupies the right half of the abdomen.
Haustra aren't usually seen	Haustra usually seen

Multiple air-fluid level seen	One air-fluid level seen
	Attached air-filled appendix can be seen

Other options:

Option B: In **large bowel obstruction due to fecolith,** the presence of **haustrations** is a feature. Haustral folds are irregular and do not cross the entire diameter of the bowel. These are absent in the cecum. The given image shows haustrations in large bowel obstruction.

The following image shows an x-ray radiograph of **large bowel obstruction:**

Option C: **Cecal volvulus** shows characteristic radiological findings like **cecal dilatation**, **single air-fluid** level that may extend into the left upper quadrant, and **absence of gas** in the distal colon. The **colonic haustrations** are well maintained.

The following image shows an x-ray radiograph of **cecal volvulus:**

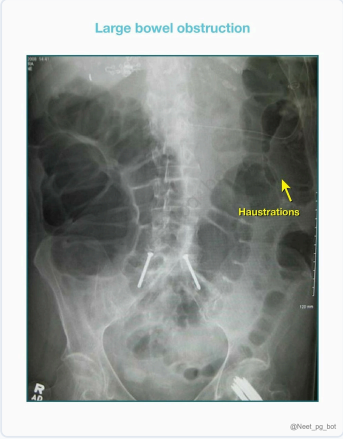
Option D: In **small bowel** obstructions, the obstructed bowel loops are **central in location and lie transversely**.
The **jejunal** obstruction shows **valvulae conniventes** (mucosal folds that cross the entire diameter of the bowel) giving it a **ladder effect.** The distended cecum may show the presence of gas shadow in the right iliac fossa.

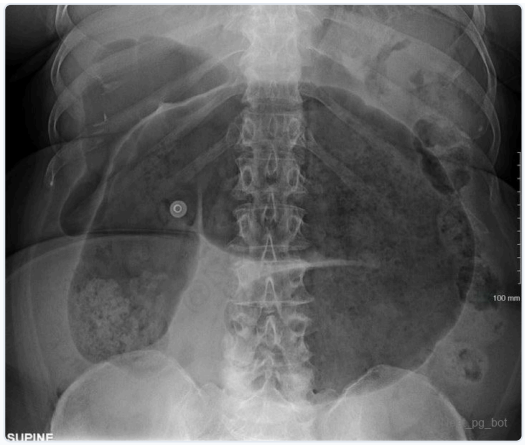
Common **causes** of **small bowel obstruction** are **adhesions (~60%),** neoplasms (~20%), hernias (~10%), Crohn's disease (~5%) and other miscellaneous conditions (~5%).

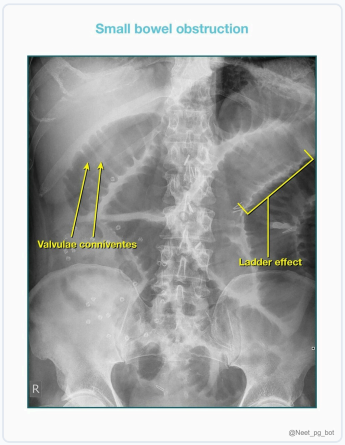
The following image shows an x-ray radiograph of **small bowel obstruction:**

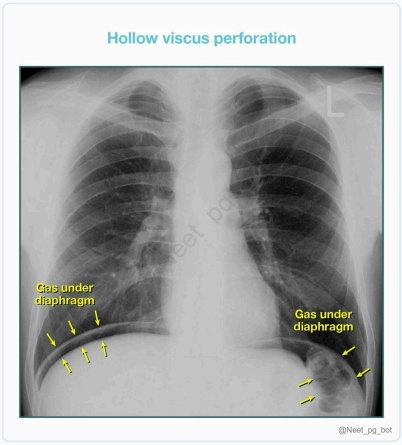
Cases of **bowel perforation** have an erect abdominal radiograph showing **free gas under** the **diaphragm**. A supine radiograph may show **dilated gas-filled loops of the bowel,** suggestive of paralytic ileus.

The below radiograph image shows a hollow viscus **perforation:**









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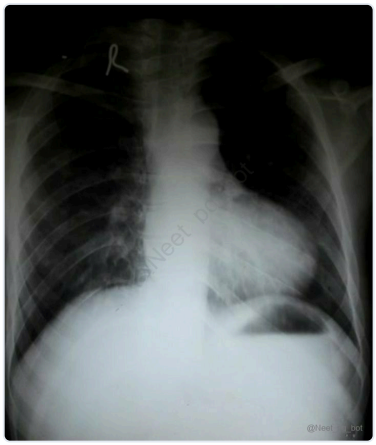
References: Online Resource, https://radiopaedia.org/articles/sigmoid-volvulus
Grainger & Allison's Diagnostic Radiology 7th edition, Page no. 472,464,468,473,544,545
Online Resource, https://radiopaedia.org/articles/caecal-volvulus
Online Resource, https://radiopaedia.org/articles/bowel-obstruction?lang=gb

Q122. Radiology INI-CET 2022 ID: ME0663

A child presents with cyanosis. His chest radiograph is shown below. What is the diagnosis?



A ✓ Tetralogy of Fallot (TOF)

B Ebstein's anomaly

C Transposition of great arteries (TGA)

D Total anomalous pulmonary venous return (TAPVC)

Correct Answer: Tetralogy of Fallot (TOF)

The given clinical scenario and chest X-ray showing a "**coeur en sabot**" or **boot-shaped heart** are suggestive of **tetralogy of Fallot (TOF)**.

TOF is a cyanotic congenital heart disease with the following four components:

- Obstruction to right ventricular outflow (**pulmonary stenosis**)
- Malalignment type of ventricular septal defect (**VSD**)
- Dextroposition of the aorta so that it overrides the ventricular septum (**overriding of the aorta**), and
- Right ventricular hypertrophy (**RVH**)

In TOF, the hypertrophied right ventricle causes the rounded apical shadow to be uptilted so that it is situated higher above the diaphragm than normal. On x-ray, this appears as a **boot-shaped heart** or **coeur en sabot,** as shown below:

Option B: **Ebstein anomaly** is a rare cyanotic congenital heart disease with diminished pulmonary blood flow due to abnormality of the tricuspid valve.

In the **Ebstein anomaly,** there is severe right-sided cardiomegaly due to an elongated and enlarged right atrium, which leads to an elevated apex giving a **box-shaped heart**, as shown below:

Option C: **Transposition of great arteries** (TGA) can present from birth to infancy with cyanosis, decreased exercise tolerance, and poor physical development **TGA** is the **most specific** heart disease in **infants** of **diabetic mothers.**

The following is a radiograph of **TGA** showing the egg-on-a-string appearance/egg-on-side appearance of the heart:

Option D: **Total anomalous pulmonary venous return** (TAPVC) occurs due to the anomalous connection of pulmonary veins instead of joining to the left atrium resulting in pulmonary venous blood reaching the right atrium.

In the supracardiac type of TAPVC, the configuration of the heart and the superior mediastinal borders resemble a **snowman**. It is also referred to as the **figure of 8 sign**, as shown below:

Boot-shaped heart
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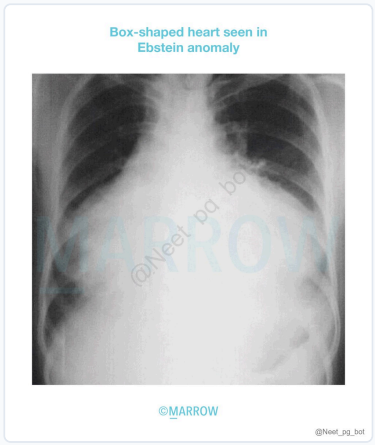
Box-shaped heart seen in
Ebstein anomaly
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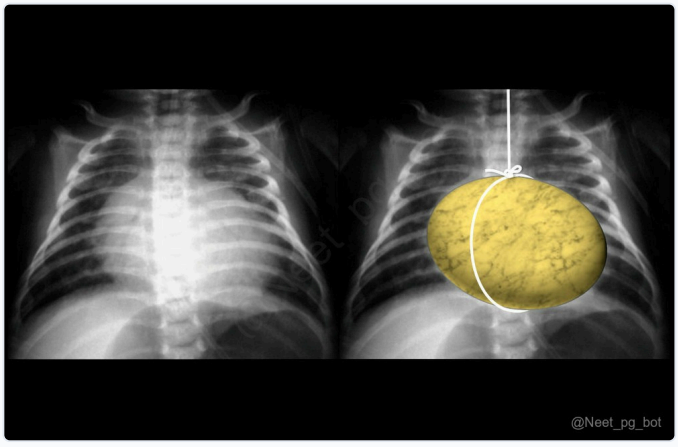
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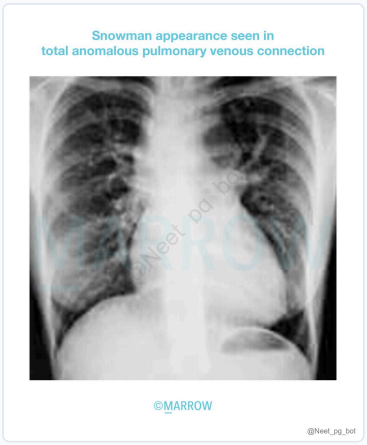
Snowman appearance seen in
total anomalous pulmonary venous connection
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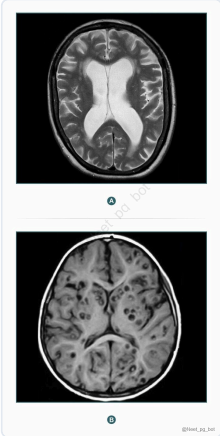
References: Nelson textbook of pediatrics 21st edition, Page no: 2406
Online Resource, https://radiopaedia.org/articles/egg-on-a-string-sign-heart#:~:text=Egg%2Don%2Da%2Dstring%20sign%2C%20also%20referred%20to,therefore%20appears%20like%20an%20egg.
Online Resource, https://radiopaedia.org/articles/total-anomalous-pulmonary-venous-return
Online Resource, https://radiopaedia.org/articles/ebstein-anomaly
Online Resource, https://radiopaedia.org/articles/tetralogy-of-fallot?lang=gb
Ghai Essential Pediatrics 9th Edition, Page no: 417, 420, 423, 424

Q123. Radiology INI-CET 2022 ID: MF7343

An HIV patient presented with a headache and altered sensorium. Consider the following statements and identify the essential criteria for the diagnosis of neurocysticercosis.

1. **Brain biopsy showing the parasite.**
2. **Muscle biopsy showing foci of calcification.**
3. **MRI sequence as seen in A.**
4. **MRI sequence as seen in B.**



A 1 and 2 only

B ✓ 1 and 4 only

C 1, 3, and 4

D 1, 2, and 4

Correct Answer: 1 and 4 only

The **MRI brain** labeled as **B** shows **multiple cysticerci** within the brain suggesting **neurocysticercosis.** Among the given statements, the essential criteria for the diagnosis of neurocysticercosis include **1 and 4** (option B), i.e.,

- Brain biopsy showing the parasite.
- Neuroradiologic demonstration of cystic lesions with scolex (MRI labeled B in this case).

Neurocysticercosis is a condition caused by the larvae of ***Taenia solium***.

Criteria for diagnosis of neurocysticercosis:

1. **Absolute criteria**

   a. Demonstration of cysticerci by **histologic** or **microscopic examination** of a **biopsy** taken from the brain or spinal cord lesion.
   b. **Visualization** of **the** parasite in the **eye** by funduscopy
   c. **Neuroradiologic demonstration** of cystic lesions containing a characteristic scolex

2. **Neuroimaging criteria**

   a. **Major** neuroimaging criteria: **Cystic lesions** without a discernible scolex, multilobulated cystic lesions in the subarachnoid space, typical small enhancing lesions, typical parenchymal brain calcifications
   b. **Confirmative** neuroimaging criteria: Resolution of cystic lesions spontaneously or after cysticidal drug therapy and migration of ventricular cysts documented on sequential neuroimaging studies
   c. **Minor** neuroimaging criteria: Obstructive hydrocephalus or abnormal enhancement of basal leptomeninges

3. **Clinical/exposure criteria**

   a. **Major clinical/exposure criteria:** Demonstration of antibodies to cysticerci or cysticerci antigen in cerebrospinal fluid by enzyme-linked immunosorbent assay; Household contact with an individual infected with *T. solium*; Evidence of cysticercosis outside the central nervous system (e.g., cigar-shaped soft-tissue calcifications)

b. **Minor clinical/exposure criteria:** Clinical manifestations suggestive of neurocysticercosis; Residence in a cysticercosis-endemic area; Frequent travel to a cysticercosis-endemic area

Diagnosis is **confirmed** by:

- **One absolute** criterion
- **Two major neuroimaging** criteria plus any **clinical/exposure** criteria
- **One major** and **one confirmative neuroimaging** criterion plus any **clinical/exposure** criteria
- **One major neuroimaging** criteria plus **two clinical/exposure** criteria (including at least one major clinical/exposure criterion), with the **exclusion** of other pathologies producing similar neuroimaging findings

A **probable diagnosis** is supported by the fulfillment of

- **One major neuroimaging** criterion plus any **two clinical/exposure** criteria
- **One minor neuroimaging** criterion plus at least **one major clinical/exposure** criterion

Other options: The MRI labeled as A shows ventriculomegaly in brain atrophy.

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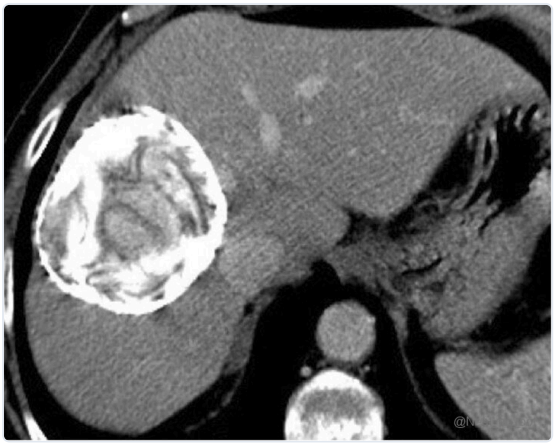
References: Harrison's Principles of Internal Medicine, 21st Edition, Page no: 1792, 1793
Online Resource, https://www.sciencedirect.com/science/article/pii/S0022510X16307481

Q124. Radiology INI-CET 2022 ID: MF7335

Identify the condition based on the non-contrast CT scan of a patient given below.



A Hepatocellular carcinoma

B Fibronodular hyperplasia

C Liver abscess

D ✓ Hydatid cyst

Correct Answer: Hydatid cyst

The given CT image shows a **cyst** in the **liver** with the **serpent sign** and areas of **calcification**, which are suggestive of **hydatid disease.**

Hydatid disease or echinococcosis is a zoonotic infection caused by the ***Echinococcus*** species of flatworms. Its life cycle consists of **dogs** as **definitive hosts** and **humans** as **accidental intermediate hosts**.

Primary hydatid cysts **most commonly** occur in the **liver.** Most patients are asymptomatic but when symptomatic, the patient presents with hepatomegaly, pain, and obstructive jaundice.

The **first/initial** imaging test is USG, and **CT scan** can be used to diagnose the condition. **CT scan** shows the presence of a space-occupying lesion with a smooth outline and septa. CT scan has an accuracy of 98% in the detection of hydatid cysts and **high sensitivity** to the detection of daughter cysts and extra-hepatic cysts.

On **CT imaging, serpent sign** is seen as wavy membranes within the cyst**:**

The USG of an **active hydatid cyst** shows multiple septations and a rosette or **honeycomb appearance:**

The **ball of wool** sign is seen on **USG** when the **contents** settle in the **dependent portion** of the cyst, as seen below:

On **USG**, the **water-lily sign** is seen in hydatid infections when there is **detachment** of the **endocyst membrane**, which results in **floating membranes** within the **pericyst** that mimic the appearance of a water lily**.**

The image below shows the structures present within a cyst.

Other options:

Option A: **Hepatocellular carcinoma** (HCC) is diagnosed with contrast-enhanced CT (CECT) or MRI. On a non-contrast CT, it appears as a low-attenuated lesion with occasional calcification. On CECT, HCC lesions show **enhancement** in the **arterial phase** and **washout** in the **portal phase.**

The following image shows the triphasic CECT of HCC:

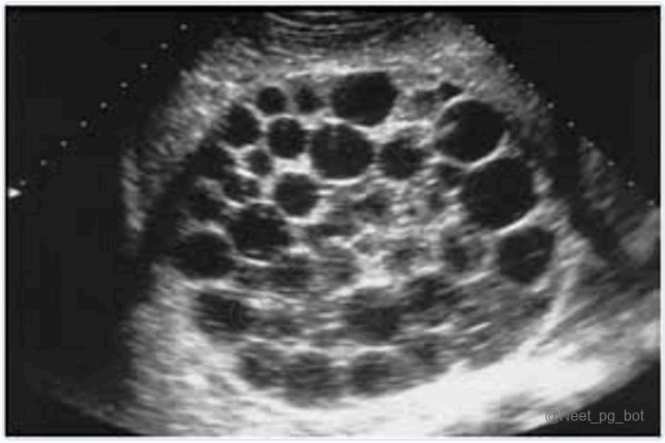
Option B: **Focal nodular hyperplasia** is a benign solid lesion in the liver. Histologically, it is composed of normal liver elements, but the biliary elements are not connected to the normal biliary tree. On **CECT**, they exhibit **homogeneous enhancement** in the **arterial phase** of IV contrast.

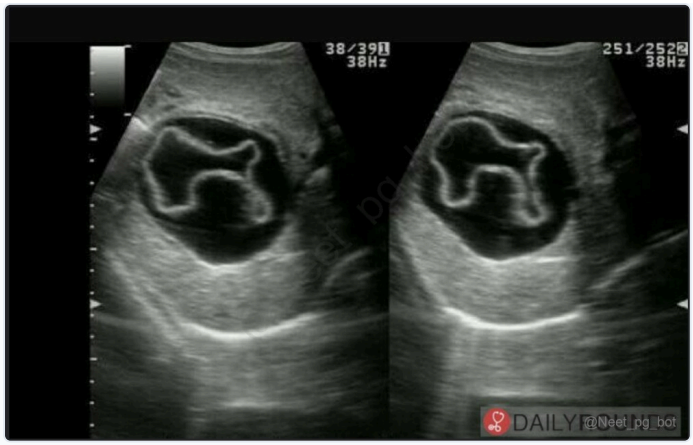
The image below shows late arterial and portal venous phase CECT of **focal nodular hyperplasia:**

Option C: **Liver abscess** may be caused by various infections. It begins as a local inflammatory reaction and later develops **liquefactive necrosis** in the center, with an inflammatory margin.

On a **non-contrast CT**, it appears as an ill-defined lesion with **low attenuation**, as seen in the image below:

The given image shows a liver abscess with **rim enhancement** on **CECT**.




38/39
38Hz
251/252
38Hz
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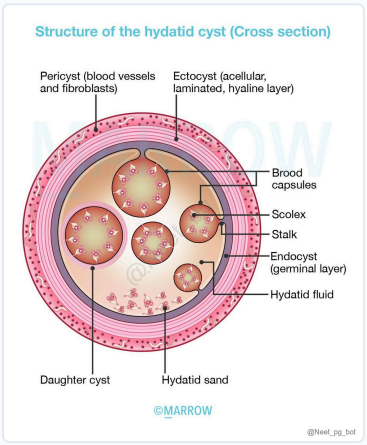
Structure of the hydatid cyst (Cross section)
Pericyst (blood vessels and fibroblasts)
Ectocyst (acellular, laminated, hyaline layer)
Brood capsules
Scolex
Stalk
Endocyst (germinal layer)
Hydatid fluid
Daughter cyst
Hydatid sand
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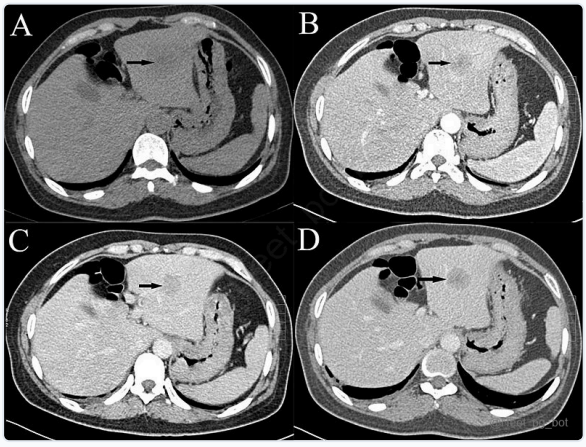
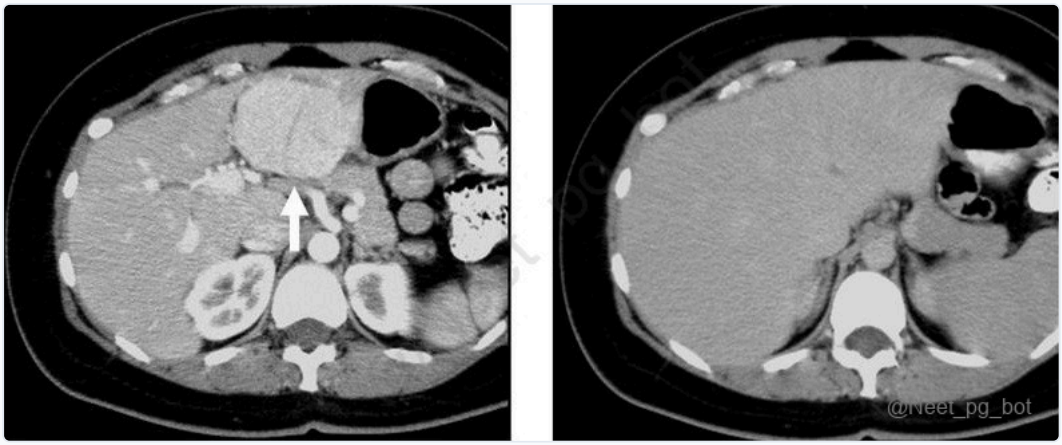

A
B
C
D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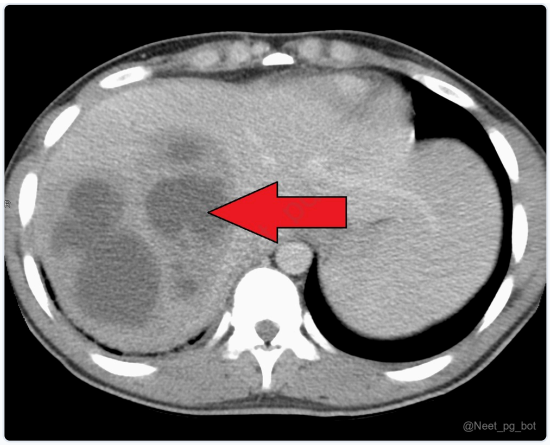

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References: Sabiston Textbook of Surgery 21st Edition, Page no. 1460
Grainger & Allison's Diagnostic Radiology 7th edition, Page no. 613, 616, 628
Online Resource, https://radiopaedia.org/articles/hydatid-cyst-signs
Paniker's Textbook of Medical Parasitology 8th edition, Page no: 130-135

Q125. Radiology INI-CET 2022 ID: ME1621

Match the following:

1. Football sign	a. Hypertrophic pyloric stenosis
2. String sign	b. Achalasia cardia
3. Rat tail sign	c. Ileocecal TB
4. Inverted umbrella sign	d. Pneumoperitoneum

A 1-b, 2-d, 3-a, 4-c

B 1-c, 2-a, 3-b, 4-d

C 1-d, 2-b, 3-a, 4-c

D ✓ 1-d, 2-a, 3-b, 4-c

Correct Answer: 1-d, 2-a, 3-b, 4-c

The correct match is **1-d, 2-a, 3-b, 4-c** (option D).

1. Football sign	d. Pneumoperitoneum
2. String sign	a. Hypertrophic pyloric stenosis
3. Rat tail sign	b. Achalasia cardia
4. Inverted umbrella sign	c. Ileocecal TB

Football sign is seen in cases of **massive pneumoperitoneum** in abdominal x-ray film taken in the supine position.

The following image shows football sign in a patient with pneumoperitoneum:

Rat tail sign refers to dilatation of the esophagus with **narrowing at the gastro-esophageal junction** in barium swallow/ fluoroscopy studies. It is also known as **bird beak sign** and is typically seen in **achalasia cardia**. In the case of a **rat tail sign with irregular contour**, **esophageal cancer** should be suspected.

The following image shows **rat tail sign** or **bird beak sign** in a patient with **achalasia cardia**:

String sign refers to the string-like appearance of severely **narrowed bowel lumen** in fluoroscopy/barium studies. It is seen in **hypertrophic pyloric stenosis**, gastrointestinal tuberculosis, colon cancer, and carcinoid tumor. **String sign of Kantor** is classically described in **Crohn's disease.**

Inverted umbrella sign, also known as **Fleischner sign,** is seen in barium studies of **ileocecal tuberculosis.** It refers to a widely gaping, thickened, patulous ileocecal valve and a narrowed, ulcerated terminal ileum.

Football sign- Pneumoperitoneum
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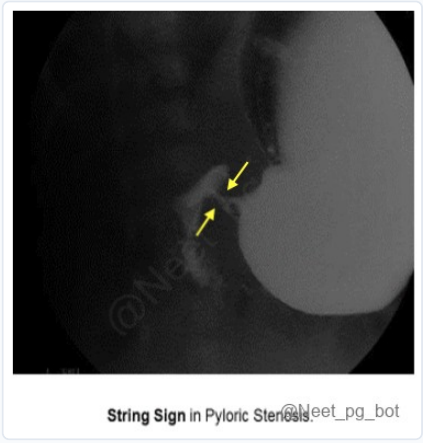
String Sign in Pyloric Stenosis.
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Fleischner sign/Inverted umbrella sign
@Neet_pg_bot

@Neet_pg_bot

References: Online Resource, https://radiopaedia.org/articles/football-sign-pneumoperitoneum?lang=us
Online Resource, https://radiopaedia.org/cases/achalasia-rat-tail-sign?lang=gb
Online Resource, https://radiopaedia.org/articles/string-sign-bowel?lang=gb
Online Resource, https://radiopaedia.org/articles/fleischner-sign-tuberculosis-of-ileocaecal-junction
Bailey & Love's Short Practice of Surgery, 27th Edition, Page no: 1244
Bailey & Love's Short Practice of Surgery - 28th Edition, Page no: 1122

Q126. Radiology INI-CET 2022 ID: MF7536

Whole-body CT scan protocol for a trauma patient includes all, except _____.

A CT Head

B CT Cervical spine

C ✓ CT Limbs

D CT Abdomen

Correct Answer: CT Limbs

The **whole-body CT** of a **trauma** patient does **not** include **CT limbs**.

Whole-body CT or trauma CT or panscan is increasingly being used in the context of trauma patients with a dangerous mechanism of injury:

1. Non-contrast CT of head
2. Non-contrast CT of neck
3. Contrast-enhanced CT of chest (arterial phase)
4. Contrast-enhanced CT of abdomen (portal venous phase)

The purpose of the scan is to rapidly evaluate life-threatening injuries and accurately diagnose known and unknown injuries.

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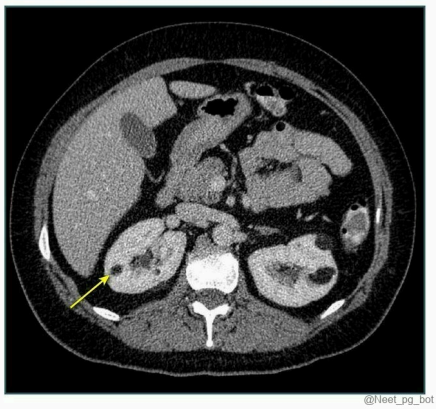
References: Online Resource, https://radiopaedia.org/articles/whole-body-ct-protocol

Q127. Radiology INI-CET 2022 ID: MF7533

A patient presents with hematuria and a mass in the right kidney on a CT scan, as shown below. What is the diagnosis?



A Renal cell carcinoma

B Renal oncocytoma

C ✓ Renal angiomyolipoma

D Renal cyst

Correct Answer: Renal angiomyolipoma

The CT image showing renal lesions that has **hypo-attenuating components consistent with macroscopic fat** is highly suggestive of **renal angiomyolipoma.**

Angiomyolipomas are **benign tumors** that occur sporadically or in association with **tuberous sclerosis**. The mean age of presentation is **43 years** and there is a female predilection. It can present as spontaneous **retroperitoneal hemorrhage**, especially with larger tumors. The patient can present with flank pain, palpable **mass**, **hematuria**, urinary tract infections, hypertension, or renal failure.

Imaging in renal angiomyolipoma: Demonstration of macroscopic fat is the main part of making a diagnosis. **USG** may show **hyperechoic lesions** in the cortex with **posterior acoustic shadowing**. In patients with tuberous sclerosis, the entire kidney is affected and produces an echogenic appearance with loss of normal corticomedullary junction.

CT findings:

- Cortical lesions that demonstrate **macroscopic fat (less than -20 HU)**
- **Absence** of **ossification or calcification**
- The presence of macroscopic fat is strongly suggestive of this condition, but not pathognomonic as RCC may rarely demonstrate macroscopic fat.
- Approximately 5% may be fat-poor.

MRI: It is an excellent tool to demonstrate intramural fat components. Two sequences are used:

- Fat-saturated techniques
- Phase-in and phase-out imaging that shows **India ink artifacts** at the junction between fat and non-fat components.

Digital subtraction angiography: They are hypervascular lesions showing a '**sunburst appearance**' in the arterial phase and an '**onion peel appearance**' in the venous phase.

They usually do not require any intervention. mTOR inhibitors like **everolimus** can be used to decrease the size of the lesion.

Other options:

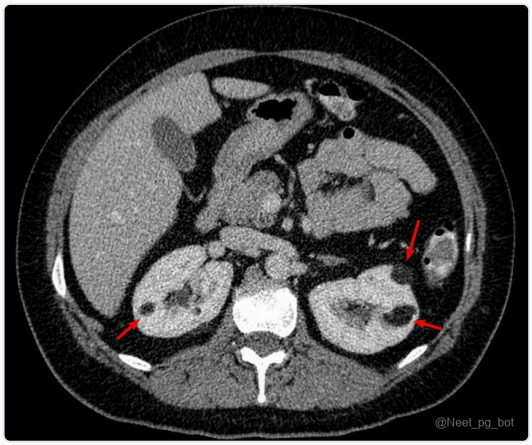
Renal cell carcinoma (RCC): It is the most common malignant tumor of the kidney. It usually occurs in the age group **50-70 years** and **macroscopic hematuria** is seen in about 60% of the cases. It is a differential for angiomyolipoma but it is important to note that **macroscopic fat** is almost always **associated with ossification or calcification.**

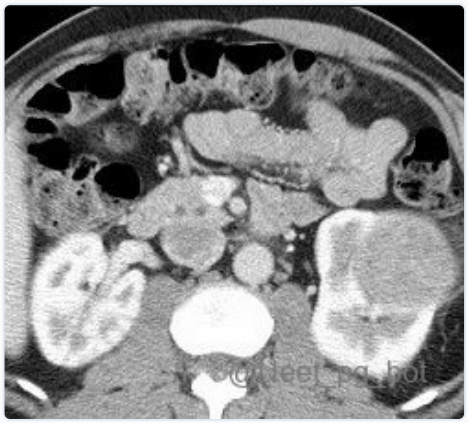
Renal oncocytoma: They are relatively benign in nature and usually present in the 6th to 7th decade. An important point regarding this tumor is its **similarity with RCC** and hence it is very difficult to differentiate between them preoperatively. A sharp **central**

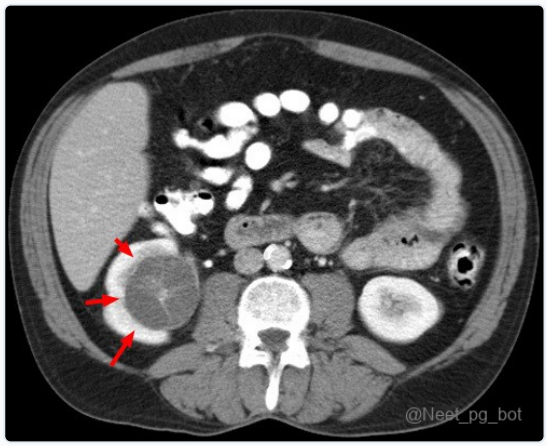
stellate scar may be seen in one-third of the cases. But RCC may also show this stellate scar appearance in some cases.

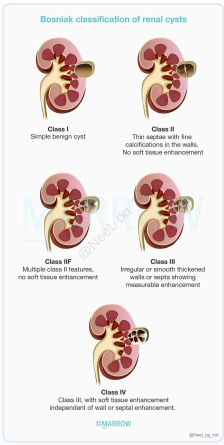
Renal cyst: It is a broader term that includes the cystic lesions of the kidney. They can be simple cysts that are benign. Complex or complicated cystic lesions needs evaluation and can be malignant. Bosniak classification of renal cysts is given above.

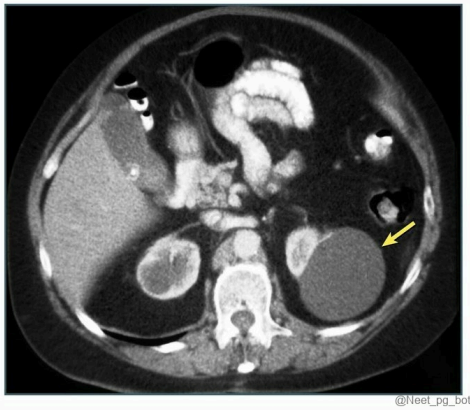
Image showing a simple renal cyst:











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References: Online Resource, https://radiopaedia.org/articles/renal-angiomyolipoma
Grainger & Allison's Diagnostic Radiology 6th edition, Page no 834,883
Online Resource, https://radiopaedia.org/articles/renal-oncocytoma
Online Resource, https://radiopaedia.org/articles/renal-cell-carcinoma-1?lang=us
Online Resource, https://radiopaedia.org/articles/renal-cyst-1

Q128. Radiology INI-CET 2022 ID: MF7531

Egg shell calcification is seen in all the following, except _____.

A Silicosis

B Sarcoidosis

C Post-radiation therapy lymph nodes

D ✓ Metastatic lymph nodes

Correct Answer: Metastatic lymph nodes

Eggshell calcification is **not** seen in the case of **metastatic lymph nodes.**

A ring of calcification at the periphery of a lymph node is known as eggshell calcification.

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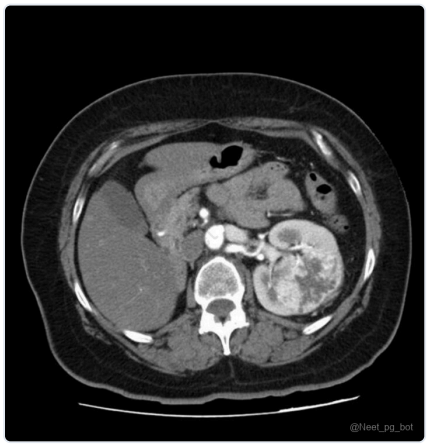
References: Online Resource, https://radiopaedia.org/articles/eggshell-calcification-lymph-nodes
Grainger and Allison, textbook of medical imaging, 7th edition, Page no. 79,82,215,227

Q129. Radiology INI-CET 2022 ID: MF7607

A 55-year-old man presents with hematuria and a mass in the left kidney on a CECT scan, as shown below. What is the diagnosis?



A Renal angiomyolipoma

B Renal oncocytoma

C ✓ Renal cell carcinoma

D Renal cyst

Correct Answer: Renal cell carcinoma

The history of an **elderly male** presenting with **painless hematuria, renal mass,** and an axial renal cortical phase CECT image showing renal lesions with **irregular enhancement** is suggestive of **renal cell carcinoma (RCC)**.

Renal cell carcinoma (RCC) is the most common malignant tumor of the kidney. It usually occurs in the age group **50–70 years** and **macroscopic hematuria** is seen in about 60% of the cases. It usually presents with painless gross hematuria, flank pain, or painless flank mass.

Contrast-enhanced CT is the investigation of choice for RCC. On CT, small lesions **enhance** homogeneously, whereas **larger** lesions have **irregular enhancement** due to areas of necrosis. The clear cell subtype may show much stronger enhancement is a differential for angiomyolipoma. Still, it is essential to note that **macroscopic fat** is almost always **associated with ossification or calcification.**

T1-weighted MRI is often heterogeneous due to necrosis, hemorrhage, and solid components. **T2-weighted MRI** is **hyperdense** for **clear cell** RCC and **hypodense** for **papillary** RCC. Thus, a T2-weighted MRI can be used for histological diagnosis. Contrast-enhanced MRI is the investigation of choice for detecting vascular invasion of RCC.

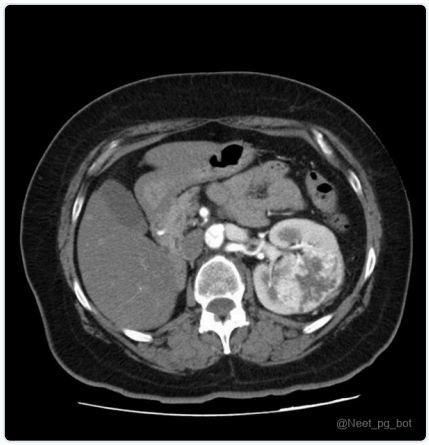
Other options:

Renal angiomyolipoma: Angiomyolipomas are **benign** tumors that occur sporadically or in association with tuberous sclerosis. The mean age of presentation is **43 years**, and there is a **female** predilection. CT findings show **cortical lesions** that demonstrate **macroscopic fat** (less than -20 HU) and **absence** of **ossification** or **calcification.**

Renal oncocytoma: They are relatively benign in nature and usually present in the 6th to 7th decade. An important point regarding this tumor is its **similarity with RCC**, and hence, it is very difficult to differentiate between them preoperatively. A sharp **central stellate scar** may be seen in one-third of the cases. But RCC may also show this stellate scar appearance in some cases.

Renal cyst: It is a broader term that includes the cystic lesions of the kidney. They can be simple cysts that are benign. Complex or complicated cystic lesions needs evaluation and can be malignant. Bosniak classification of renal cysts is given above.

Image showing a simple renal cyst: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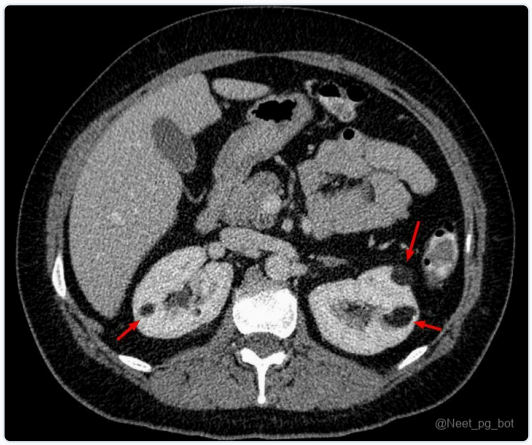
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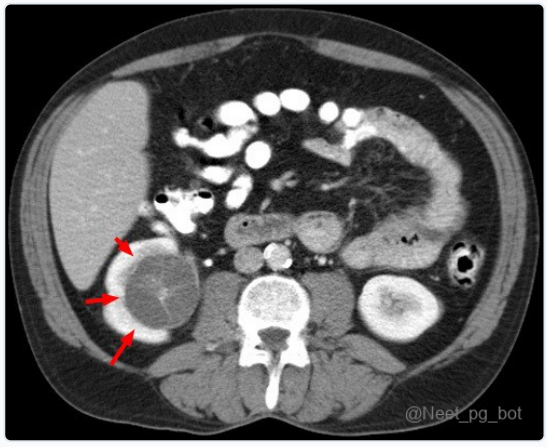
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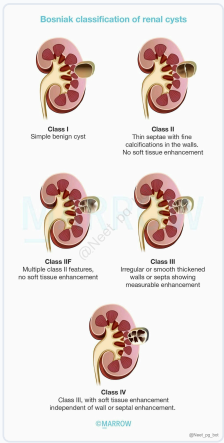
Bosniak classification of renal cysts
Class I
Simple benign cyst
Class II
Thin septae with fine calcifications in the walls. No soft tissue enhancement
Class IIF
Multiple class II features, no soft tissue enhancement
Class III
Irregular or smooth thickened walls or septa showing measurable enhancement
Class IV
Class III, with soft tissue enhancement independent of wall or septal enhancement.
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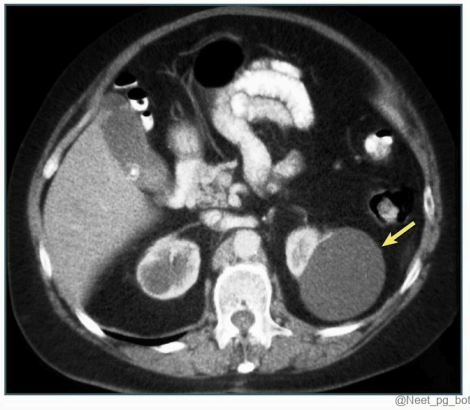

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References: Online Resource, https://radiopaedia.org/articles/renal-angiomyolipoma
Grainger & Allison's Diagnostic Radiology 6th edition, Page no 726,727
Online Resource, https://radiopaedia.org/articles/renal-oncocytoma
Online Resource, https://radiopaedia.org/articles/renal-cell-carcinoma-1?lang=us
Online Resource, https://radiopaedia.org/articles/renal-cyst-1

Q130. Radiology INI-CET 2022 ID: MF7530

A patient who was diagnosed with prostate cancer is being investigated. The bone scan is reported as a super scan with increased uptake in the bones and reduced activity in the kidney and soft tissue. What is the reason for this super scan appearance?

A ✓ Increased uptake due to diffuse metastasis

B Increased uptake by carcinoma prostate

C Increased uptake by the bone

D Decreased uptake by adrenal glands and kidney

Correct Answer: Increased uptake due to diffuse metastasis

The **super scan** appearance is due to increased uptake due to **diffuse metastasis**.

Diffuse osteoblastic metastatic diseases such as carcinoma breast and prostate carcinoma lead to a generalized **increase in the skeletal uptake** relative to soft tissue and absent or **faint renal activity** on a bone scan. This is known as a super scan. It may also be seen in generalized bone diseases such as **hyperparathyroidism**.

Carcinoma of the prostate is the most common malignant tumor in men over the age of 65 years. Most cancers are clinically silent and are detected on routine rectal examination or PSA measurements. Grossly, malignant tissue is **gritty and firm** in consistency (in contrast to the spongy appearance of benign tissue) and thus more readily palpable on **digital rectal examination.** They usually arise in the **peripheral zone** of the gland.

Local extension most commonly involves the periprostatic tissue, seminal vesicles, and the base of the urinary bladder. Metastasis spreads via the lymphatics to the obturator nodes and the para-aortic nodes. Hematogenous spread occurs chiefly through the **bones** to the axial skeleton and rarely to the viscera. The bony metastasis is **blastic in origin** and involves the **lumbar spine, proximal femur, pelvis, thoracic spine, and ribs** in decreasing order of frequency.

Prognosis and staging: The **Gleason grading system** correlates with the pathological stage, spread, and prognosis. Measurement of **serum PSA** levels may be useful to assist with the diagnosis and management.

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References: Grainger & Allison's Diagnostic Radiology 6th edition, Page no: 1115
Online Resource, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5314662/
Robbins and Cotran Pathologic Basis of Disease, 10e, Page no: 979
Bailey & Love's Short Practice of Surgery, 27th Edition, Page no 1468-1474